

Automatic construction of fractal structures with locally controlled lacunarity



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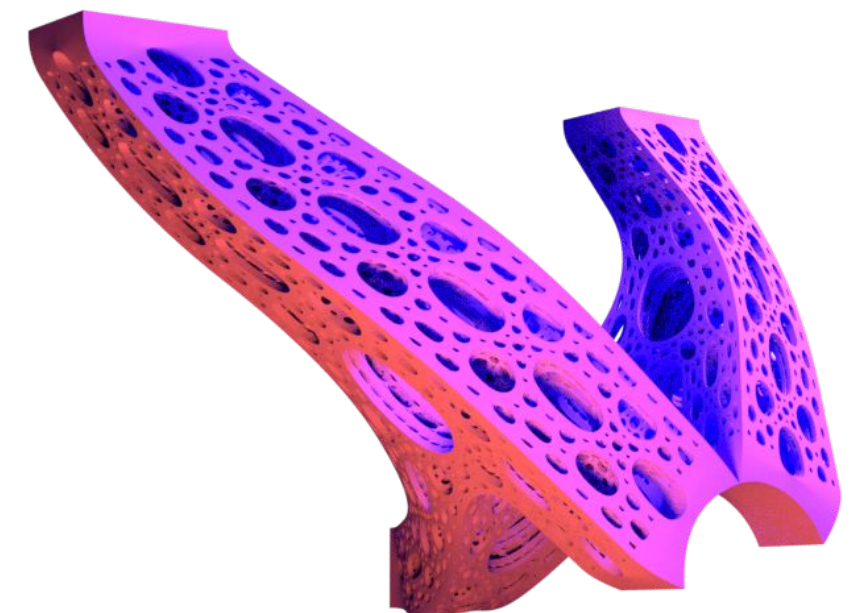
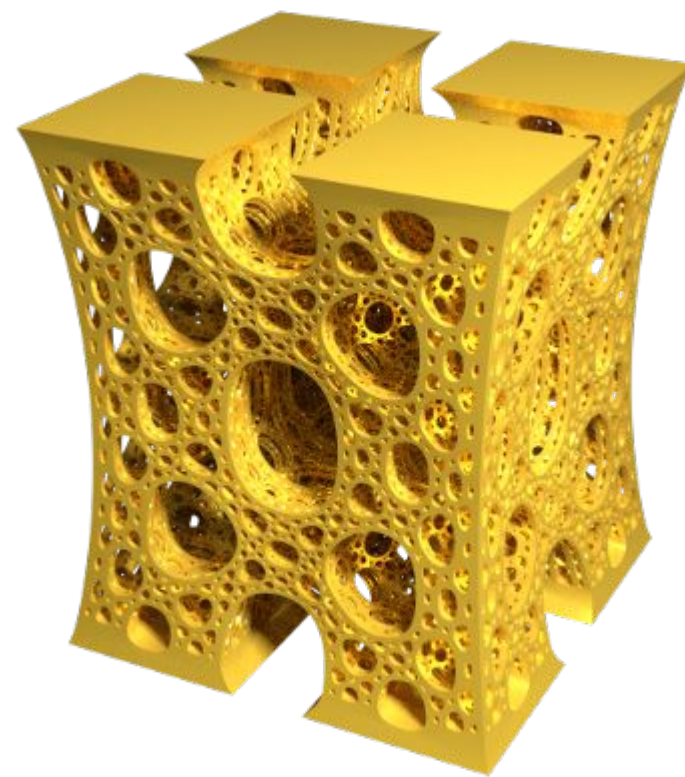
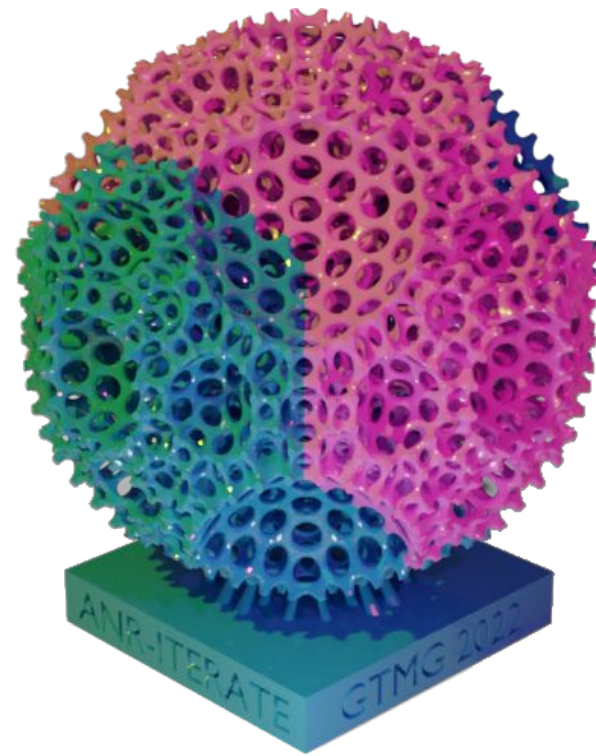
ANR ITERATE – ANR-20-CE46-0011 (2021-2025)

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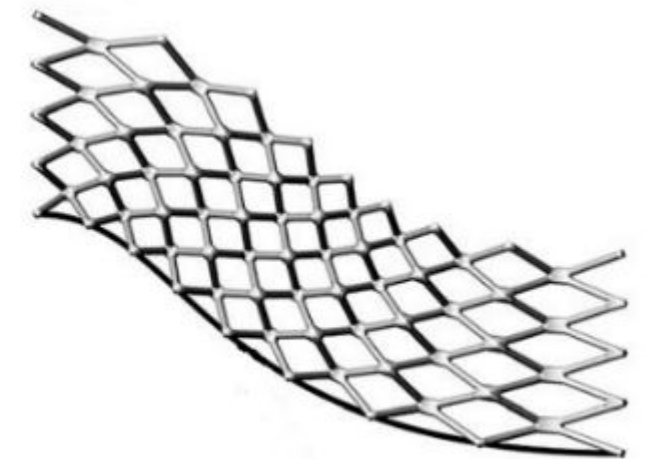
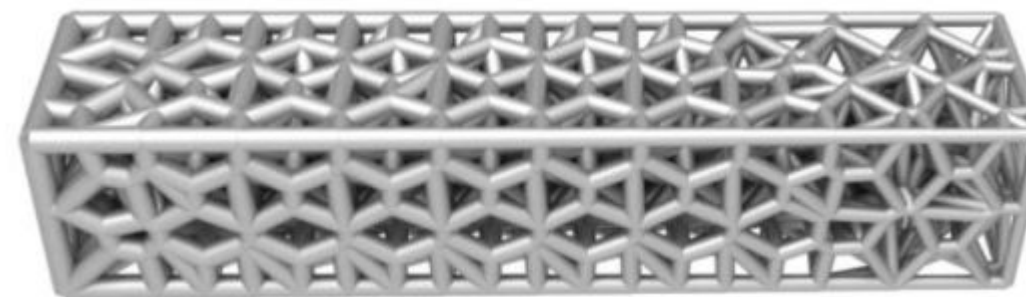
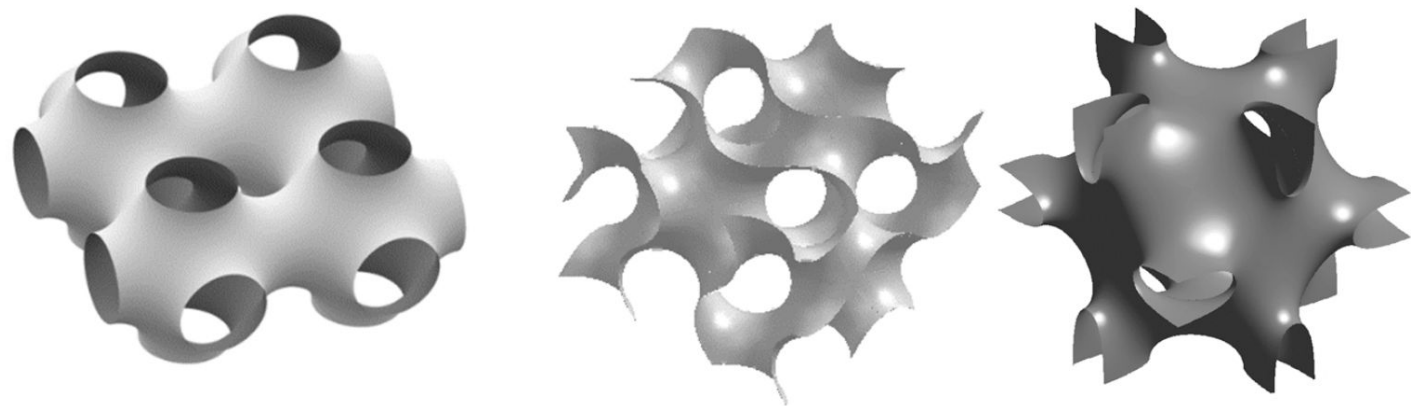
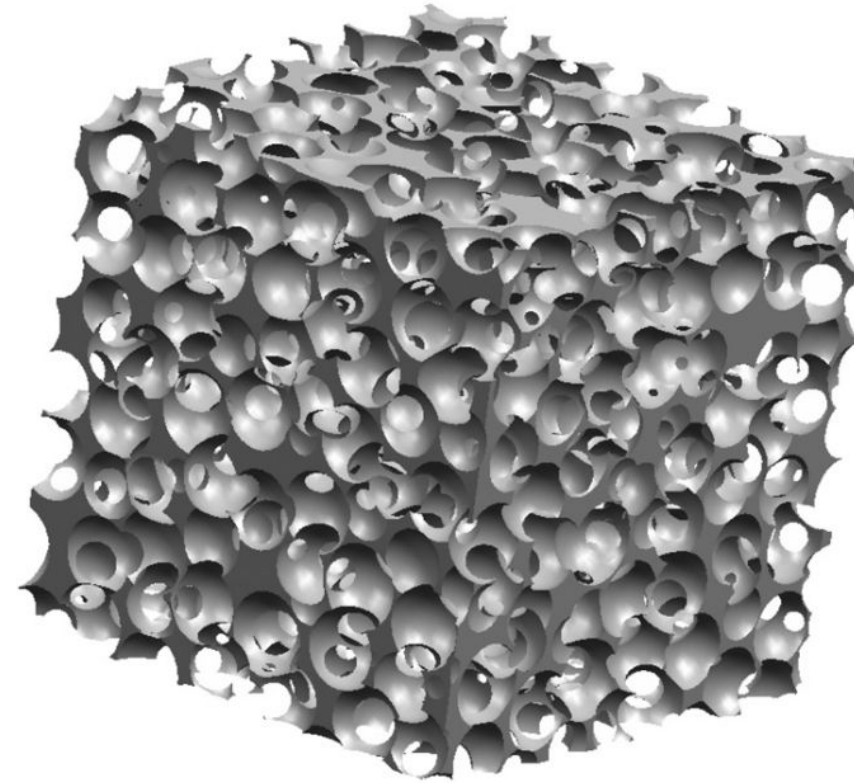
WSCG 2024 - June 3-6, 2024, Prague/Pilsen, Czech Republic

COLLÈGE DOCTORAL
VOTRE DOCTORAT
EN BOURGOGNE-
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- Foam
 - Random
- Lattice structures
 - Regular topology
 - Variation de géométrie
- Minimal surfaces
 - Triply periodic
 - Fixed geometry



- Multi-scale
- Determinist
- Good in several physics properties
 - Mechanic absorption
 - Heat transfer
 - Acoustical properties
- Generalize lattice structures



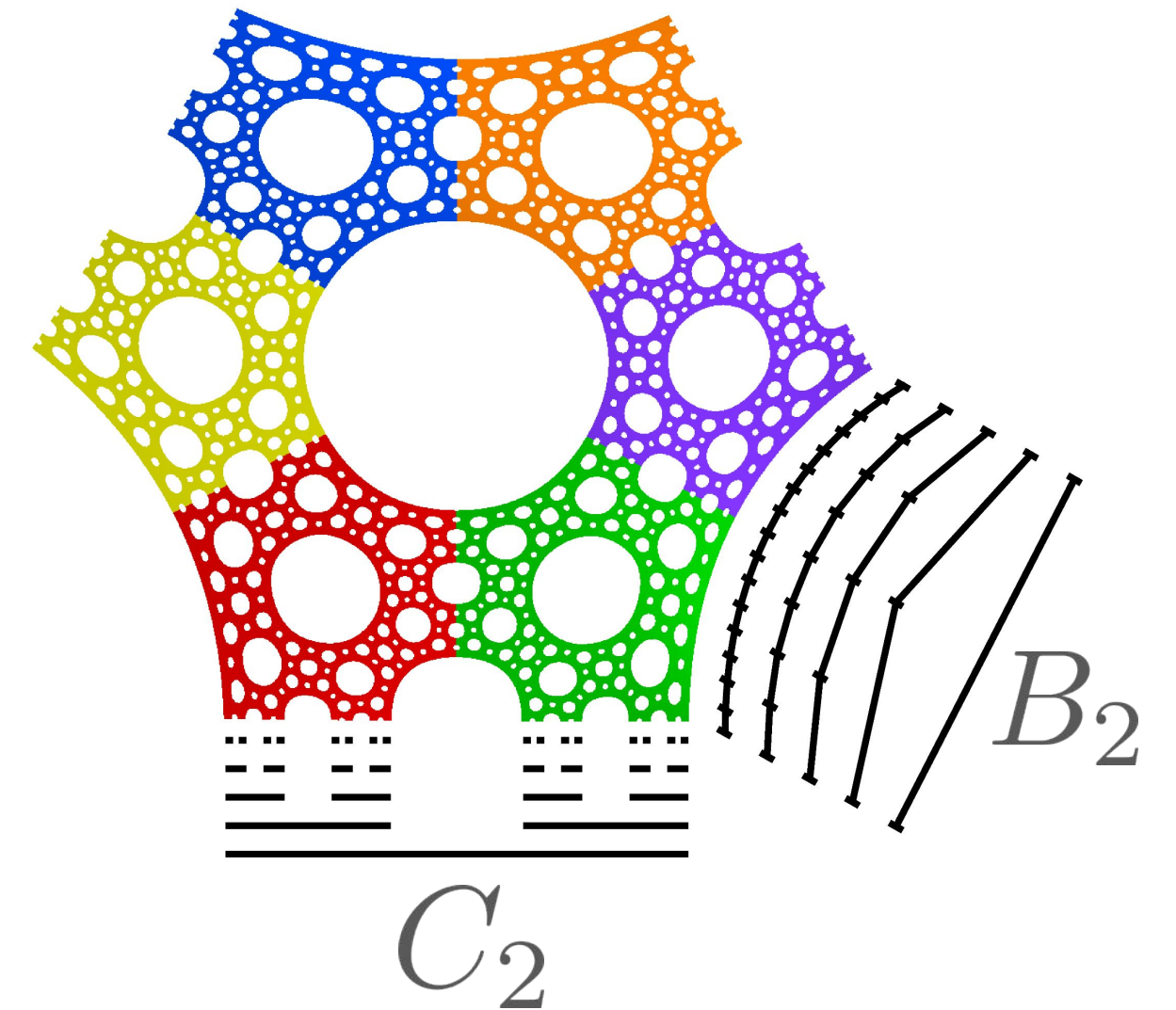
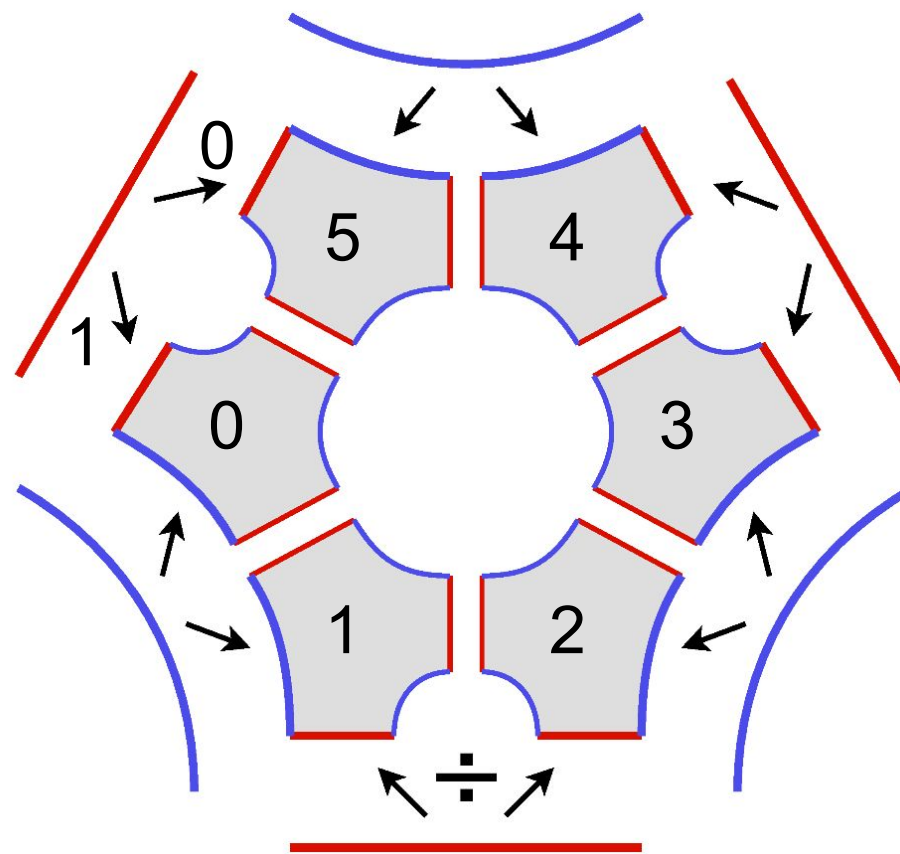
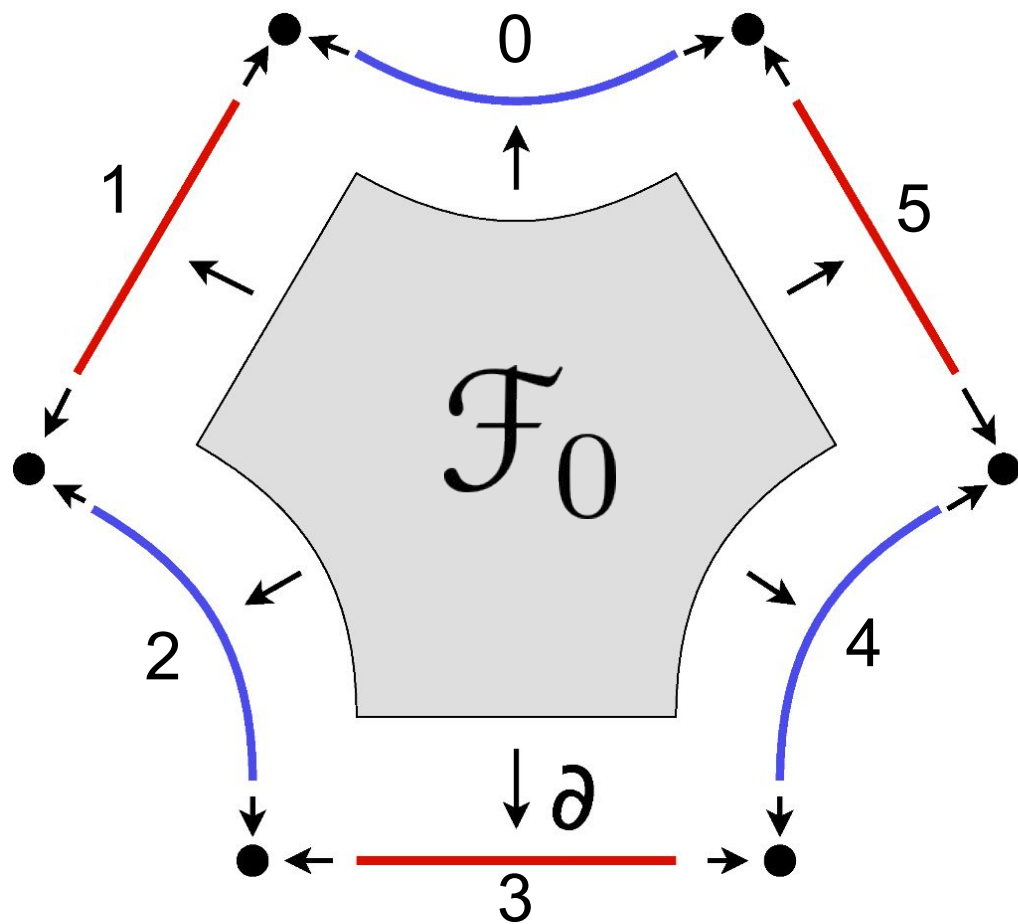
- BC-IFS Model
- Automating subdivision rules
- Local lacunarity control
- Lacunarity quantification

Boundary Controlled Iterated Function System



■ BC-IFS

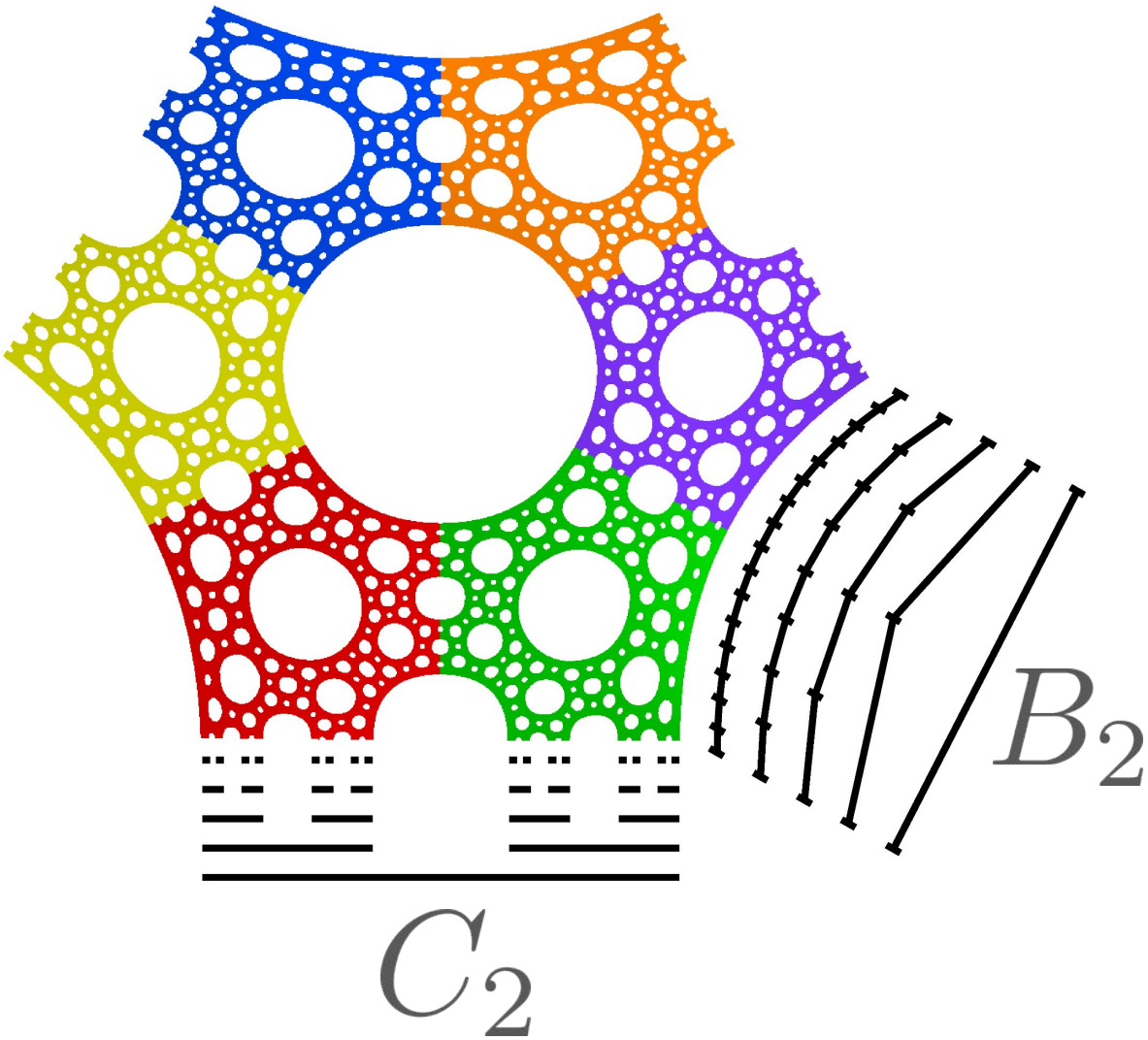
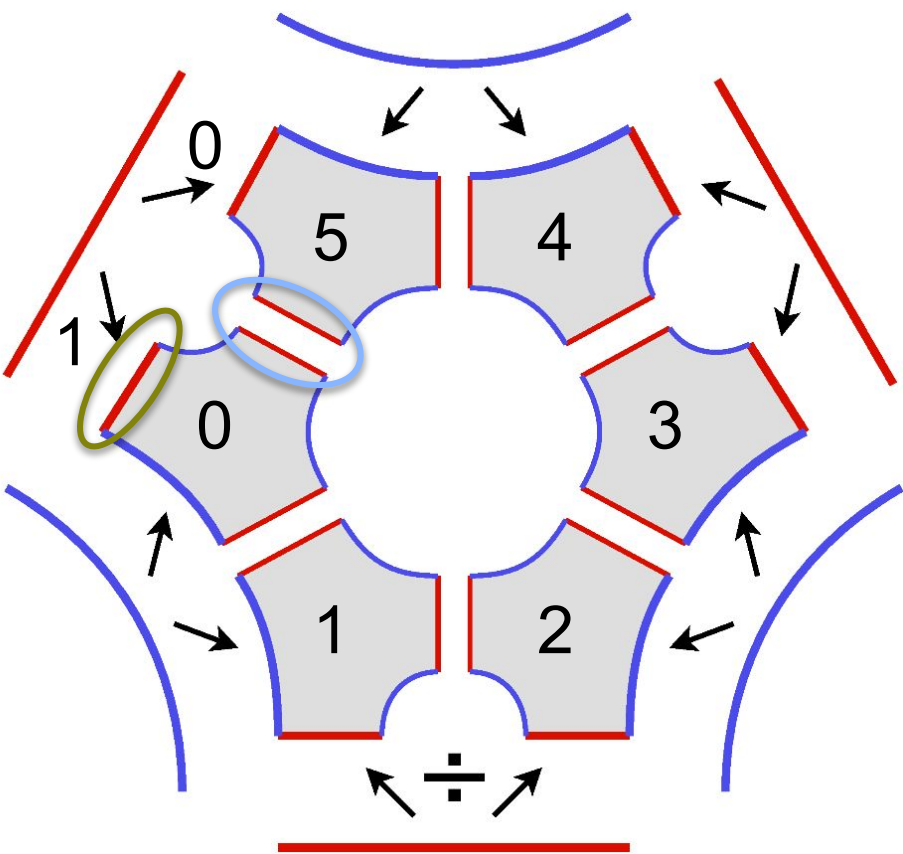
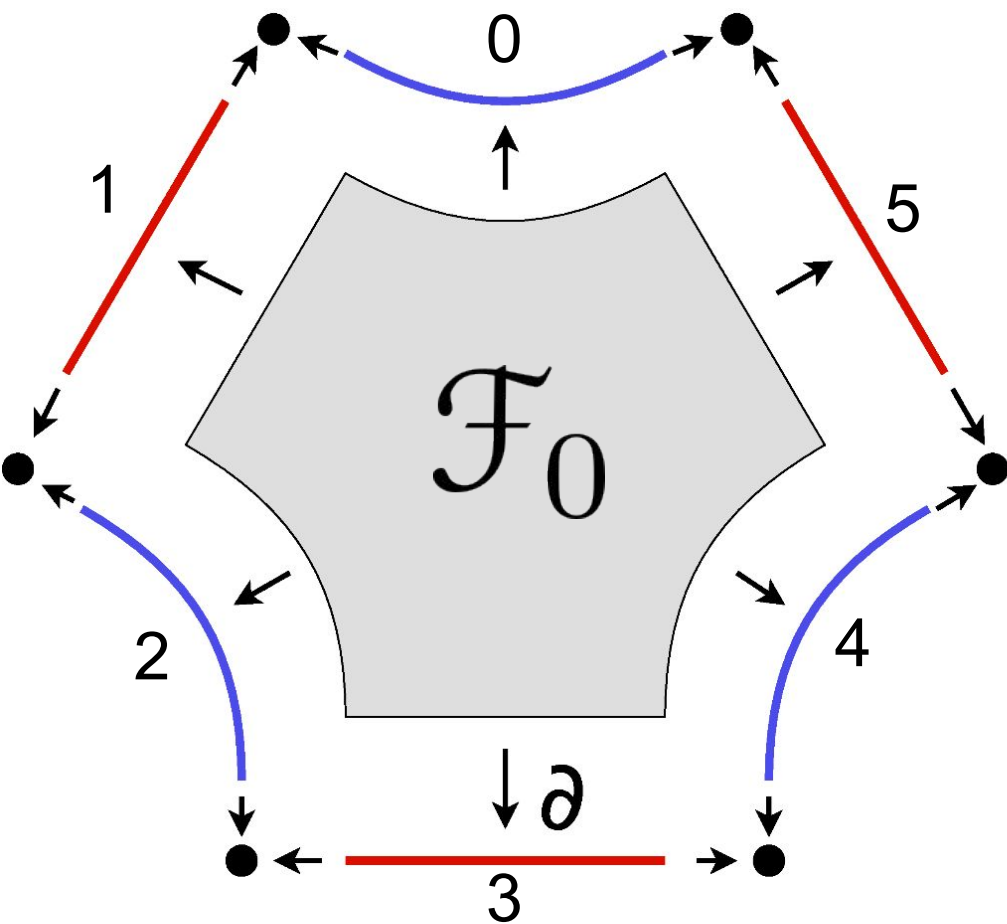
- Cellular decomposition
- Definition of fractal topology
- Incidence and adjacency constraints

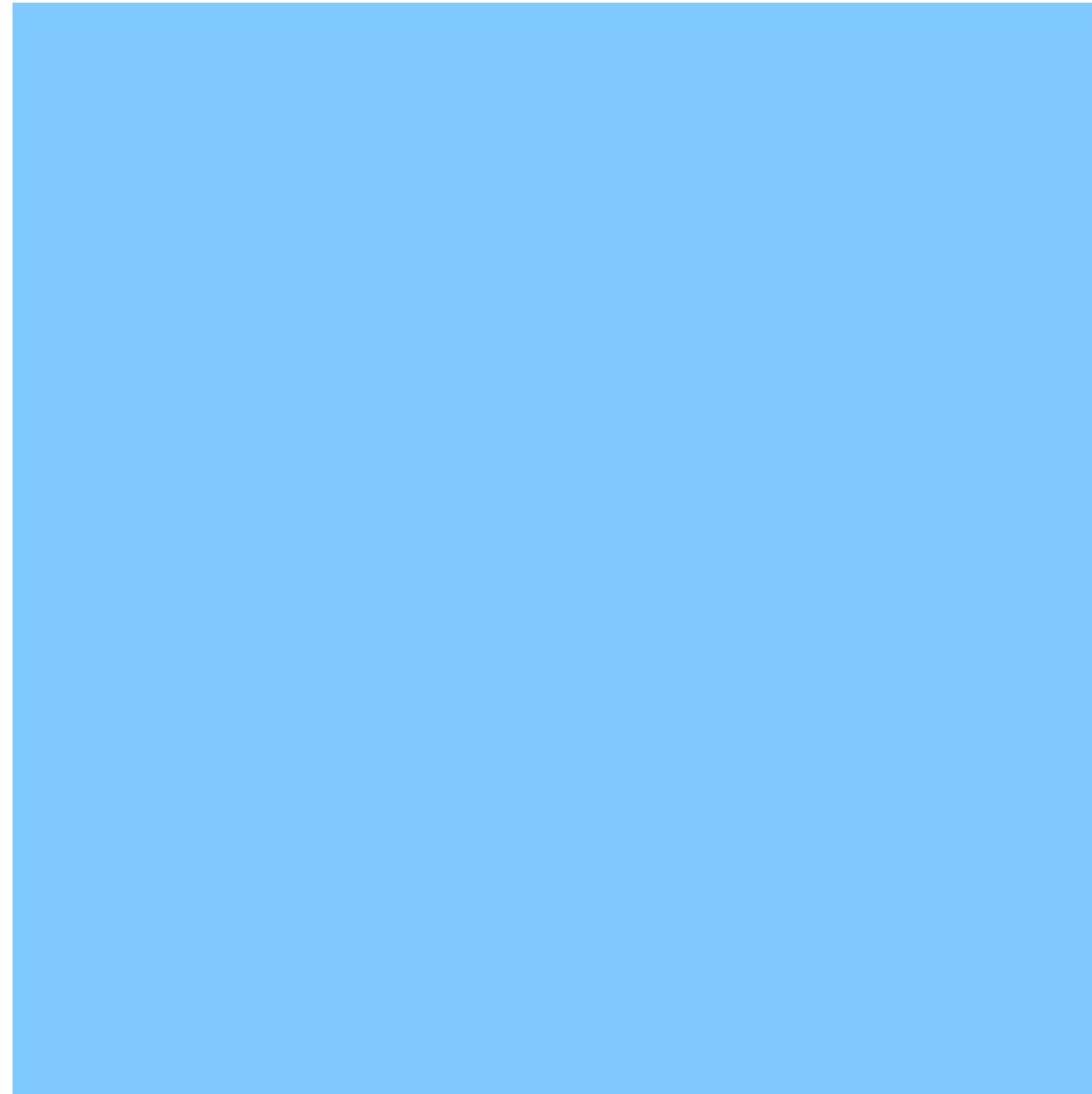


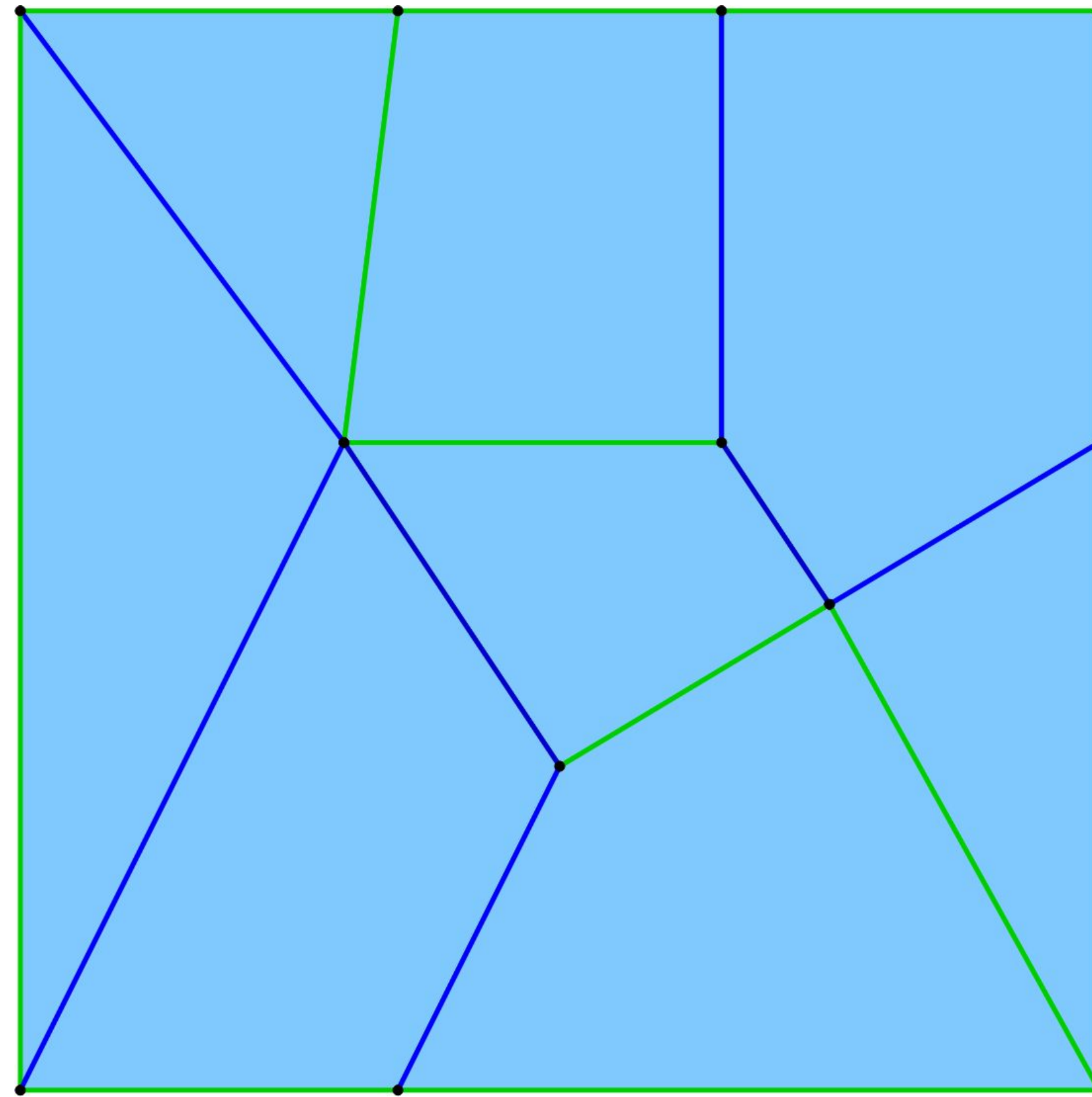
- BC-IFS
 - Cellular decomposition
 - Definition of fractal topology
 - Incidence and adjacency constraints

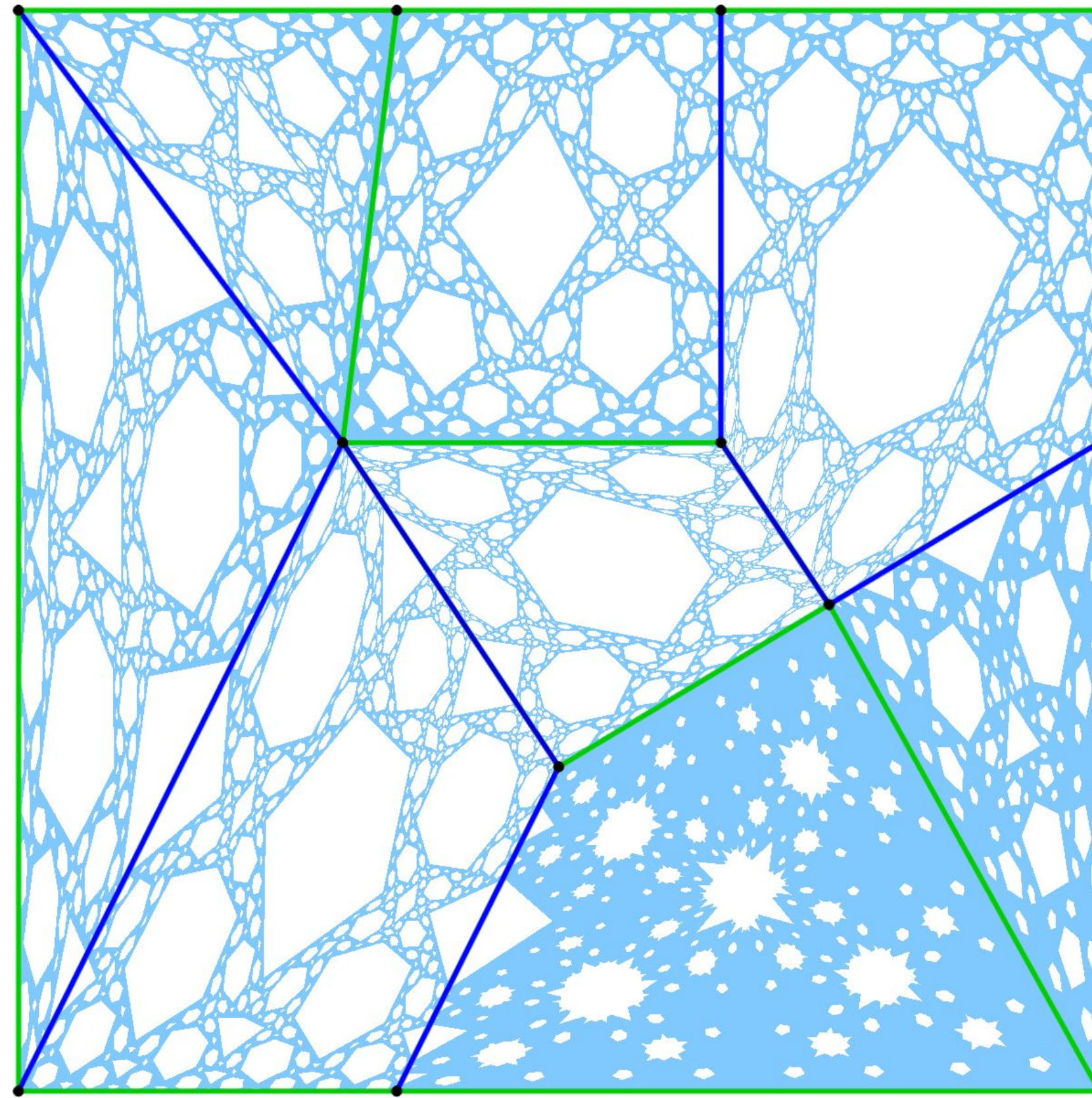
$$\div_0^{\mathcal{F}_0} \partial_5^{\mathcal{F}_0} = \div_5^{\mathcal{F}_0} \partial_3^{\mathcal{F}_0}$$

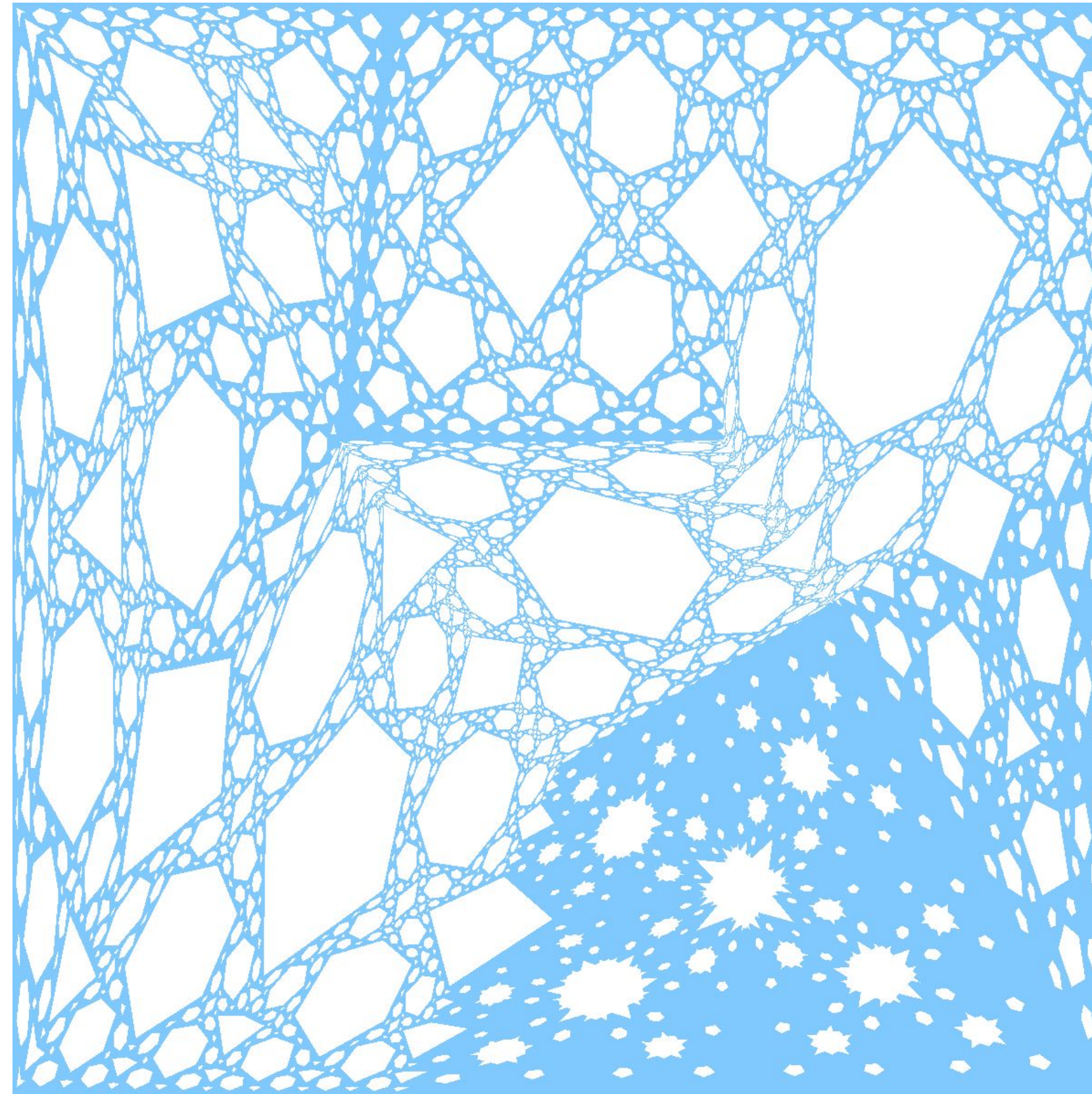
$$\div_0^{\mathcal{F}_0} \partial_1^{\mathcal{F}_0} = \partial_1^{\mathcal{F}_0} \div_1^{C_2}$$





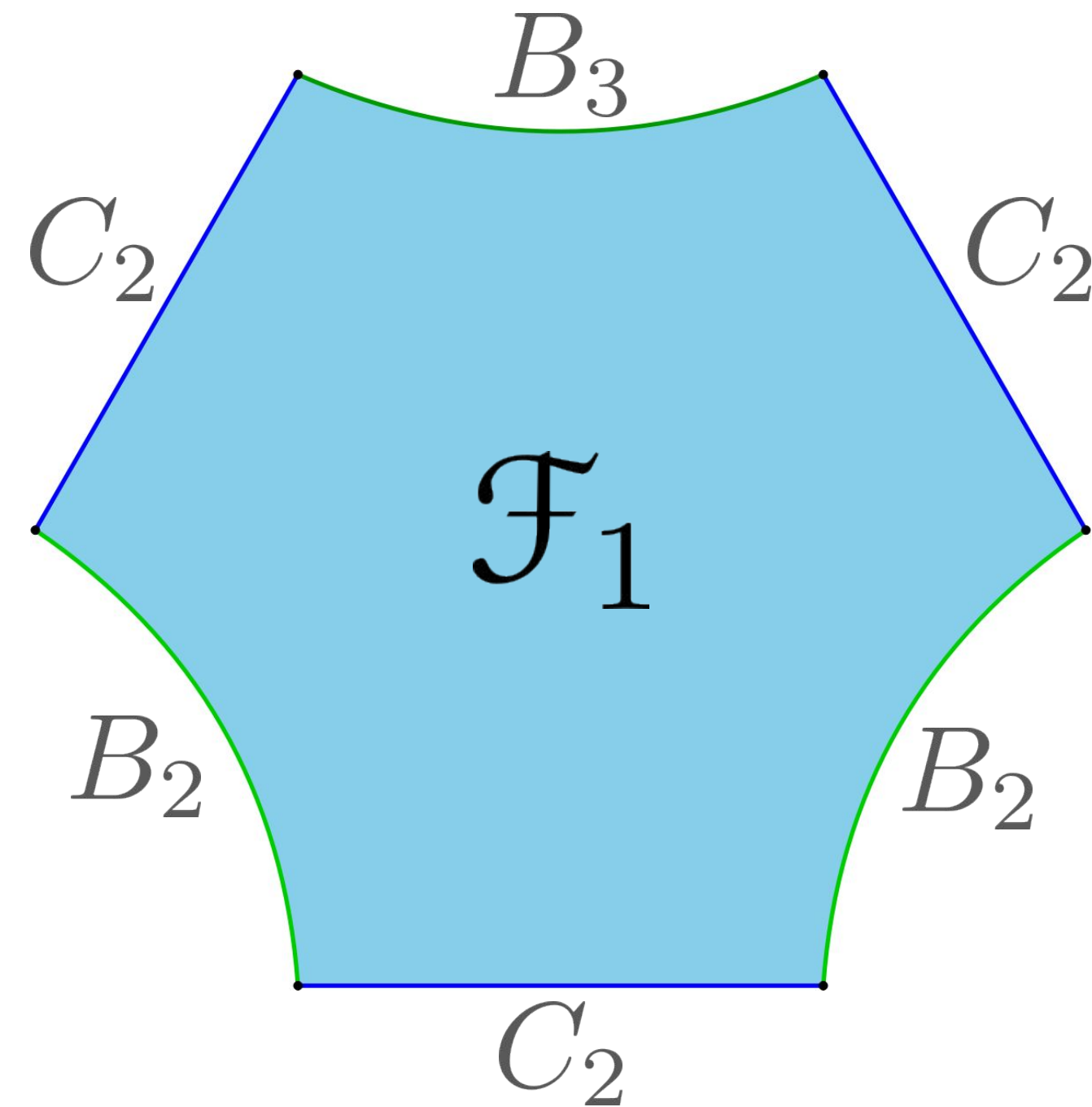


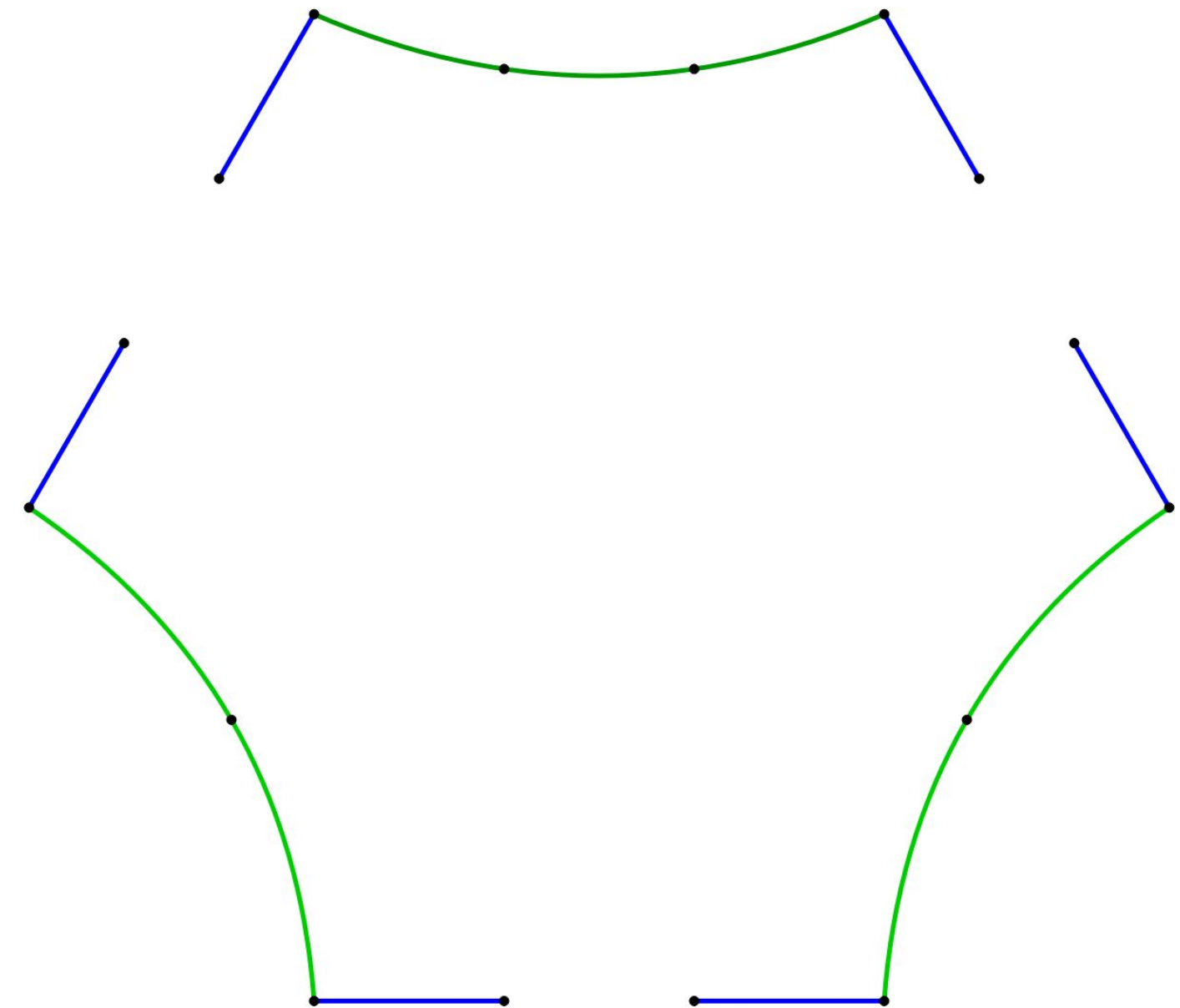
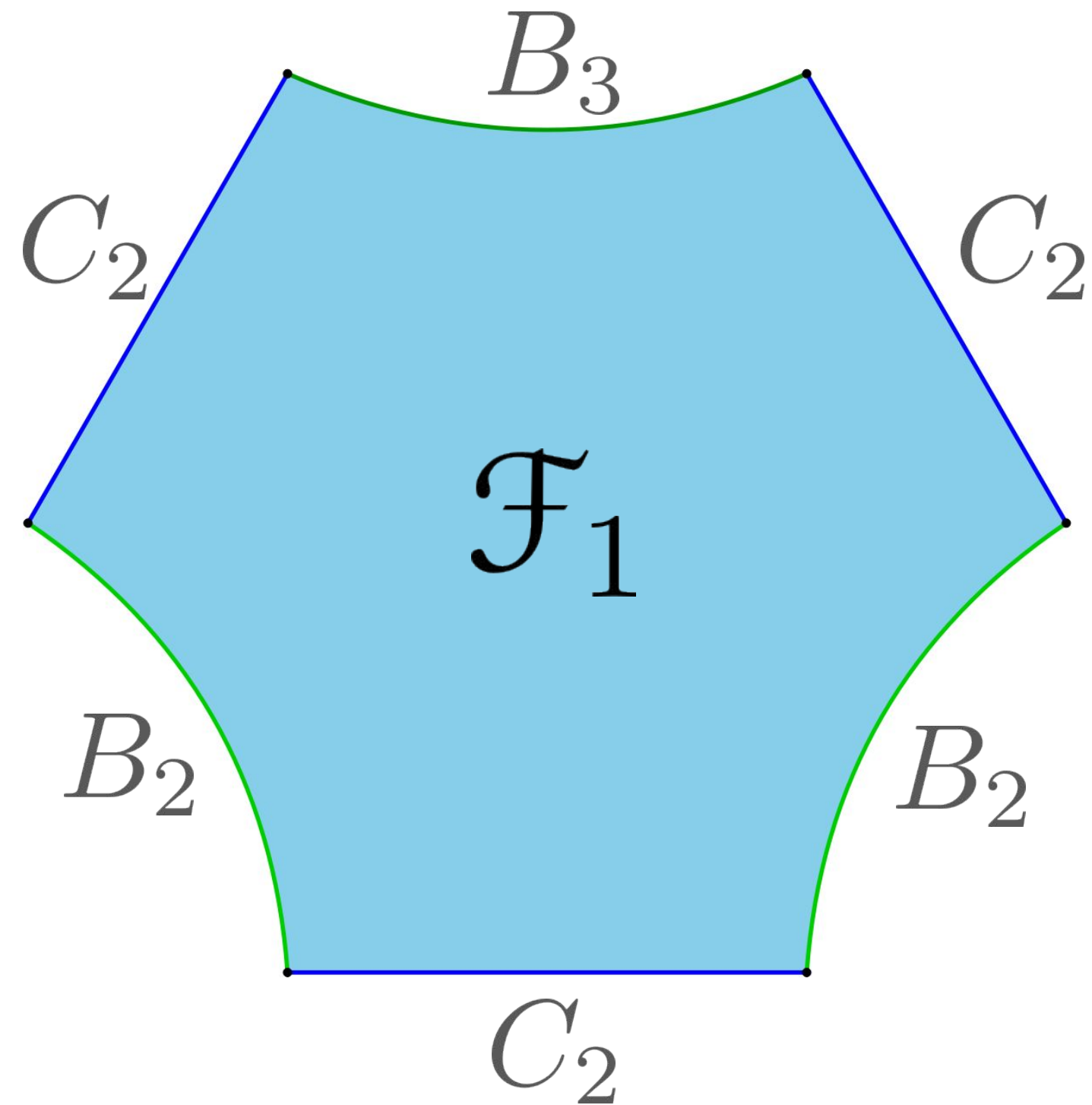


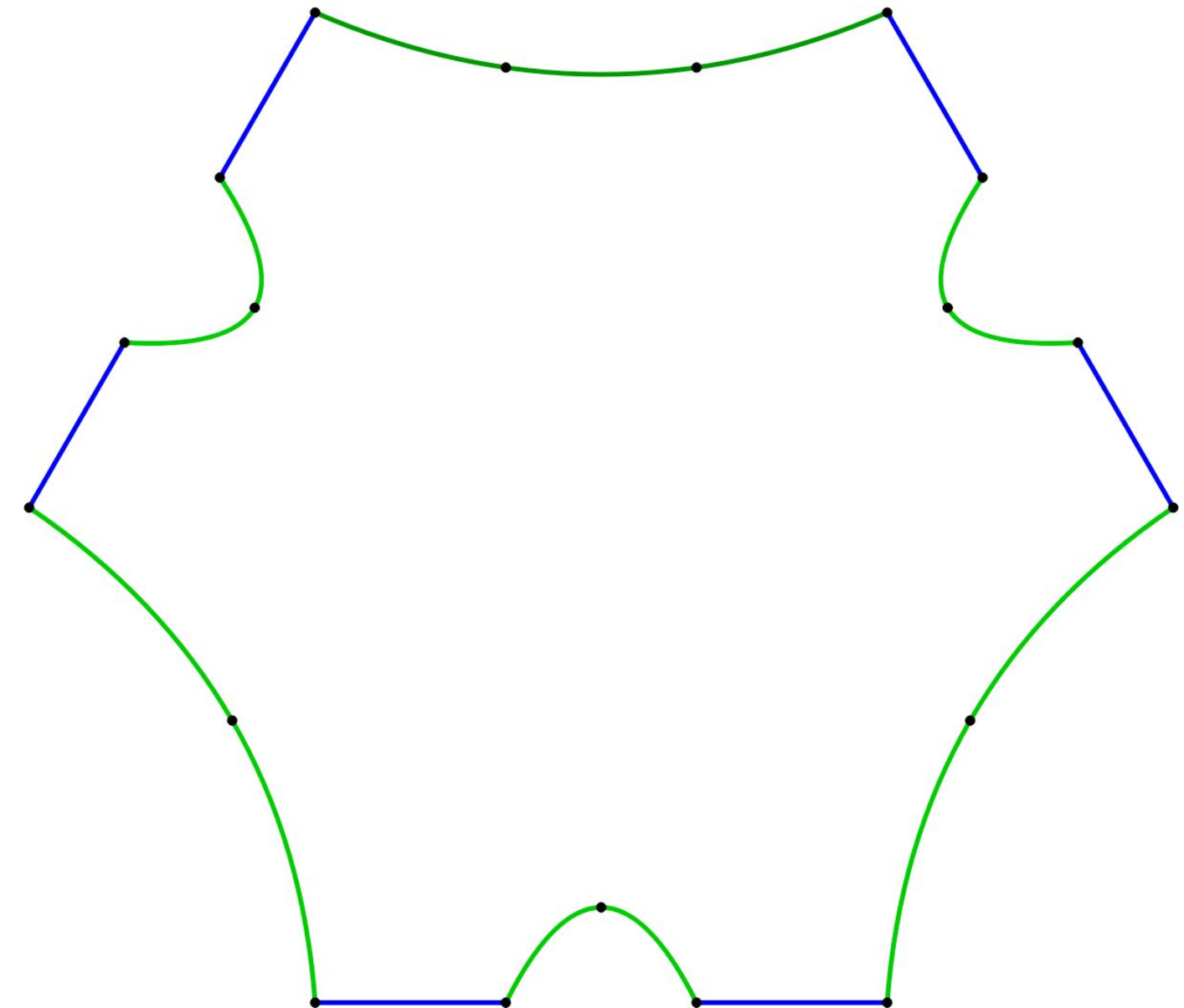
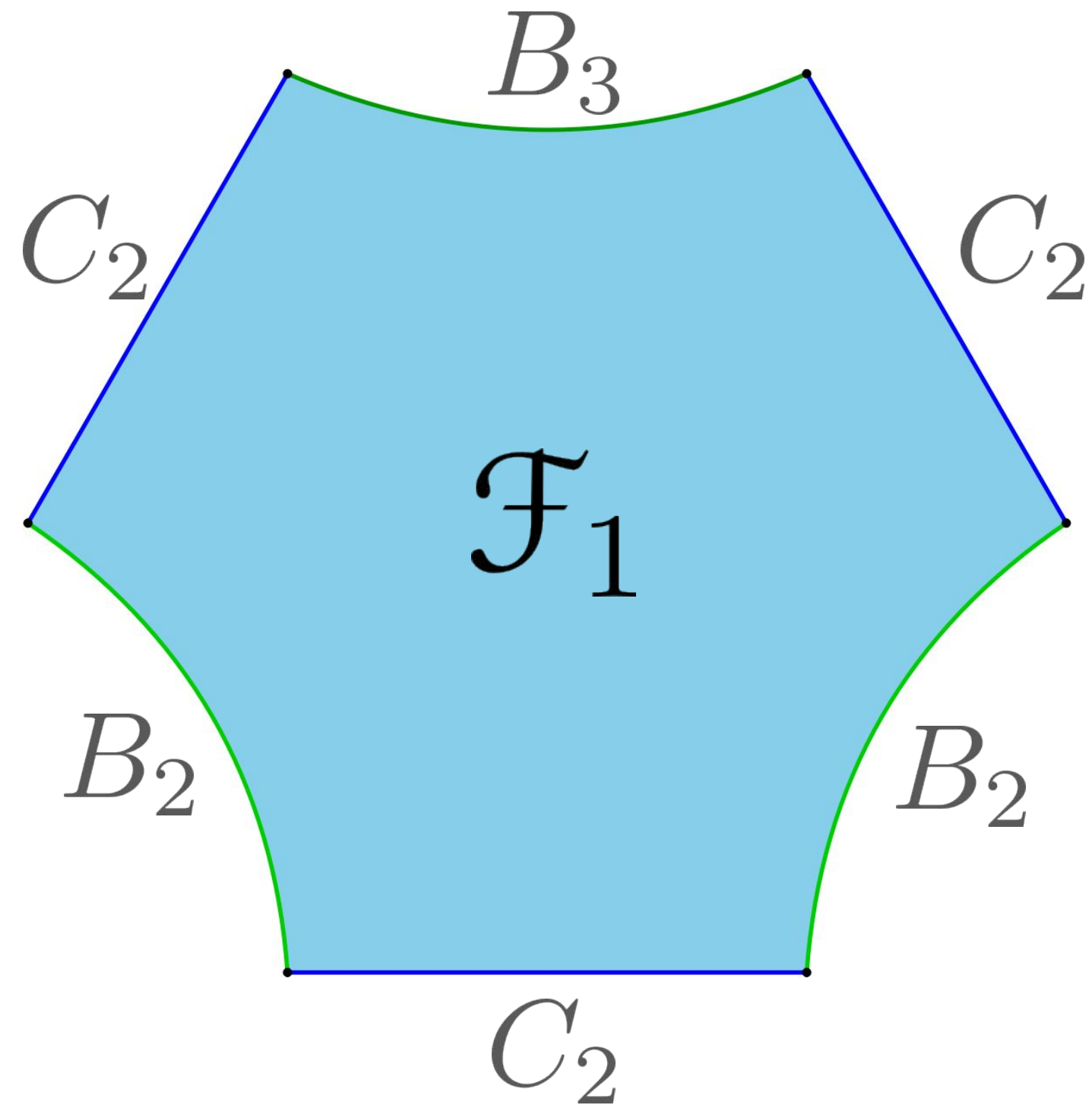


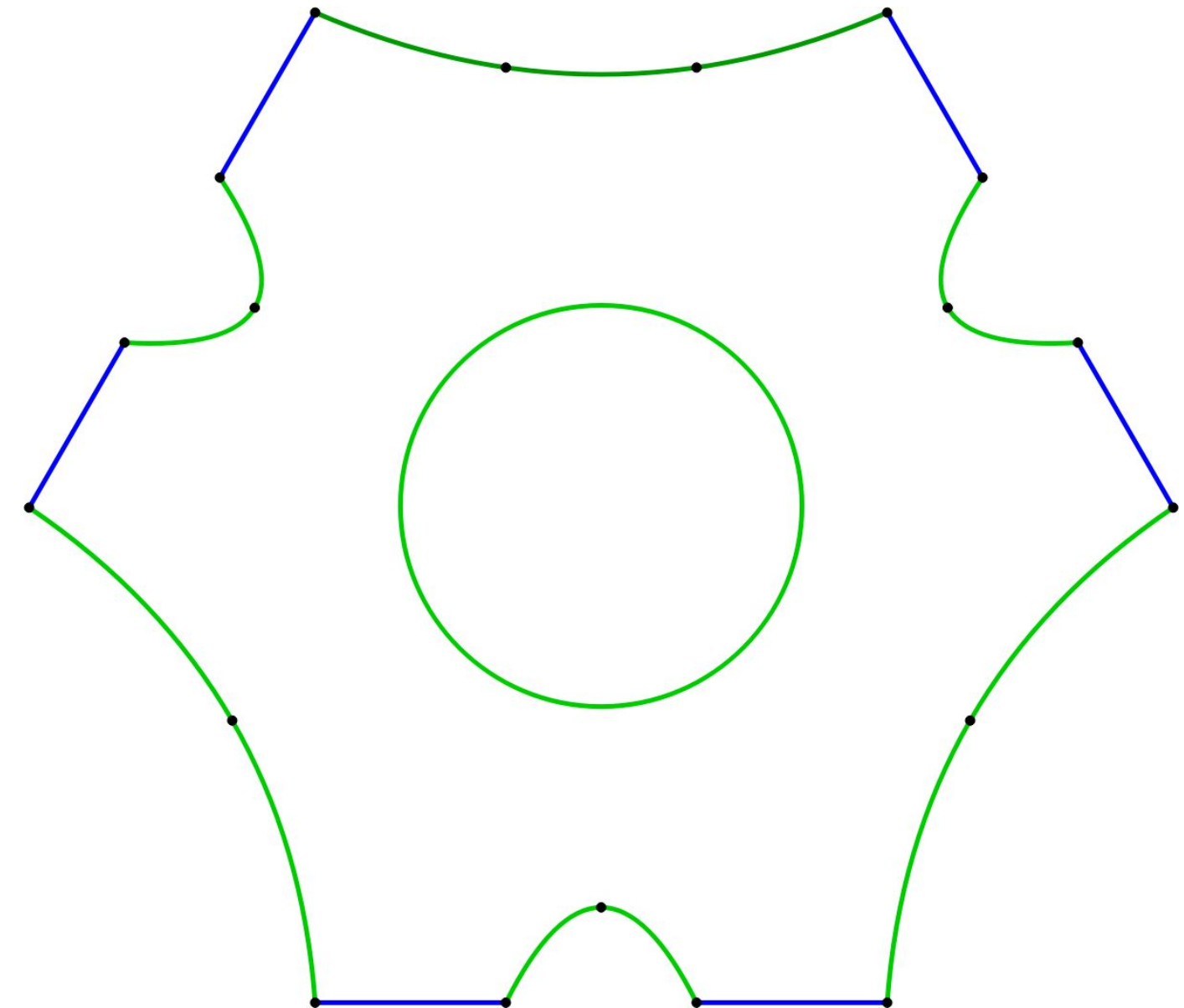
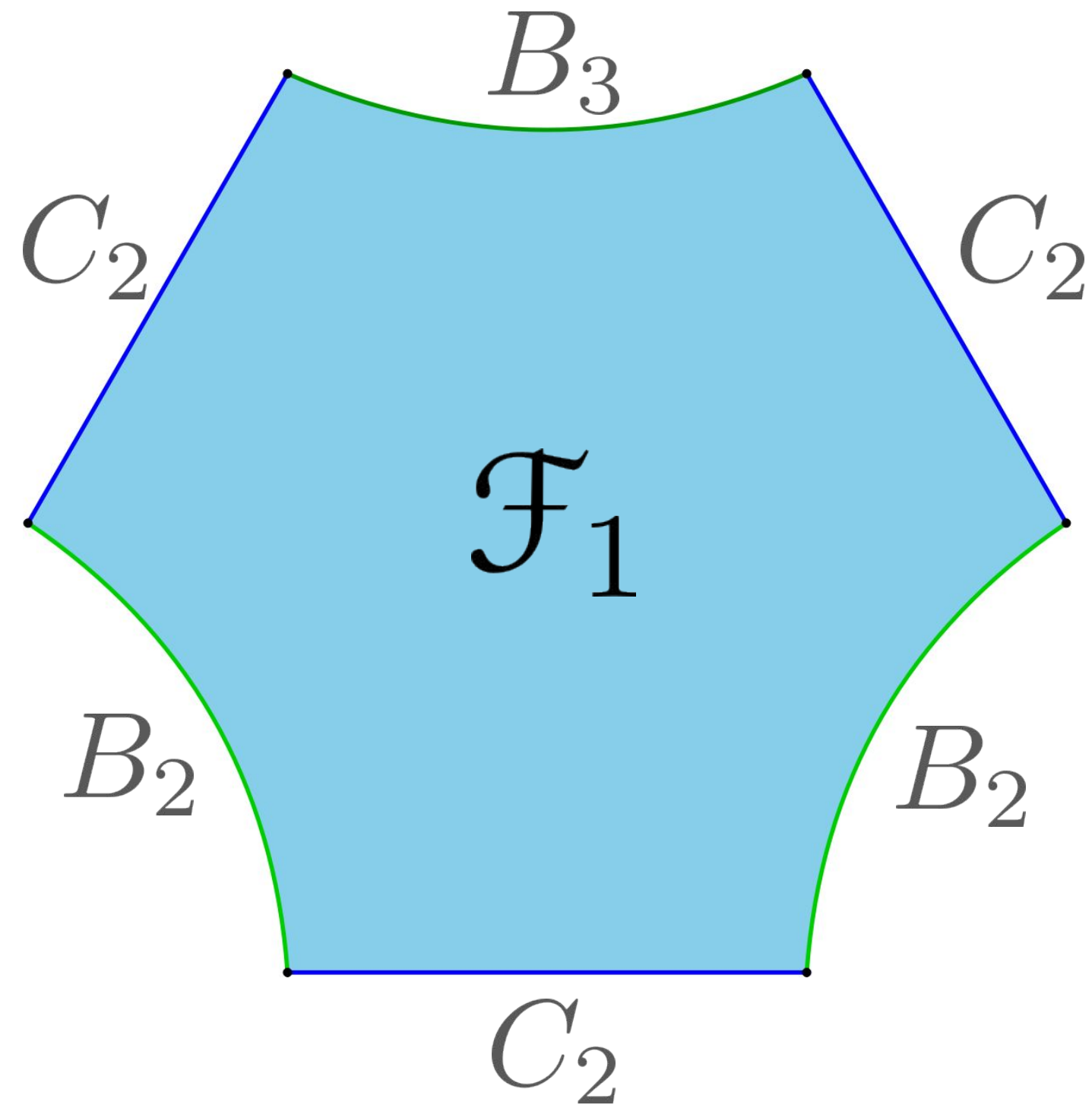
Automating subdivision rules

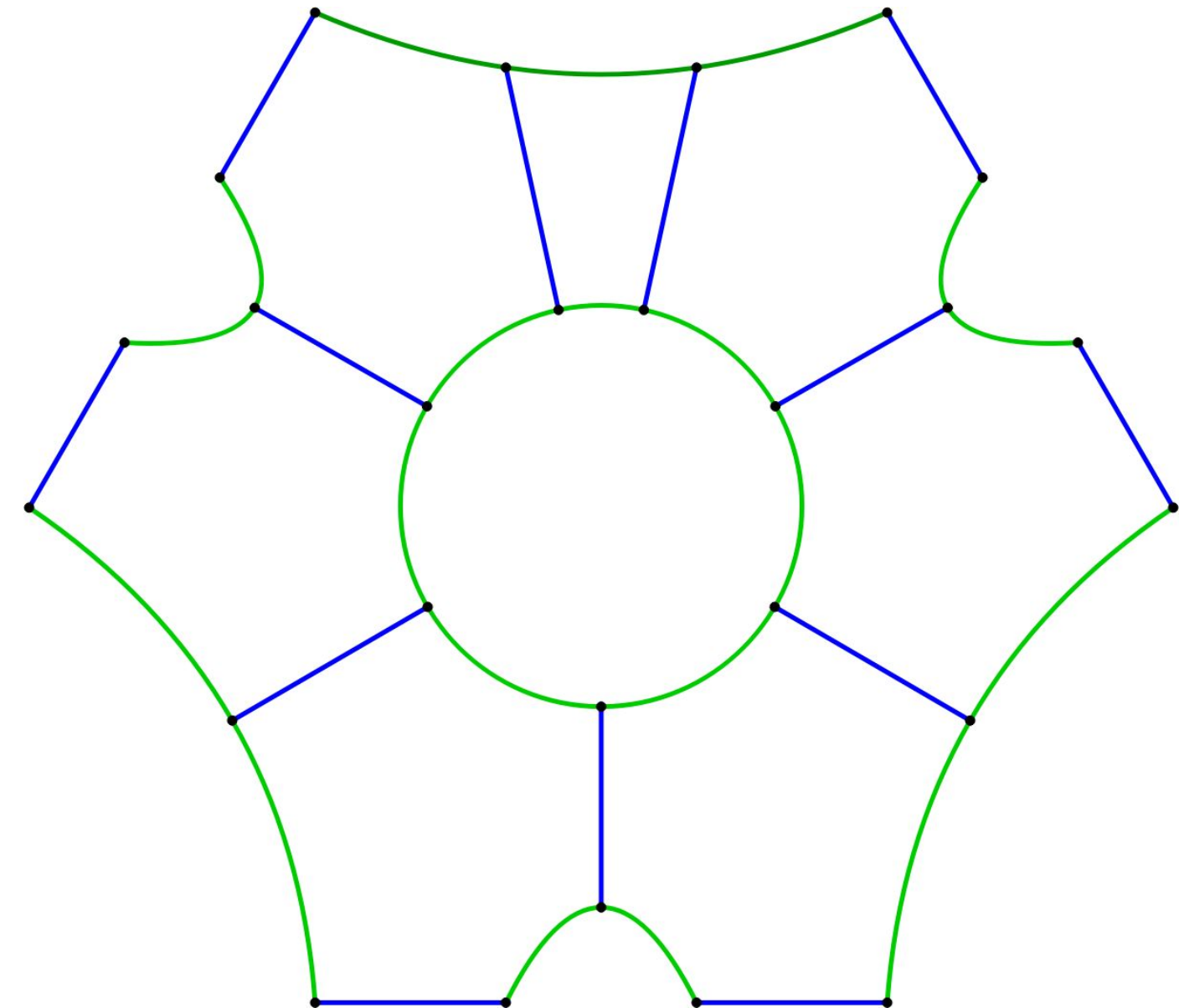
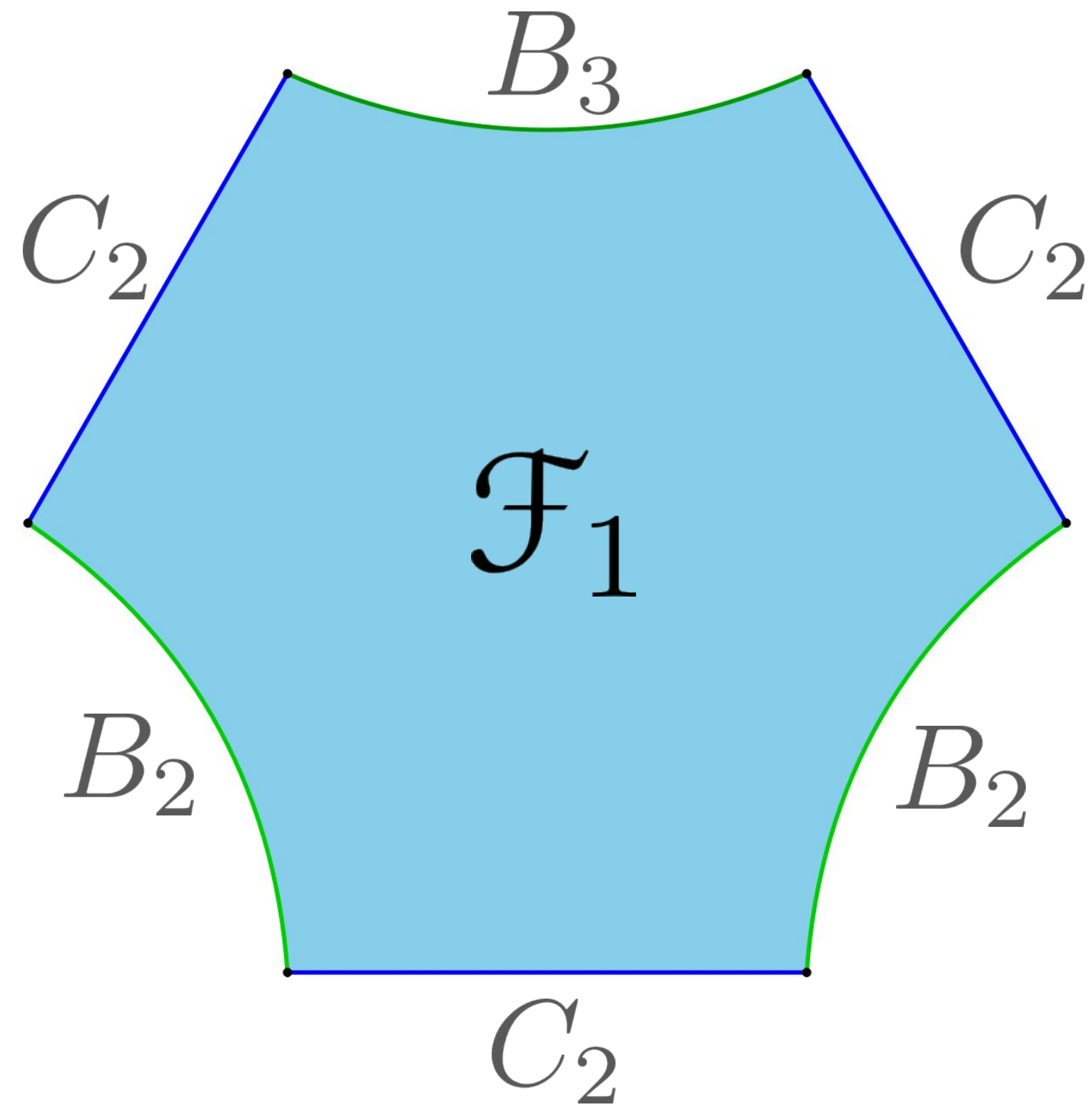


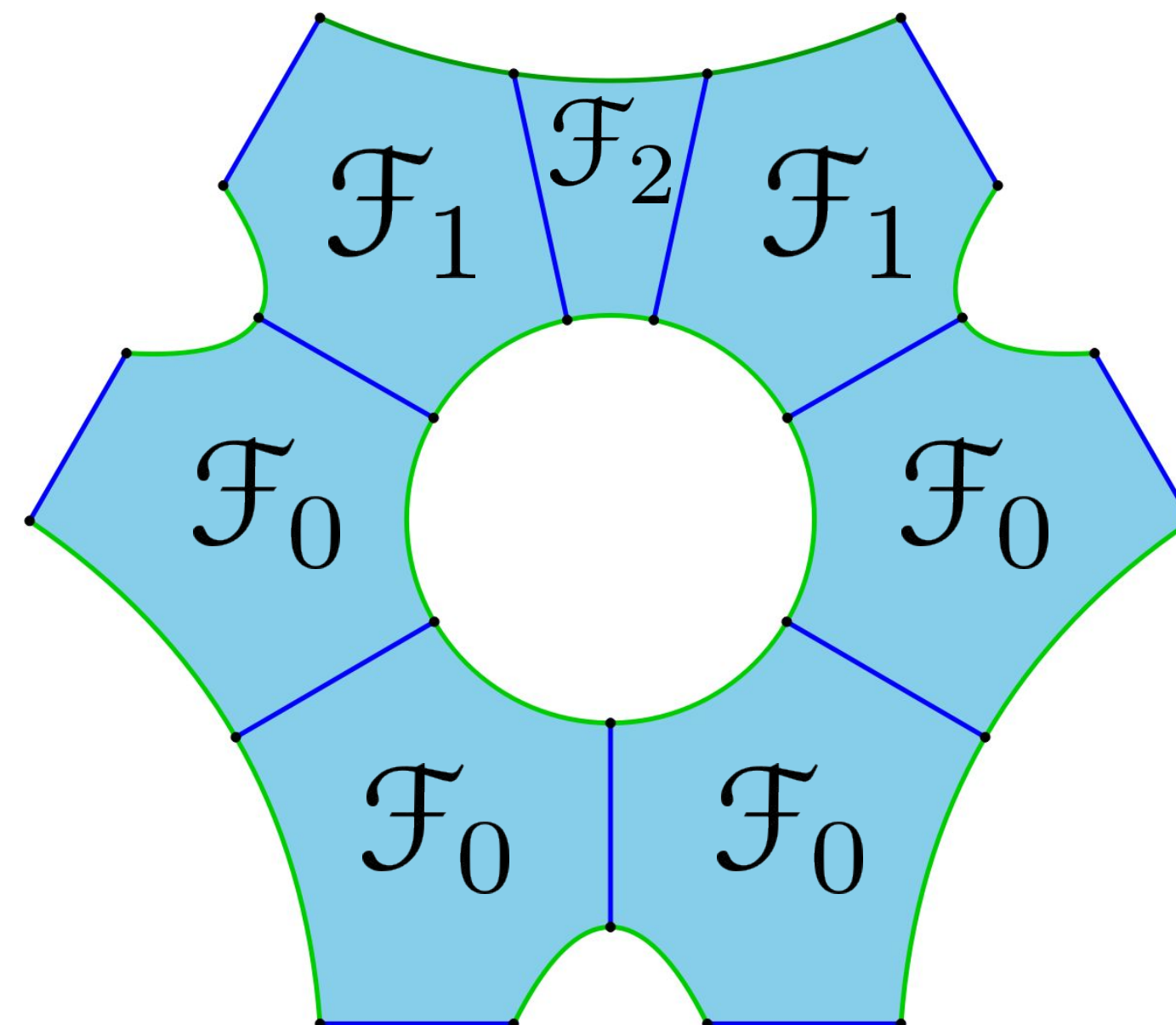
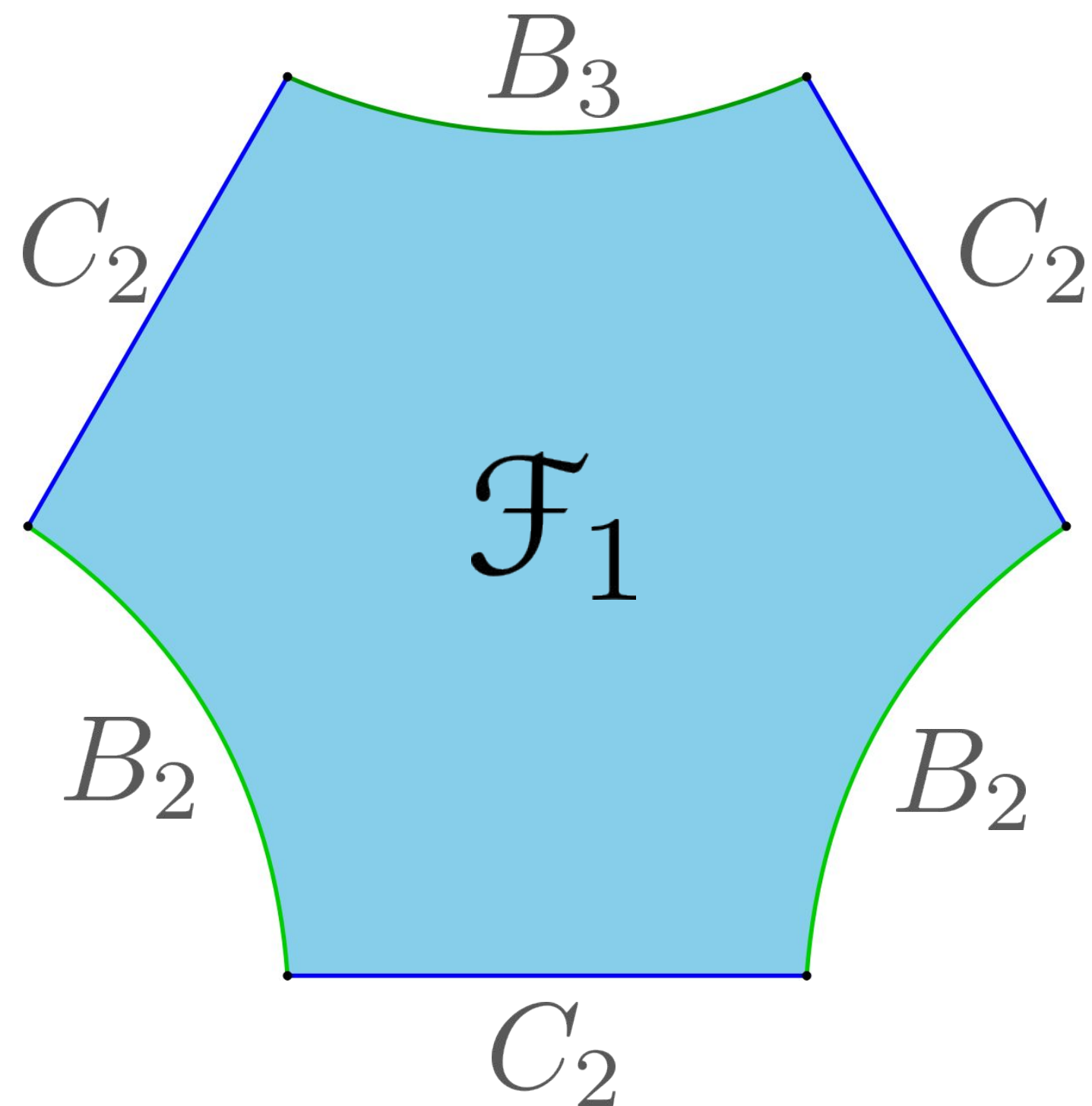


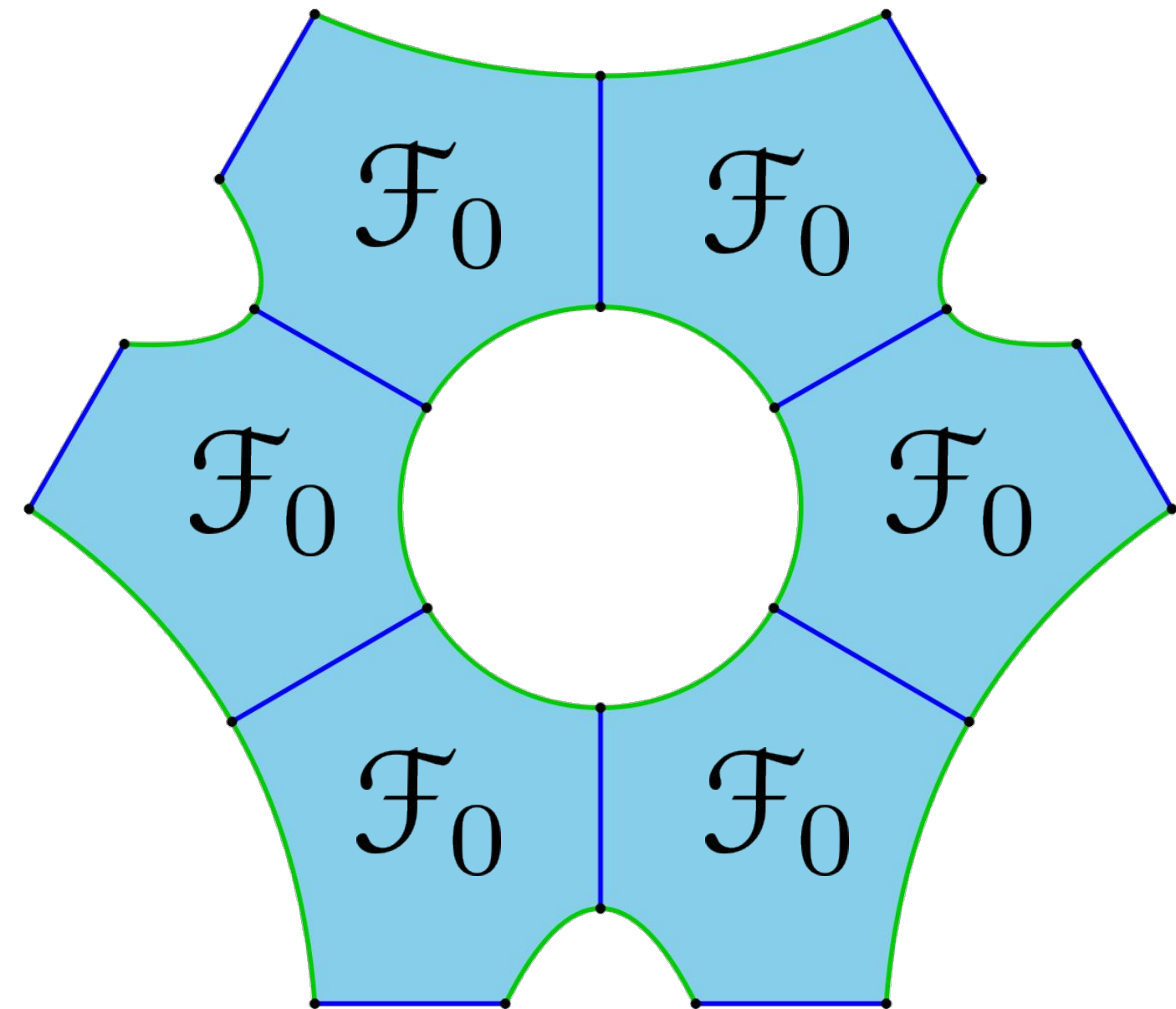
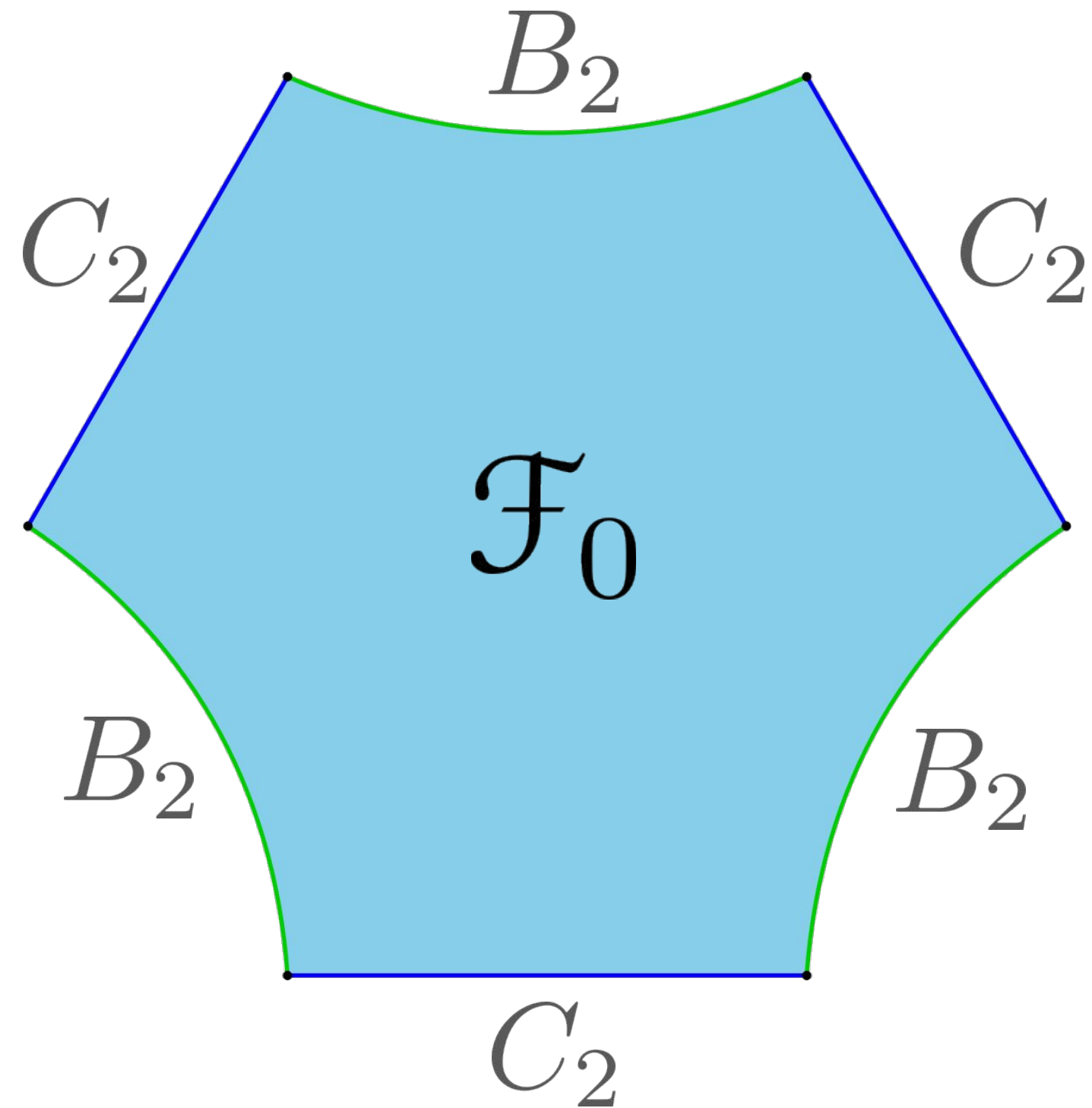


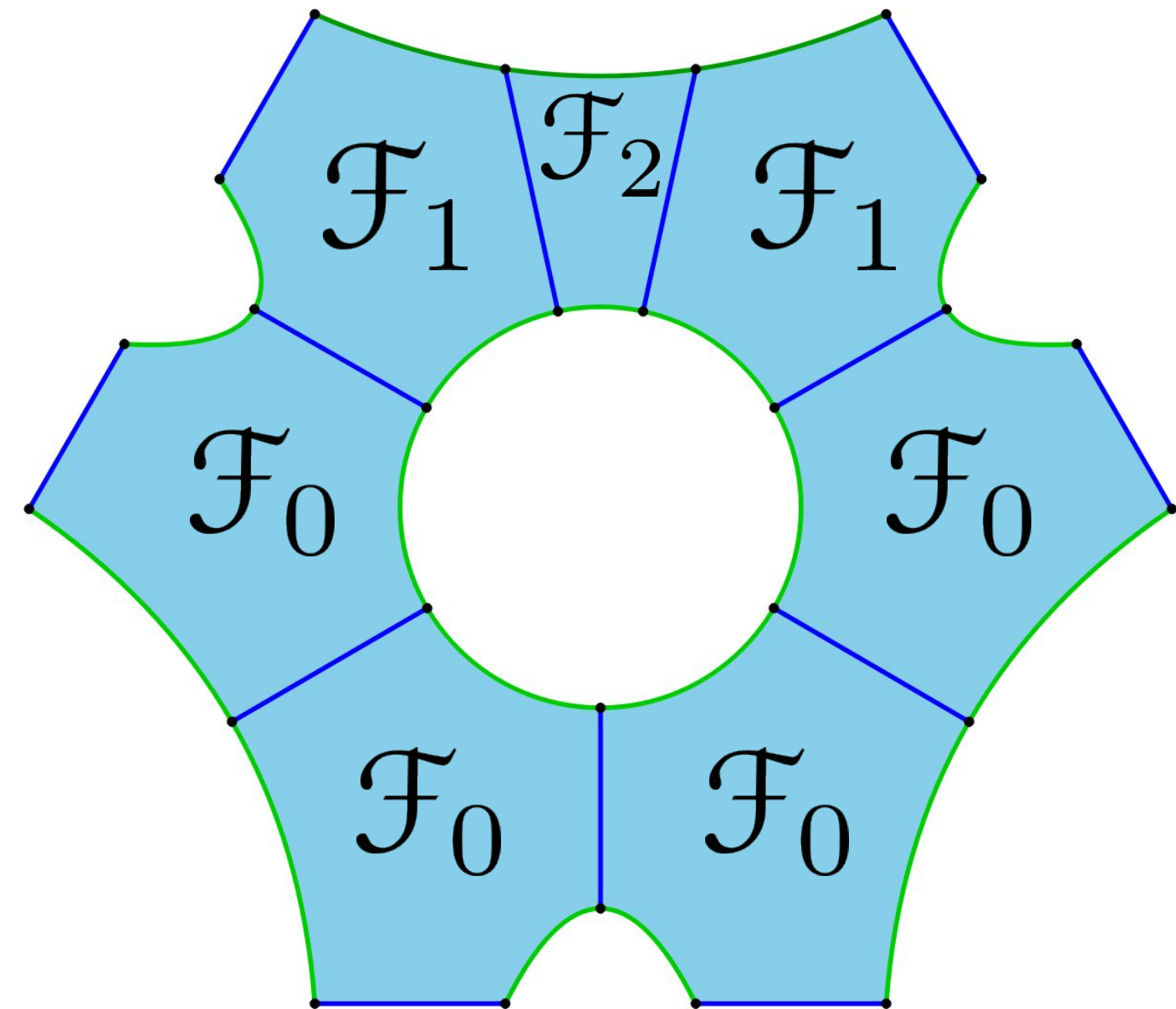
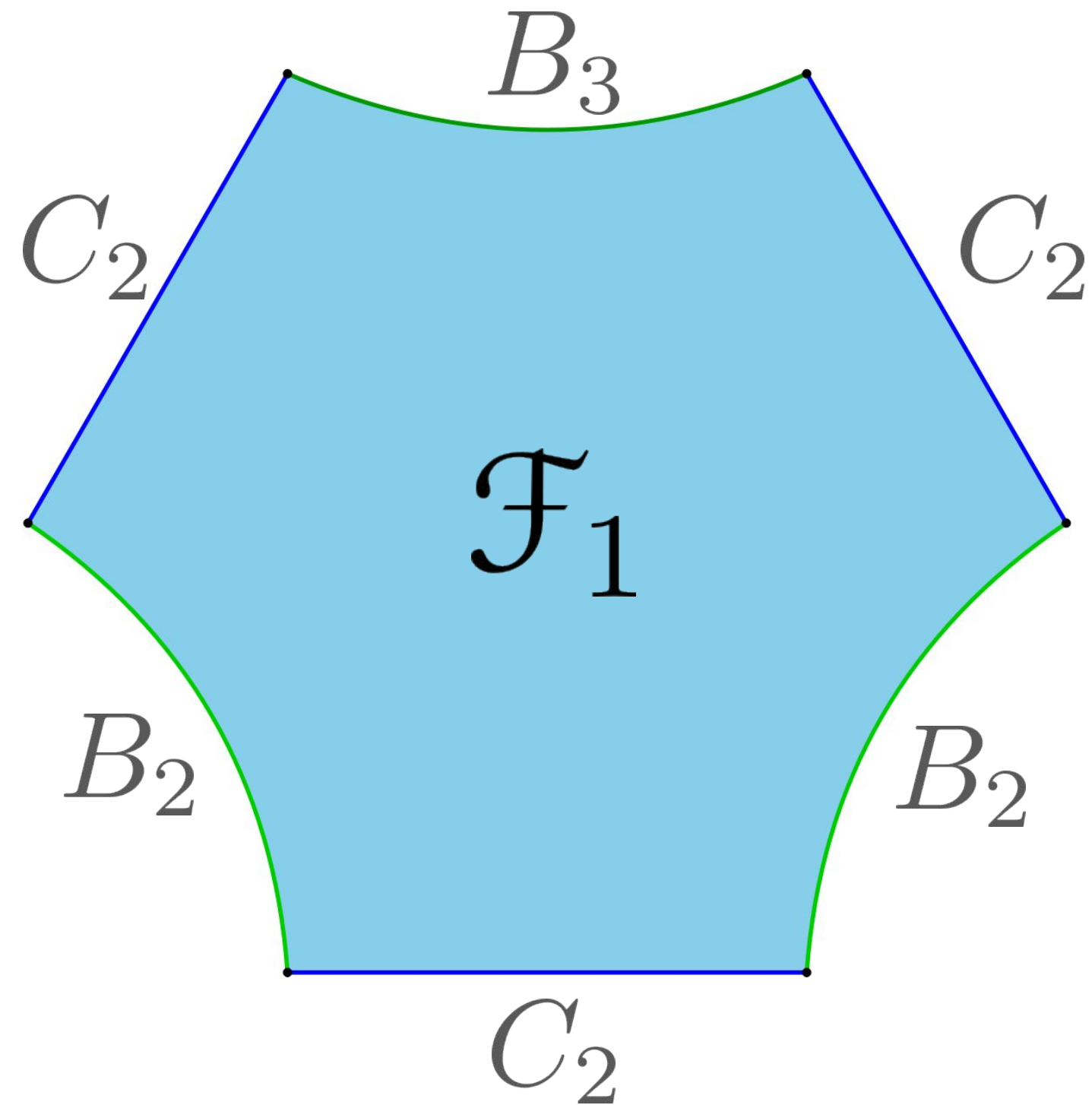


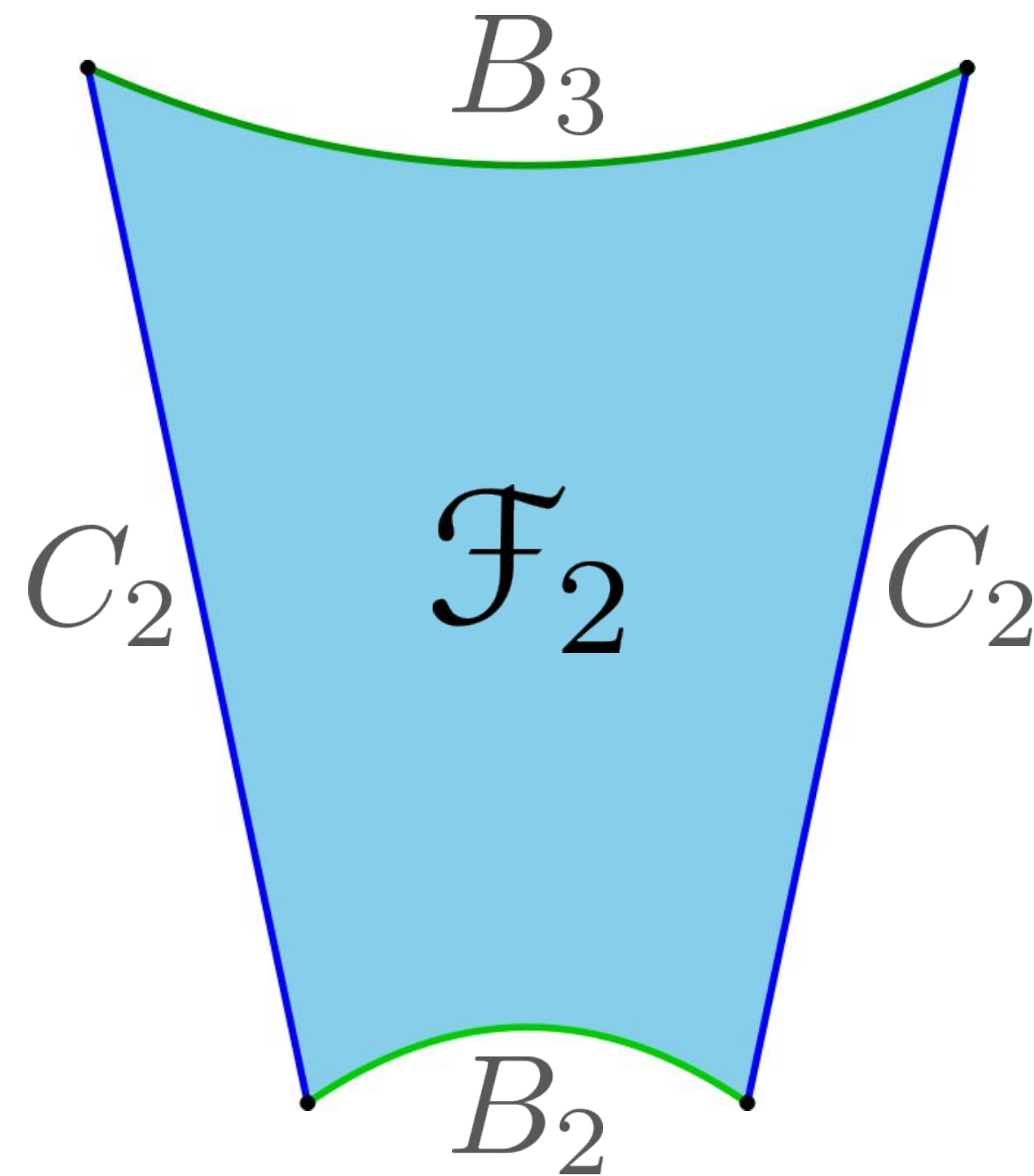


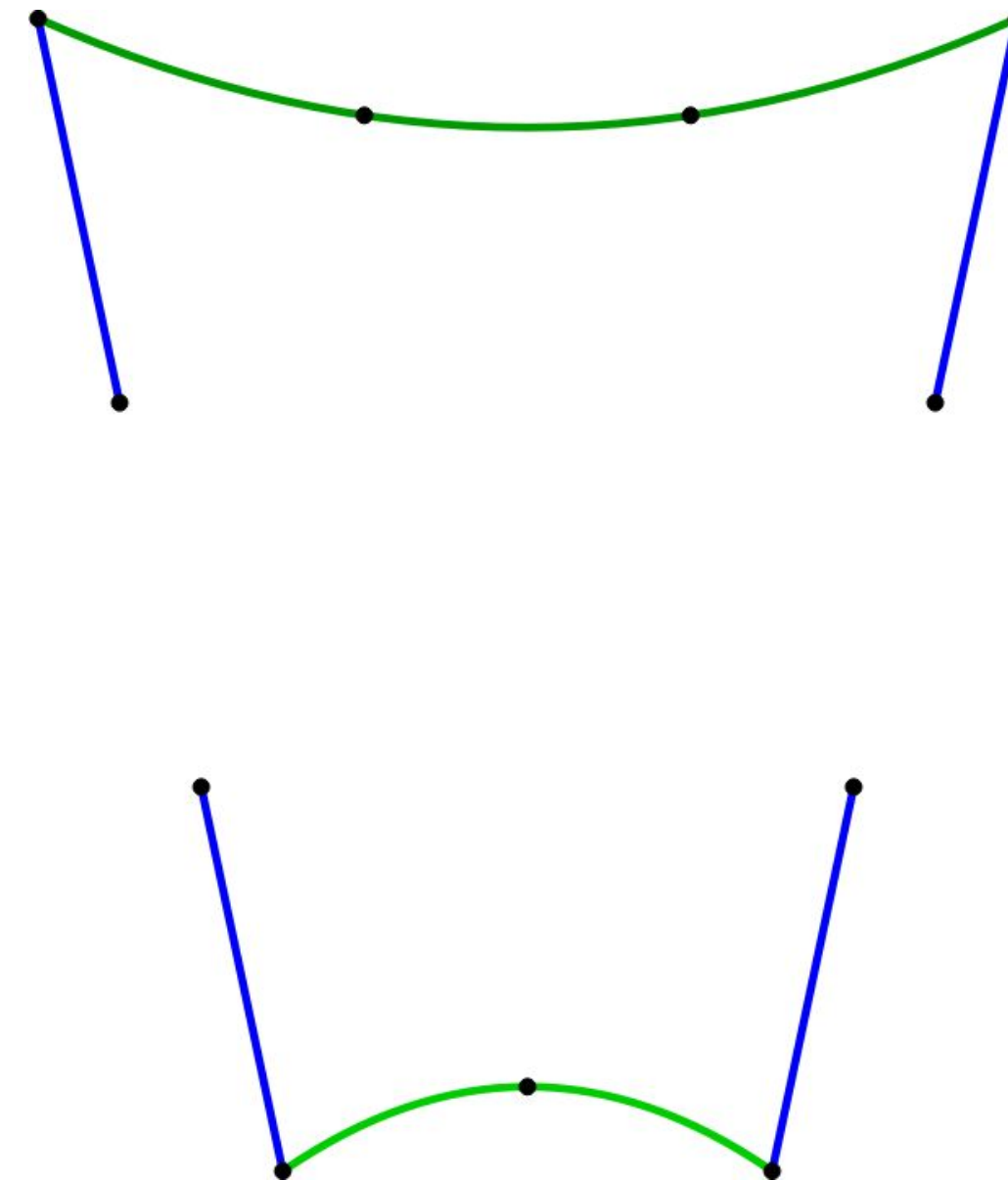
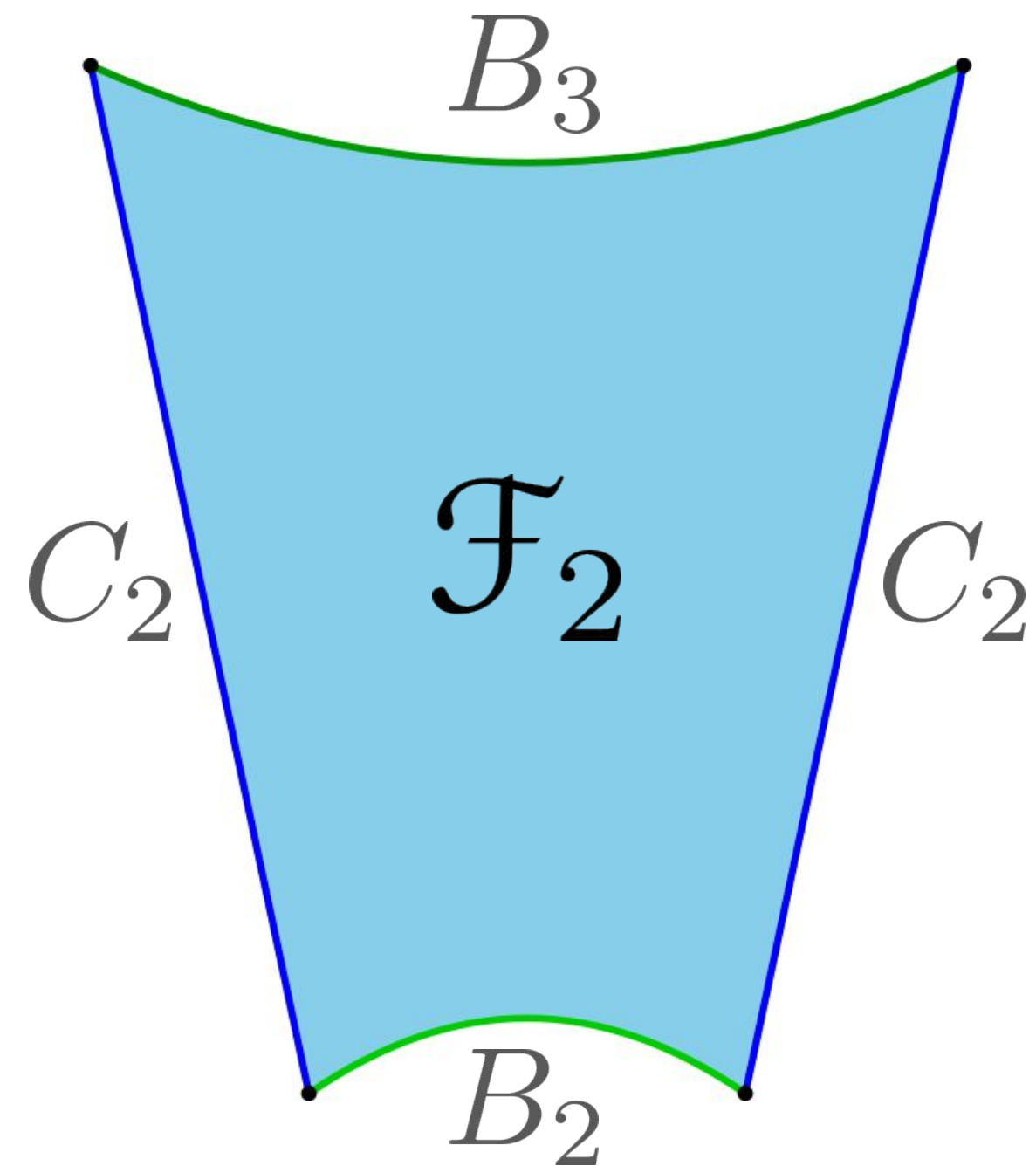


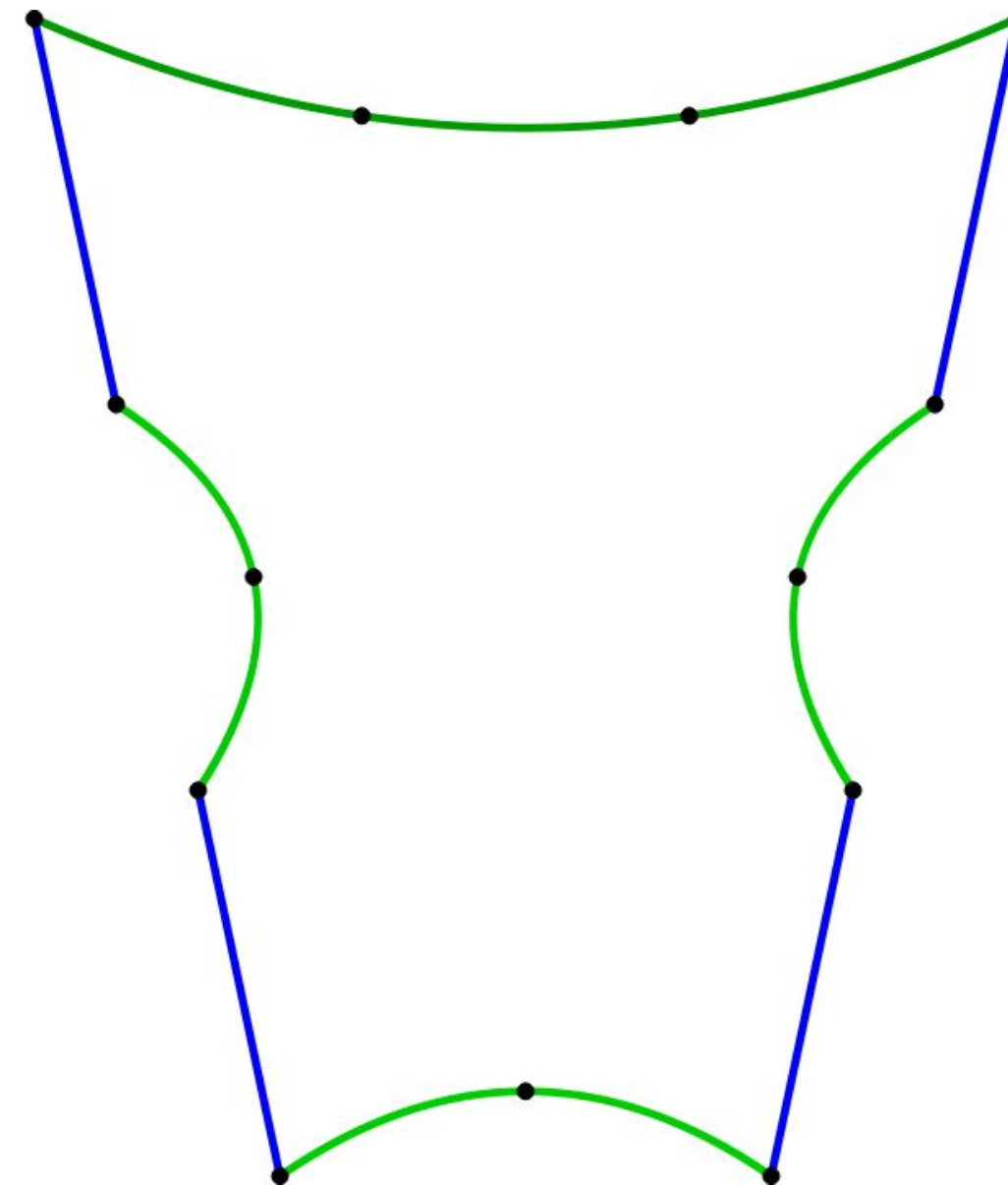
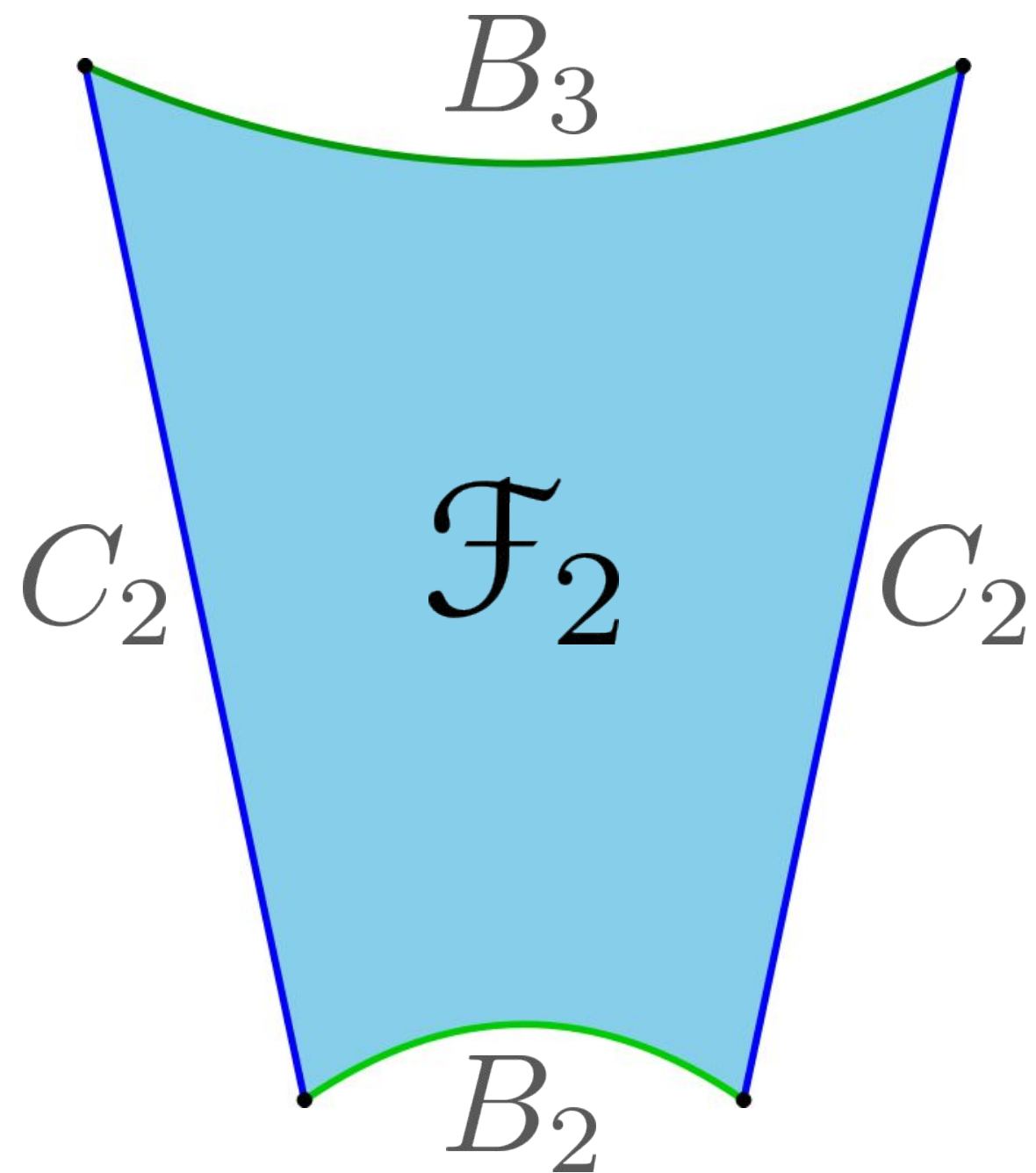


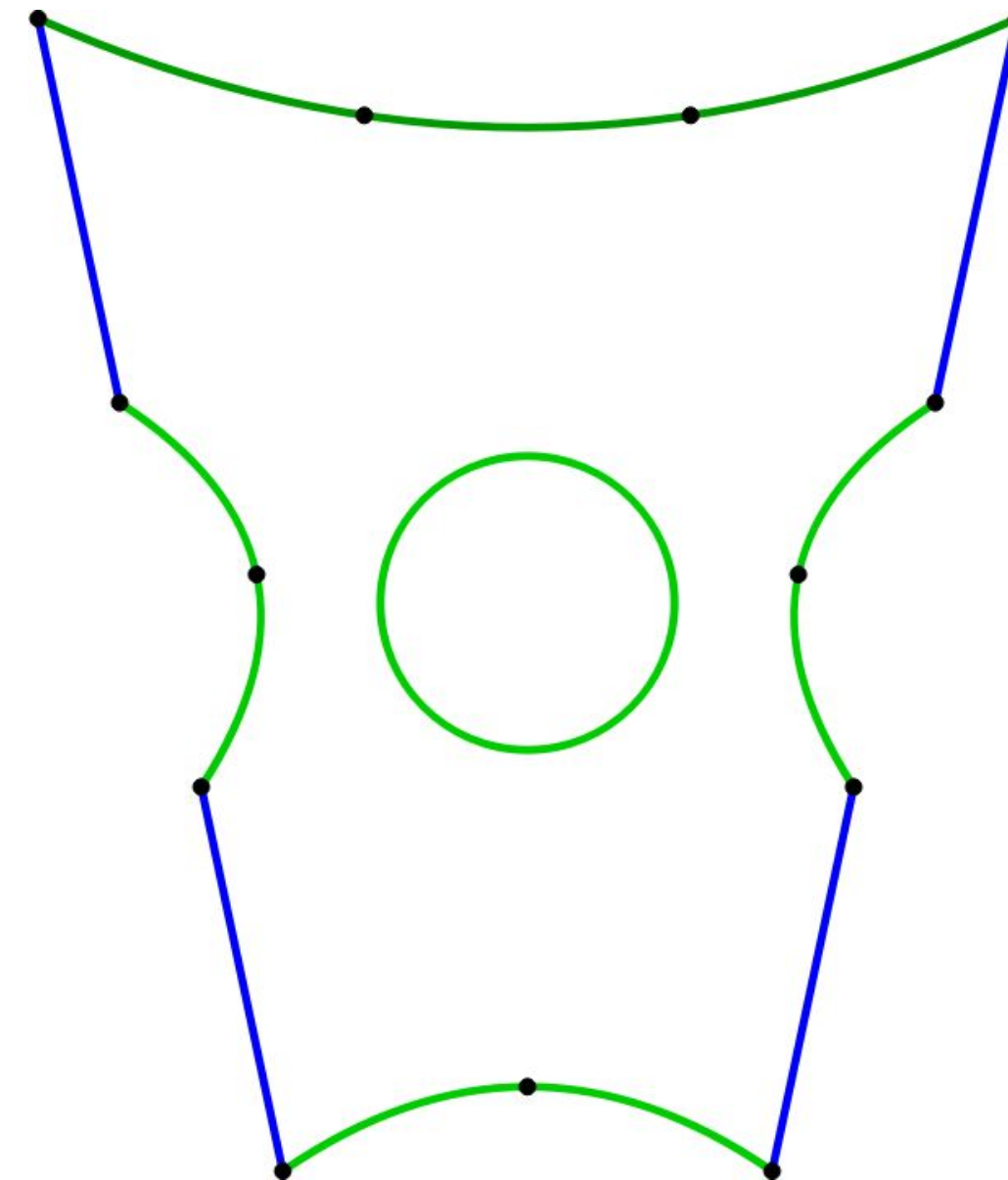
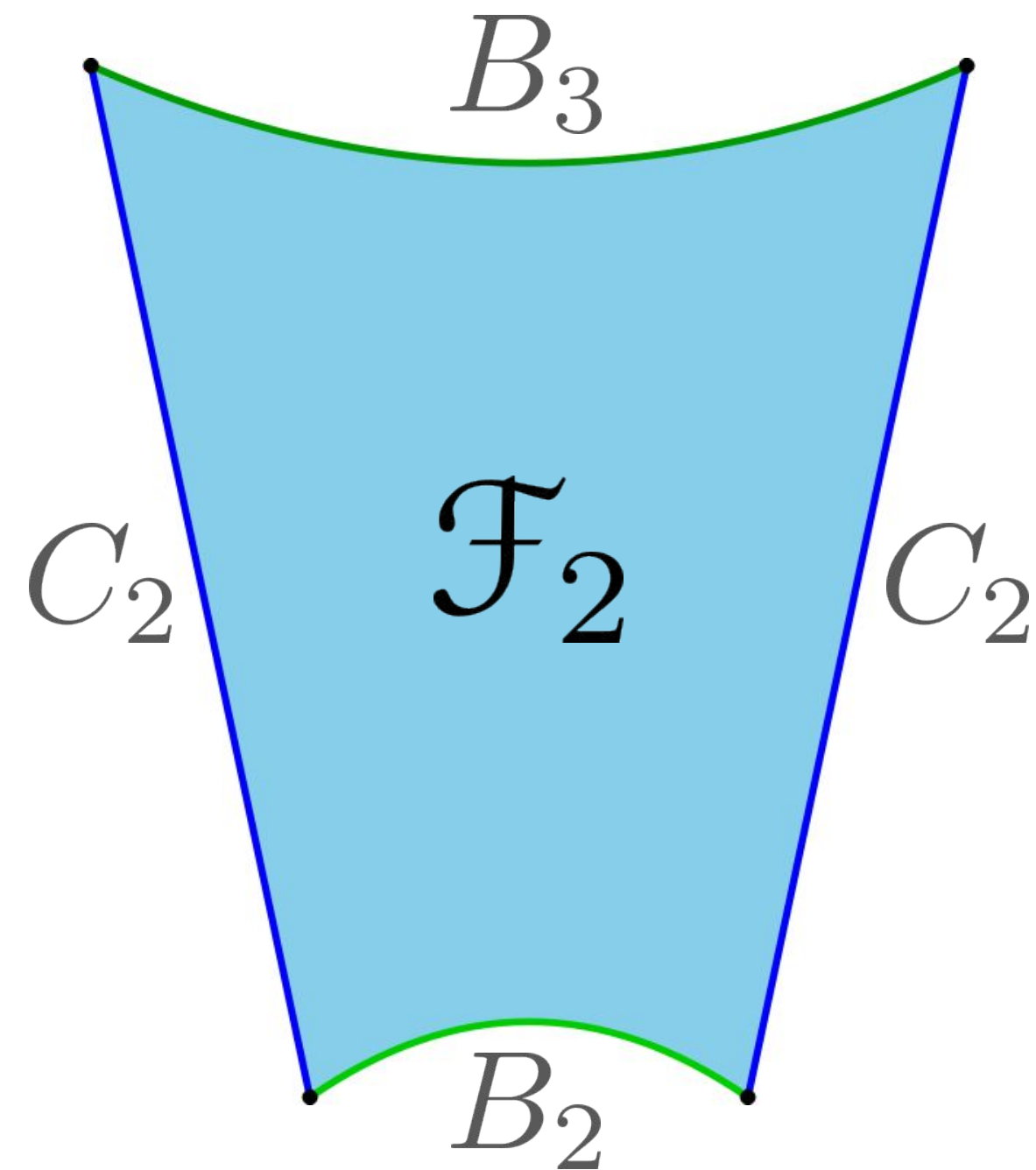


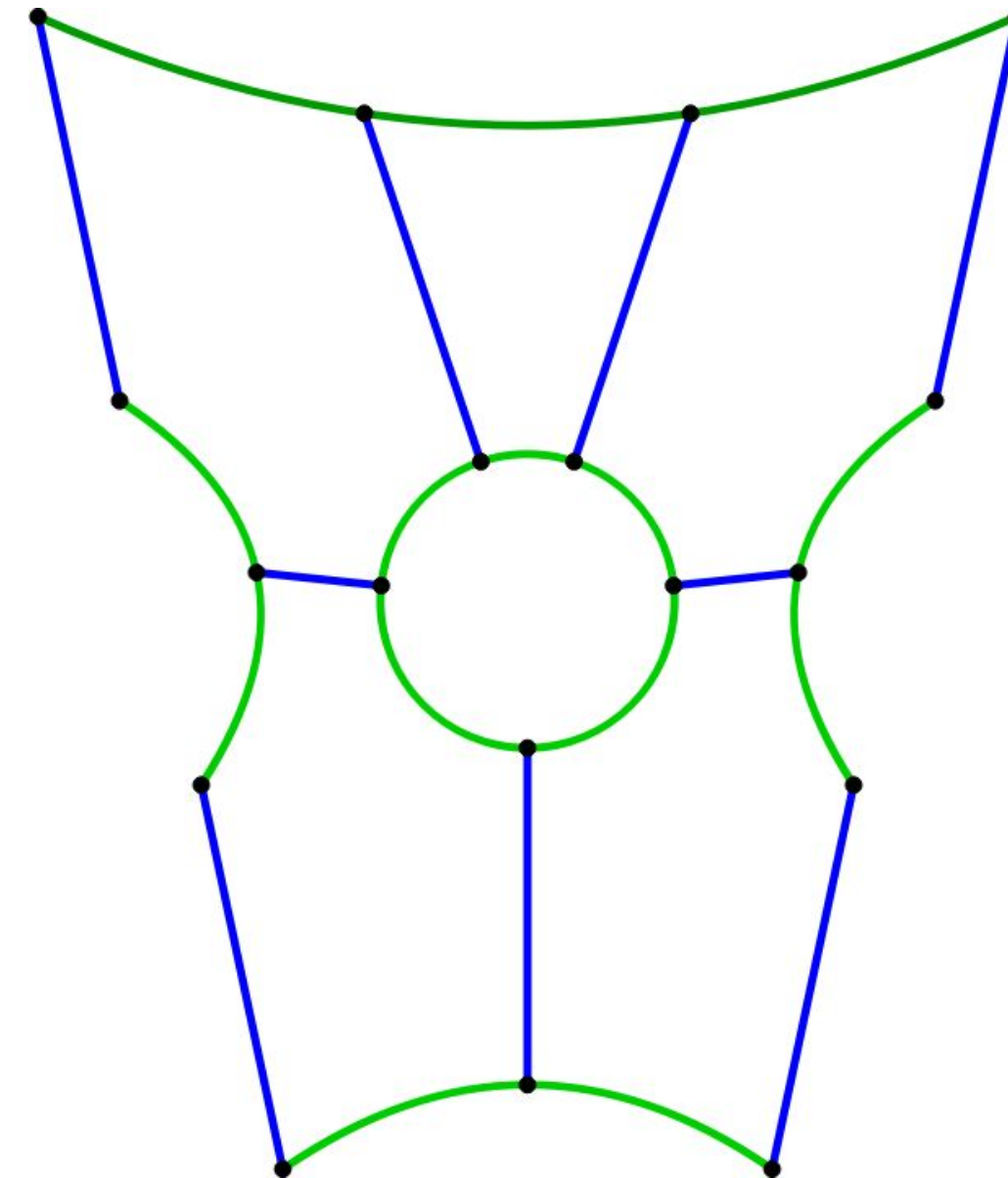
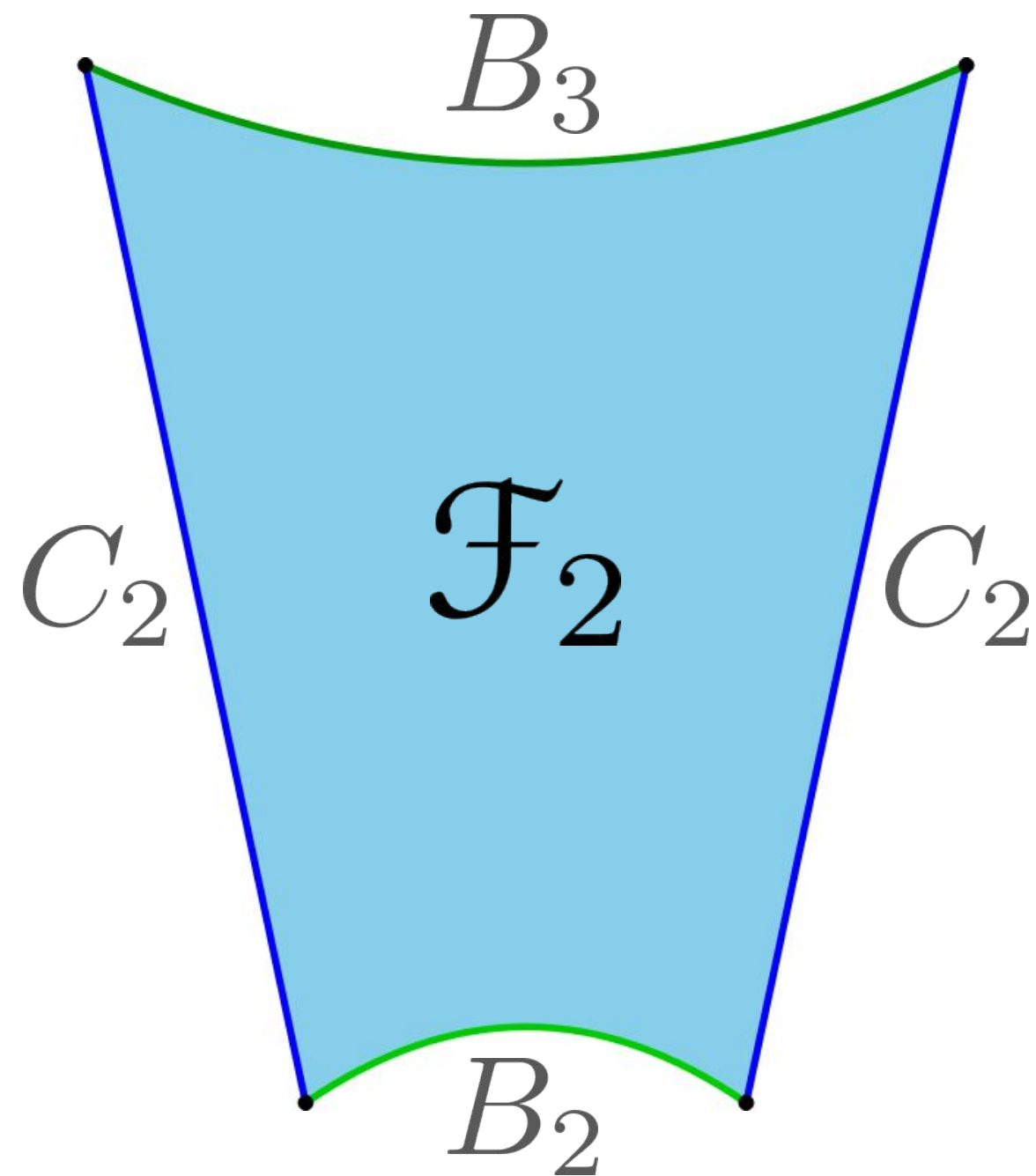


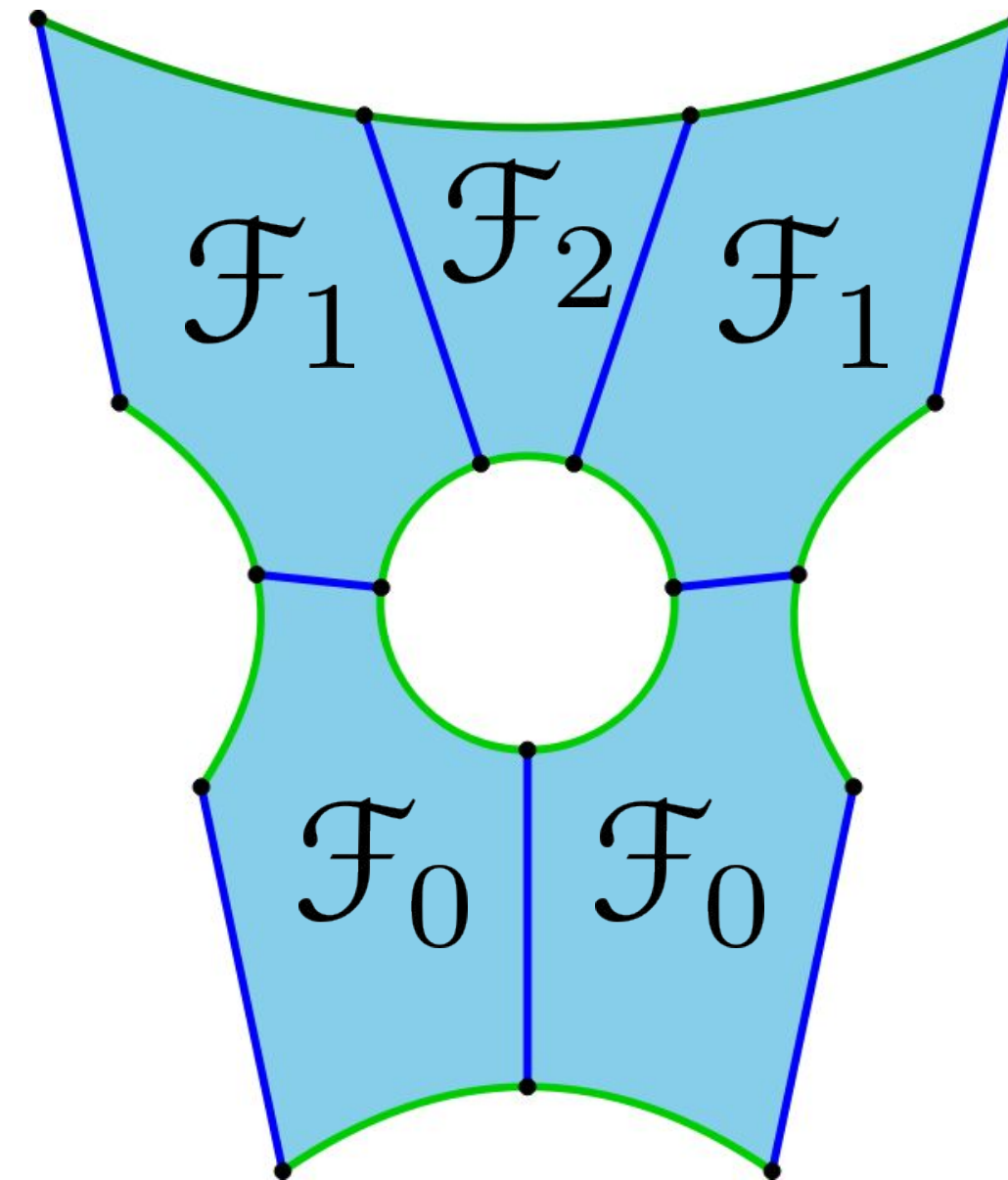
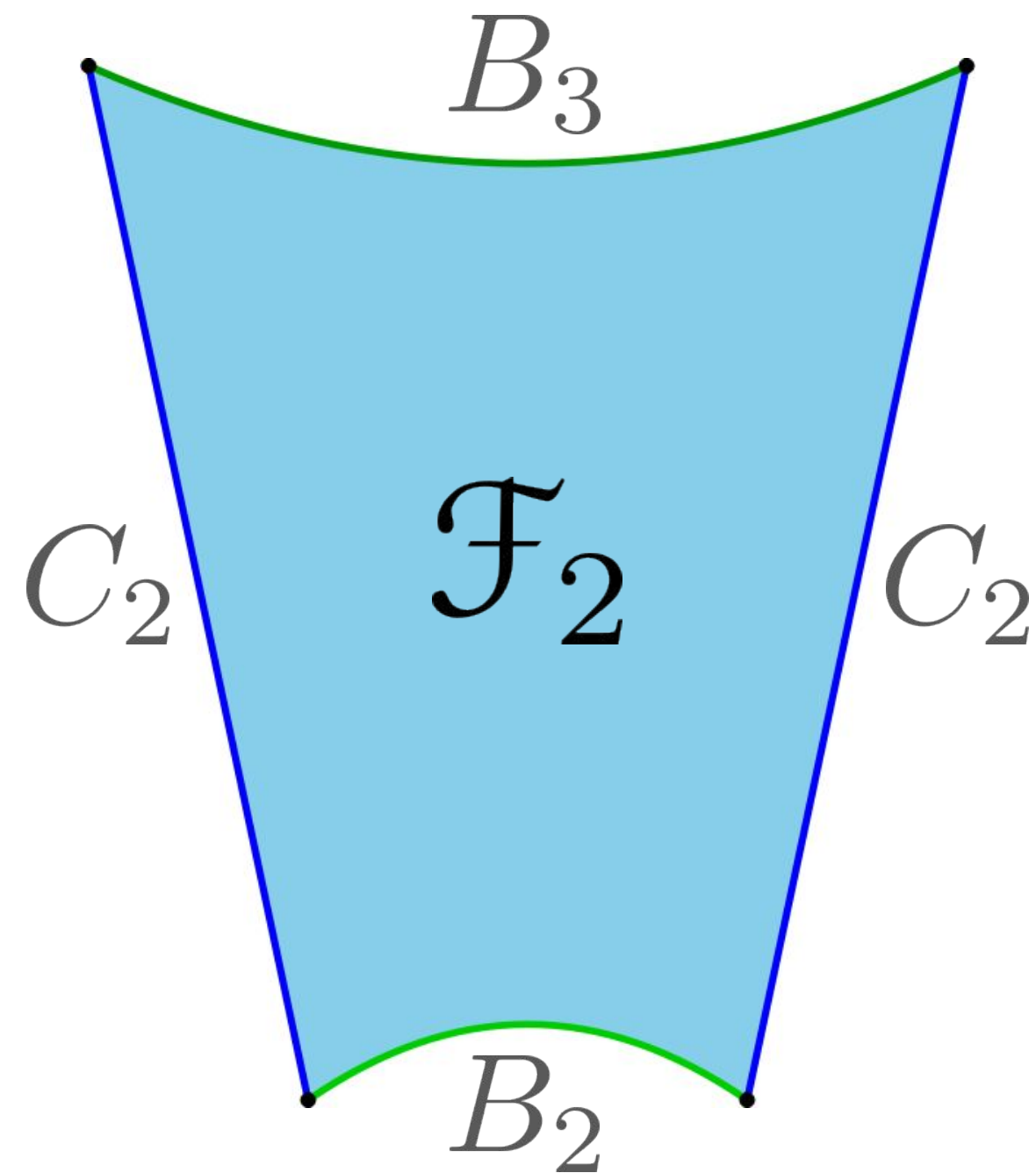


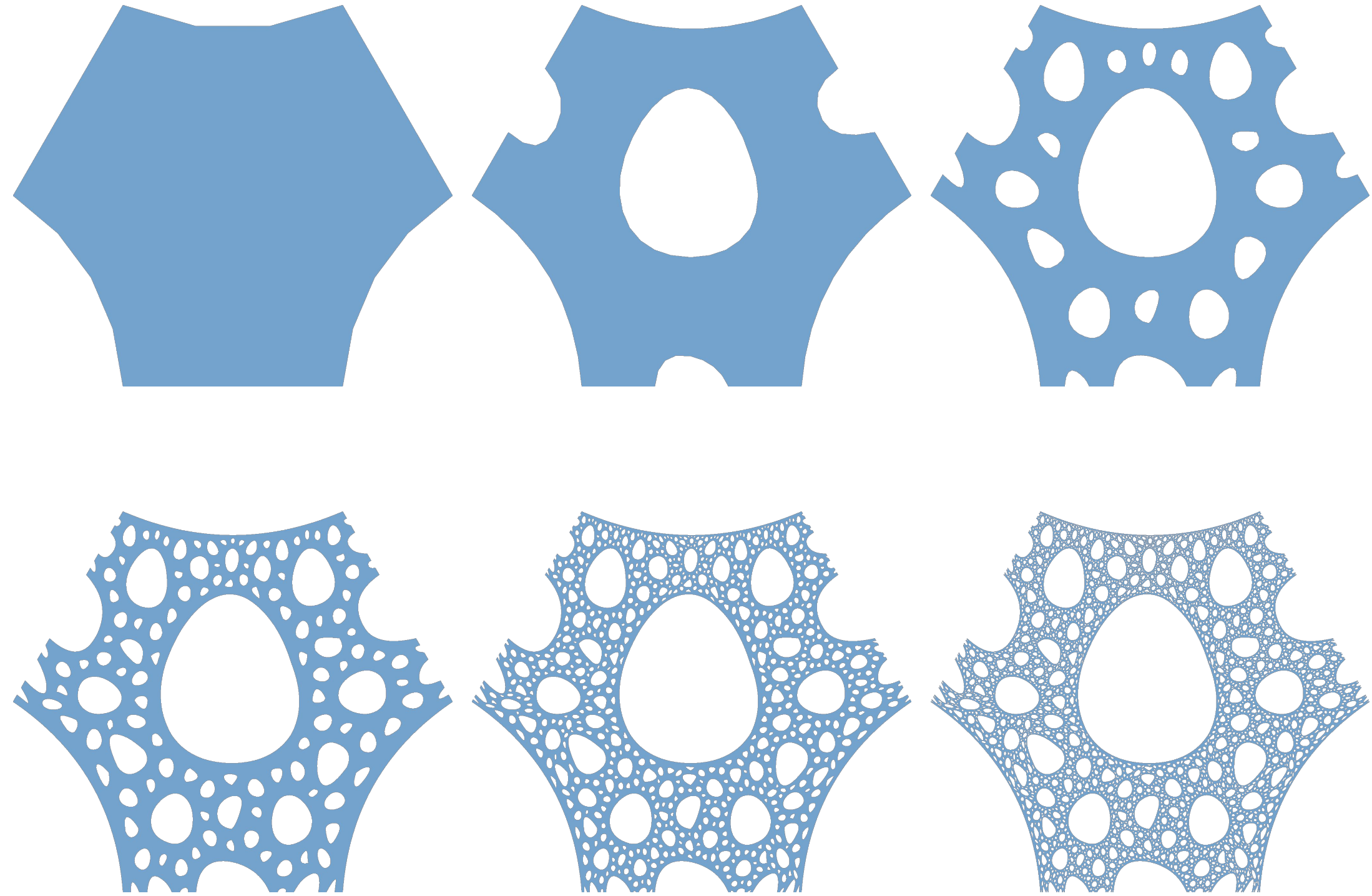
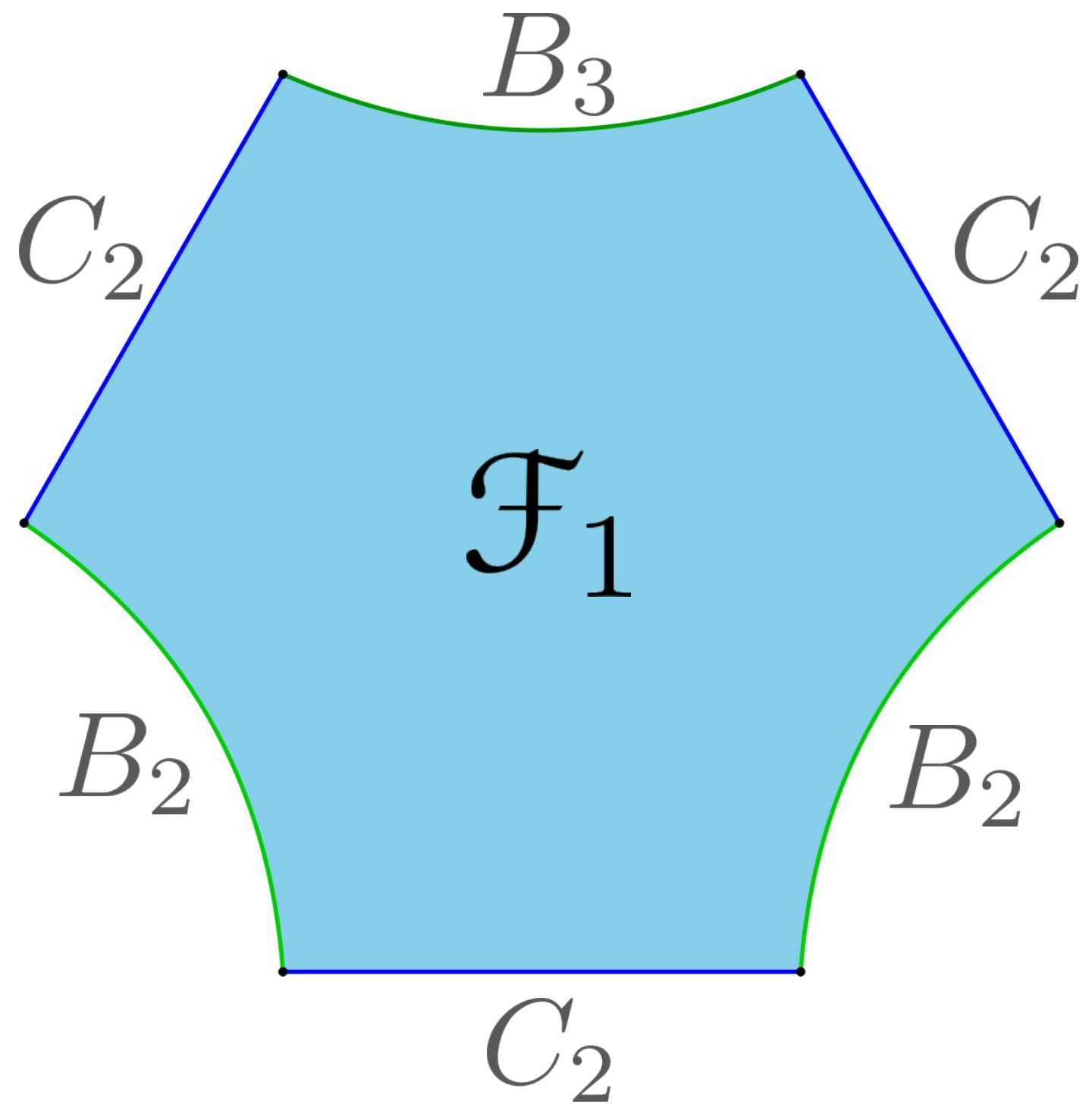








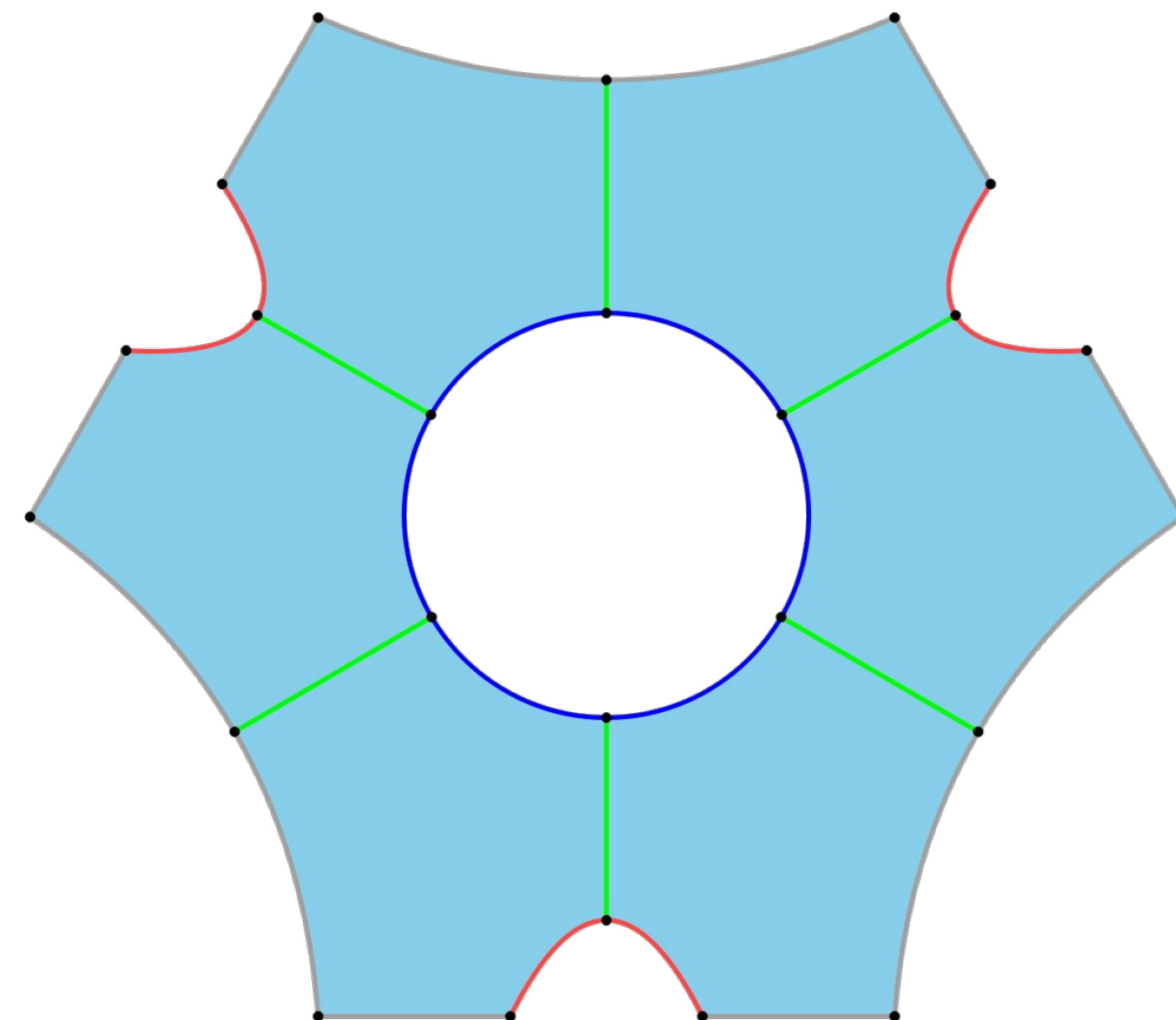
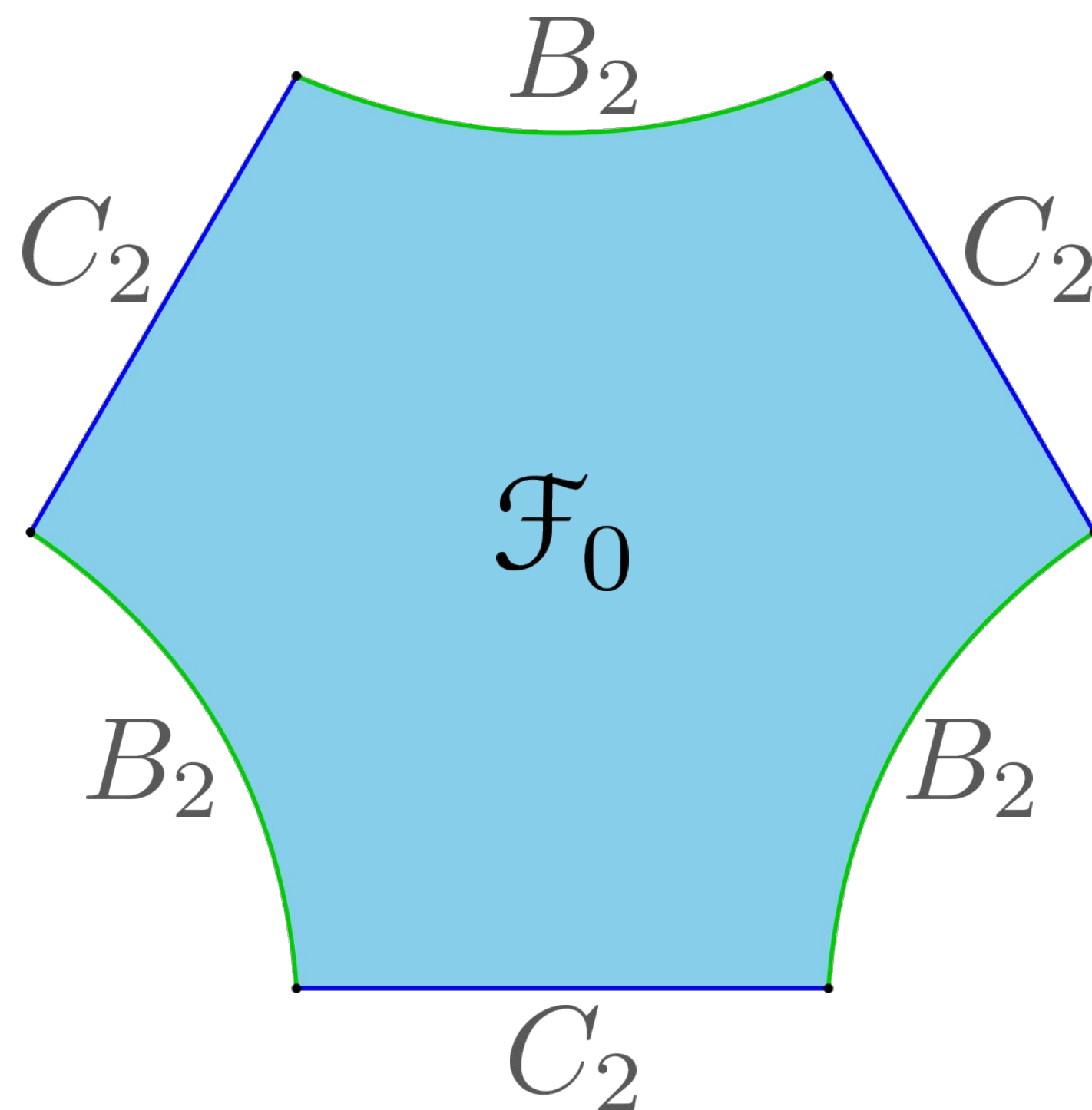




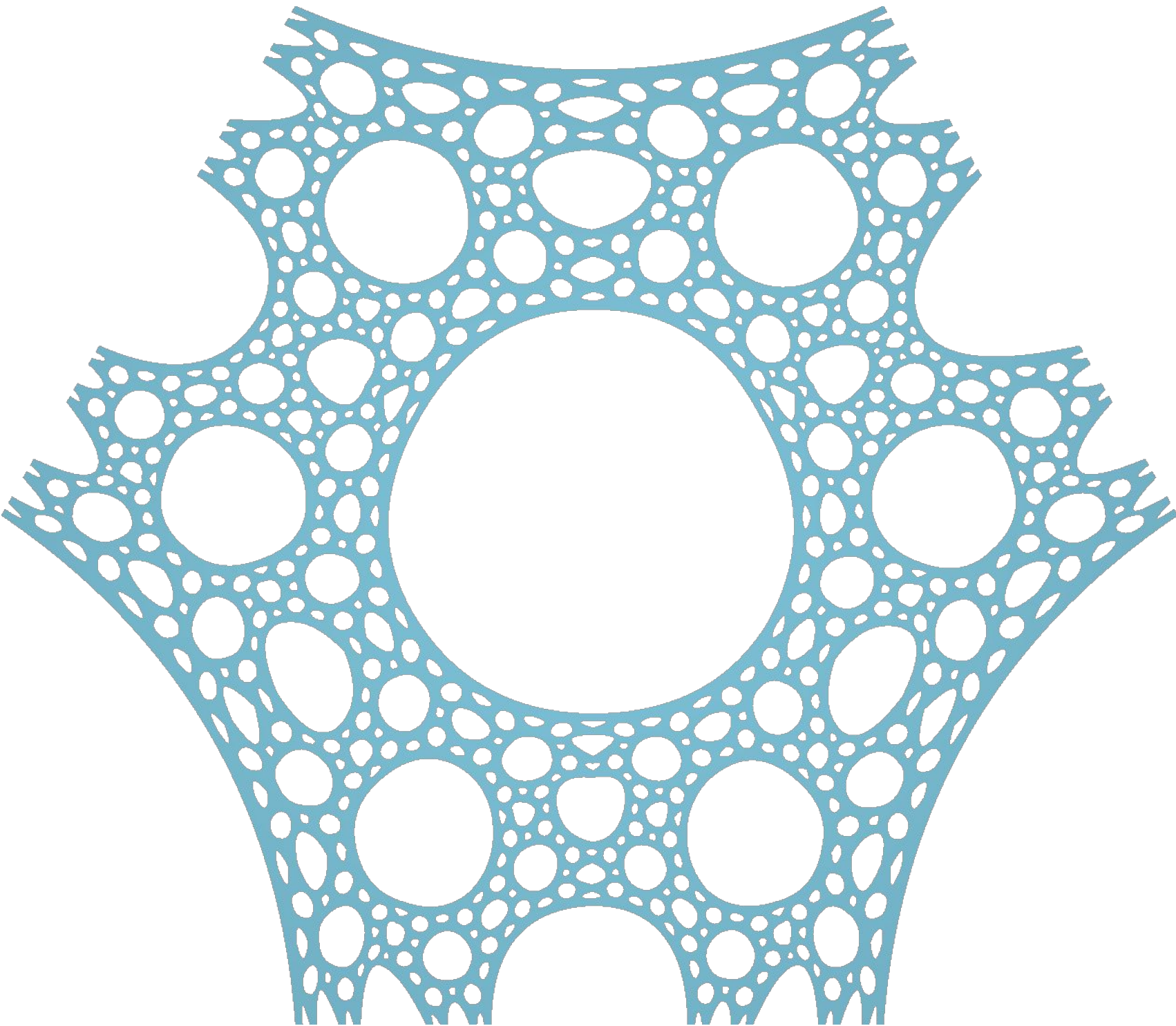
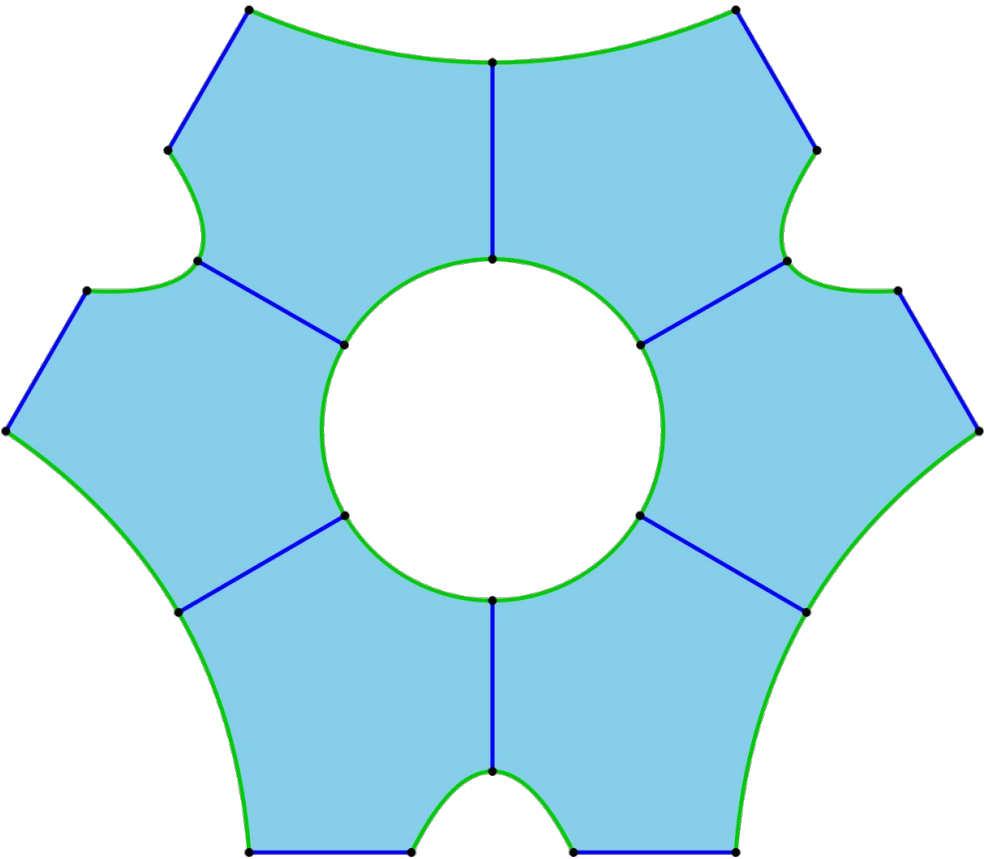
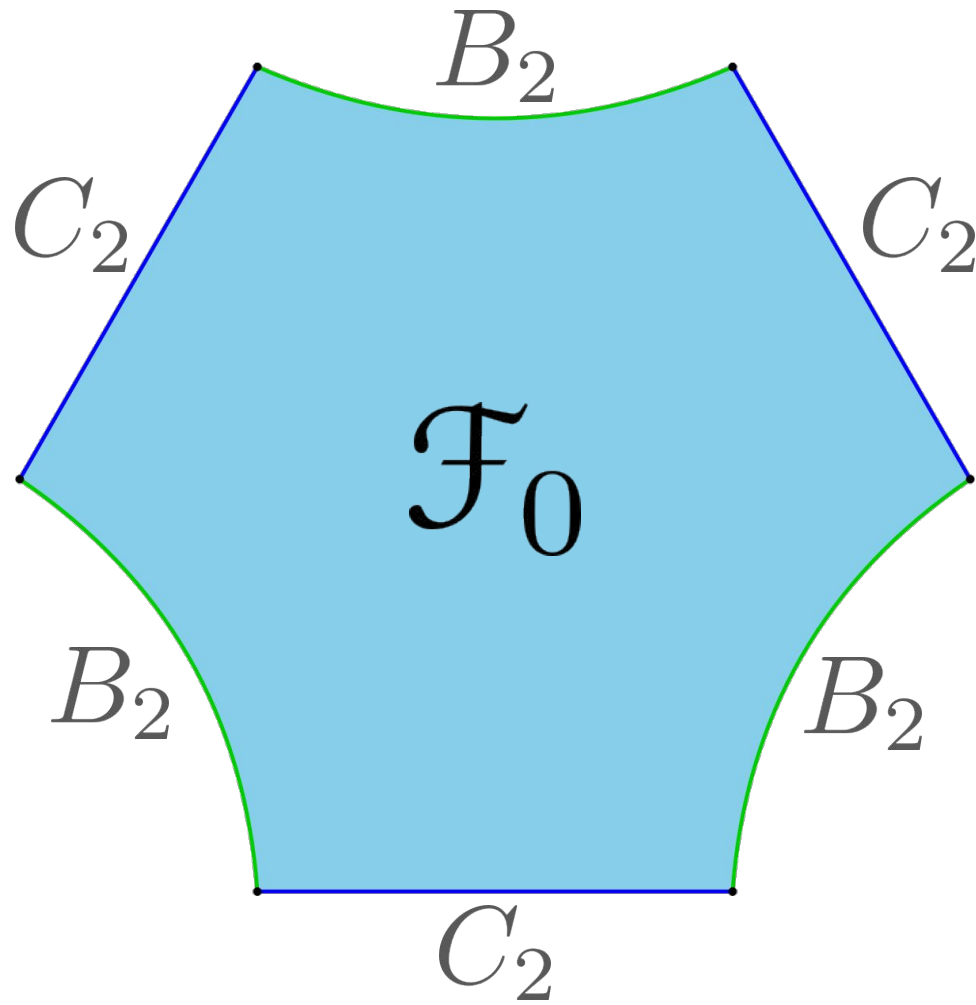
Local lacunarity control



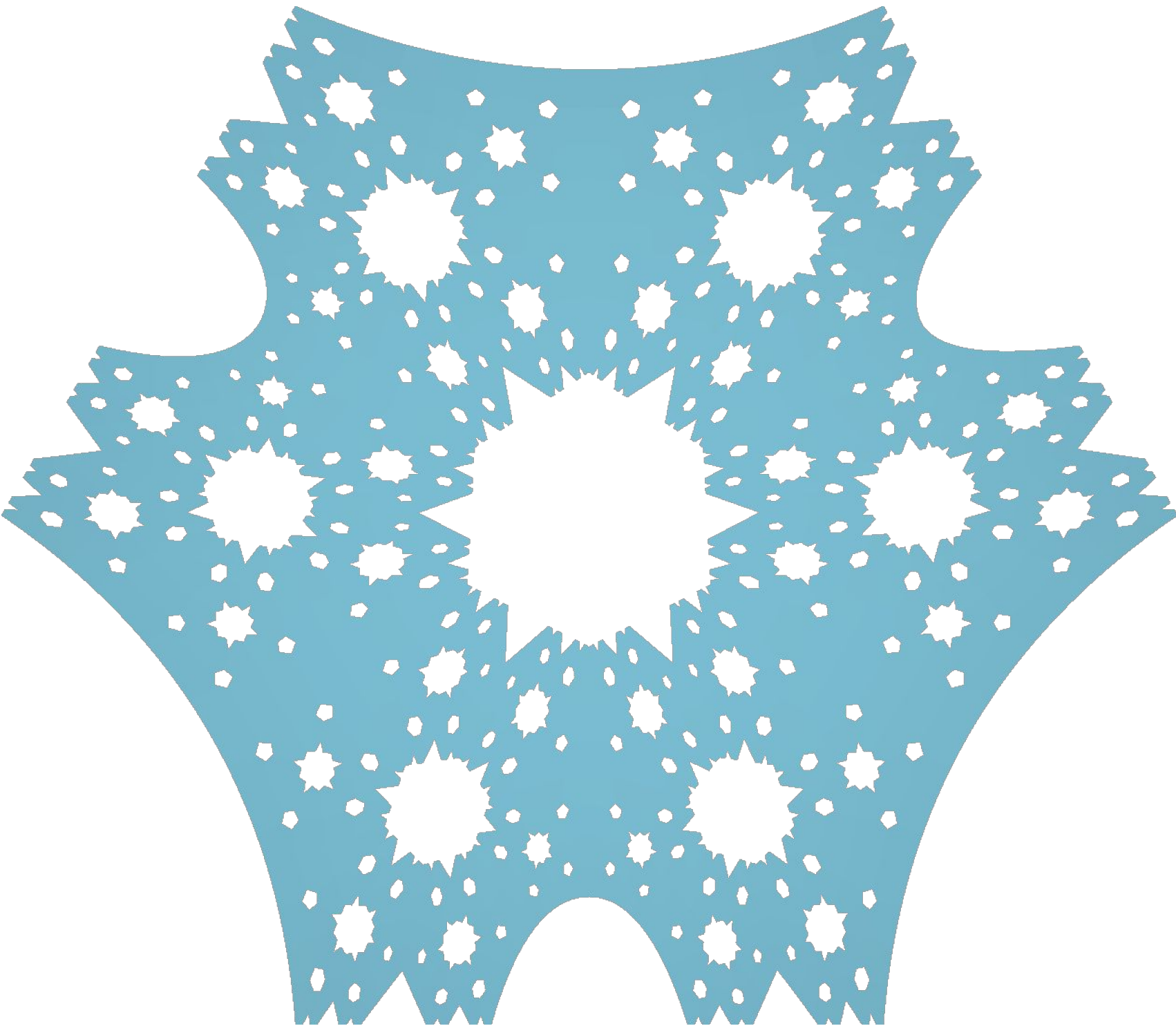
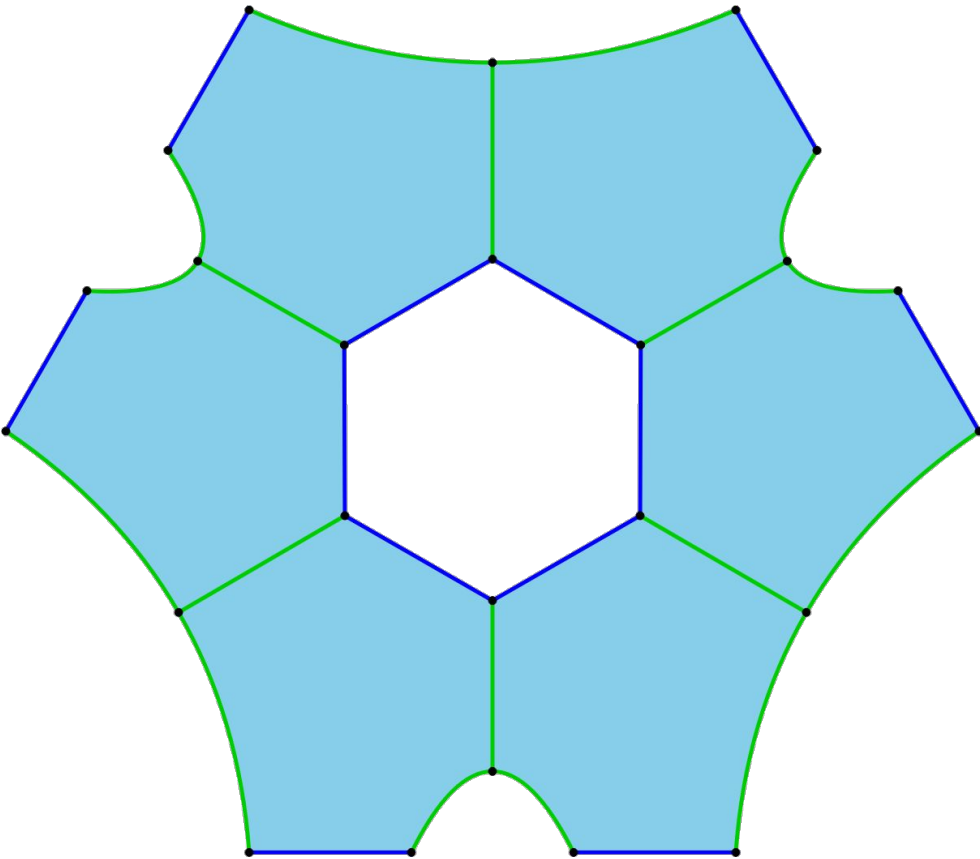
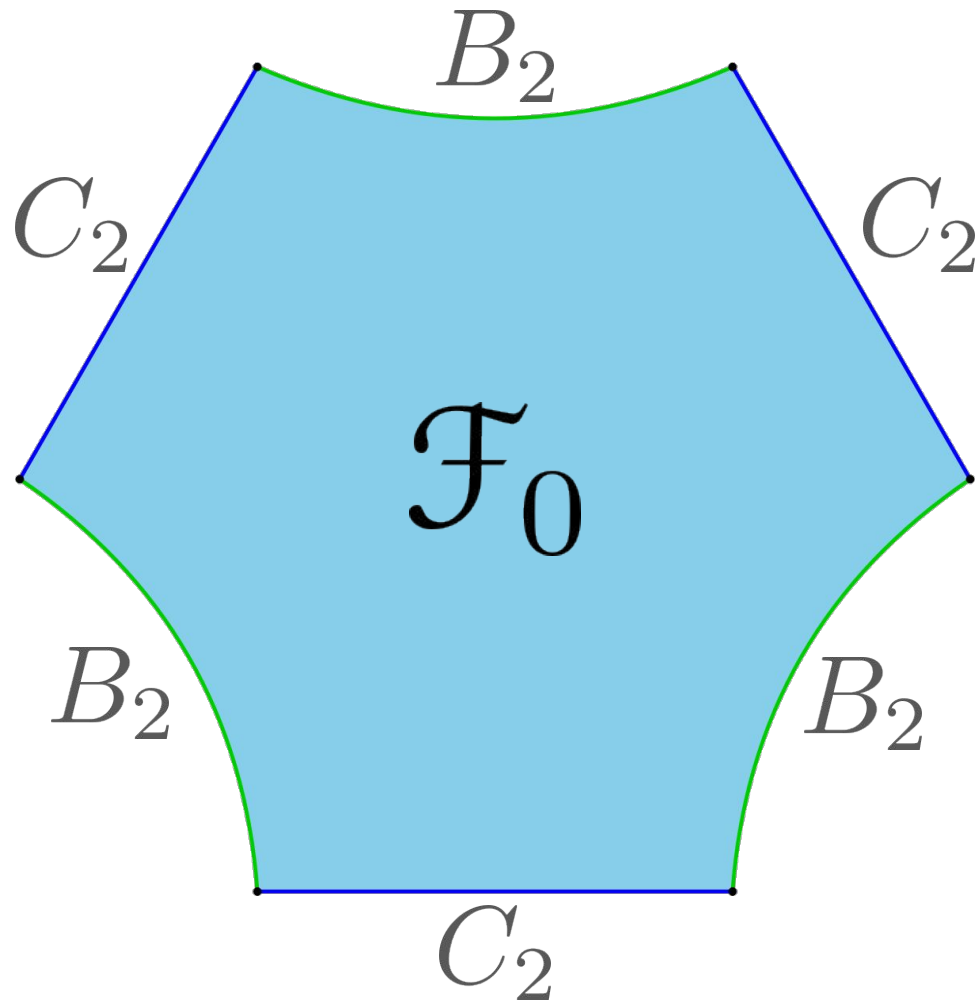
- E_a (green)
- E_l (blue)
- E_c (red)



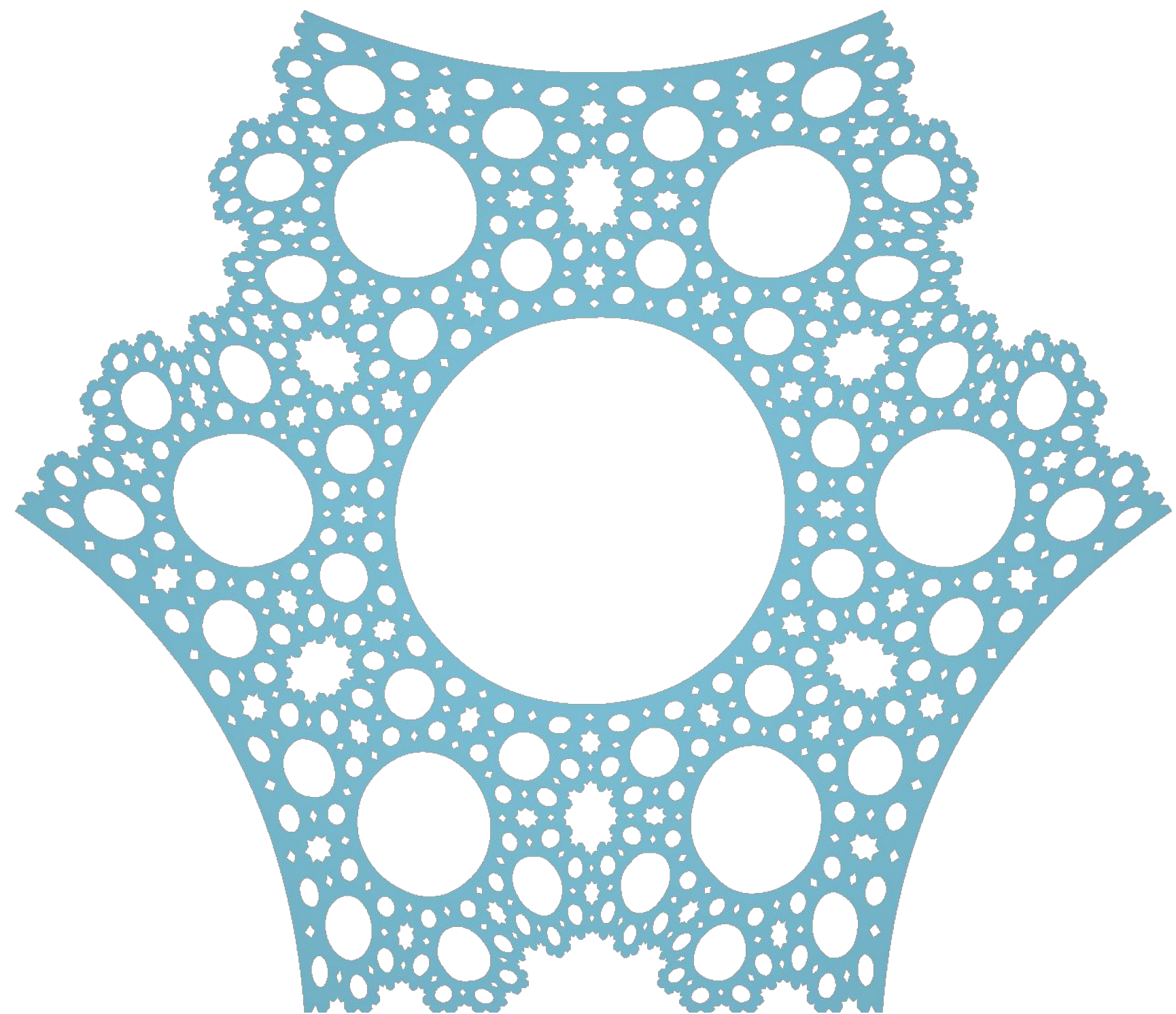
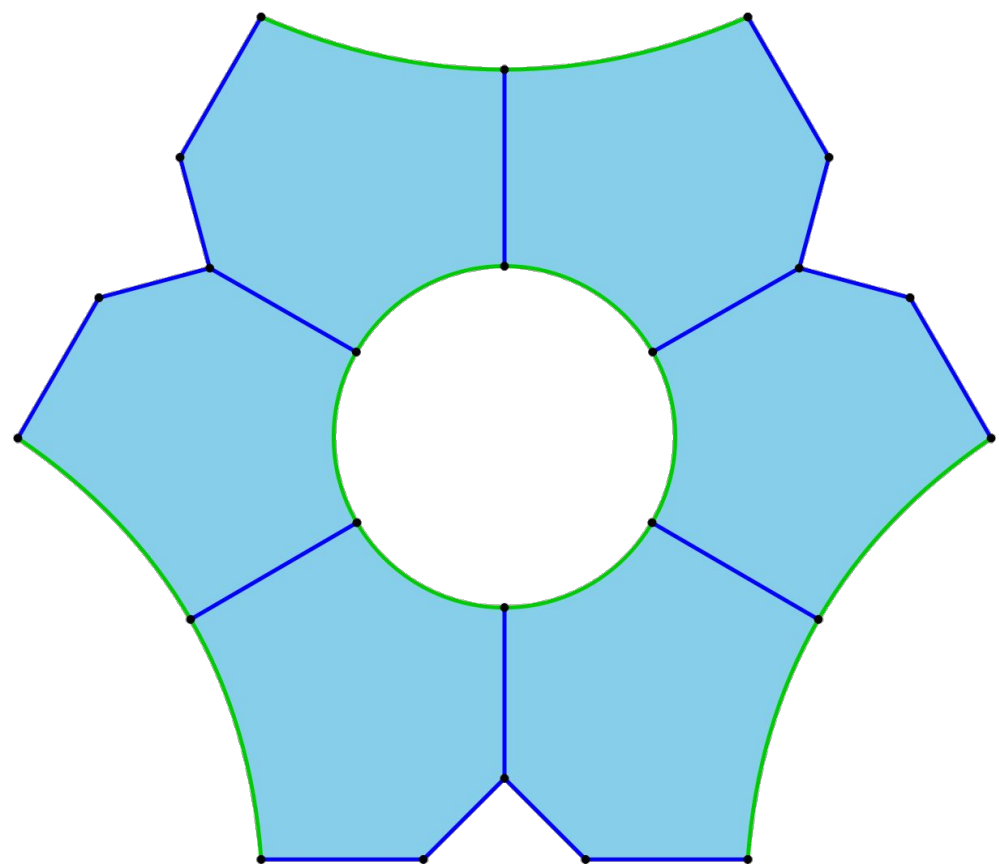
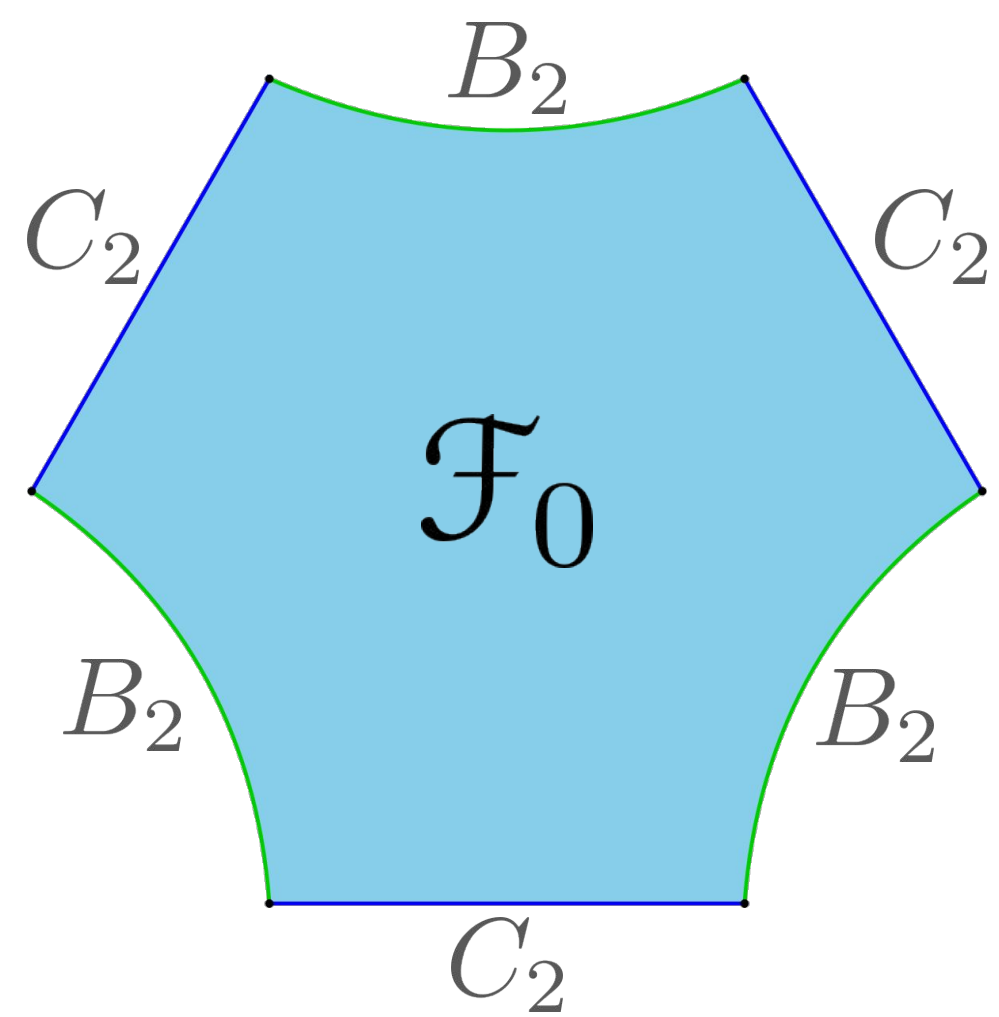
- $E_a: C_2$
- $E_l: B_2$
- $E_c: B_2$



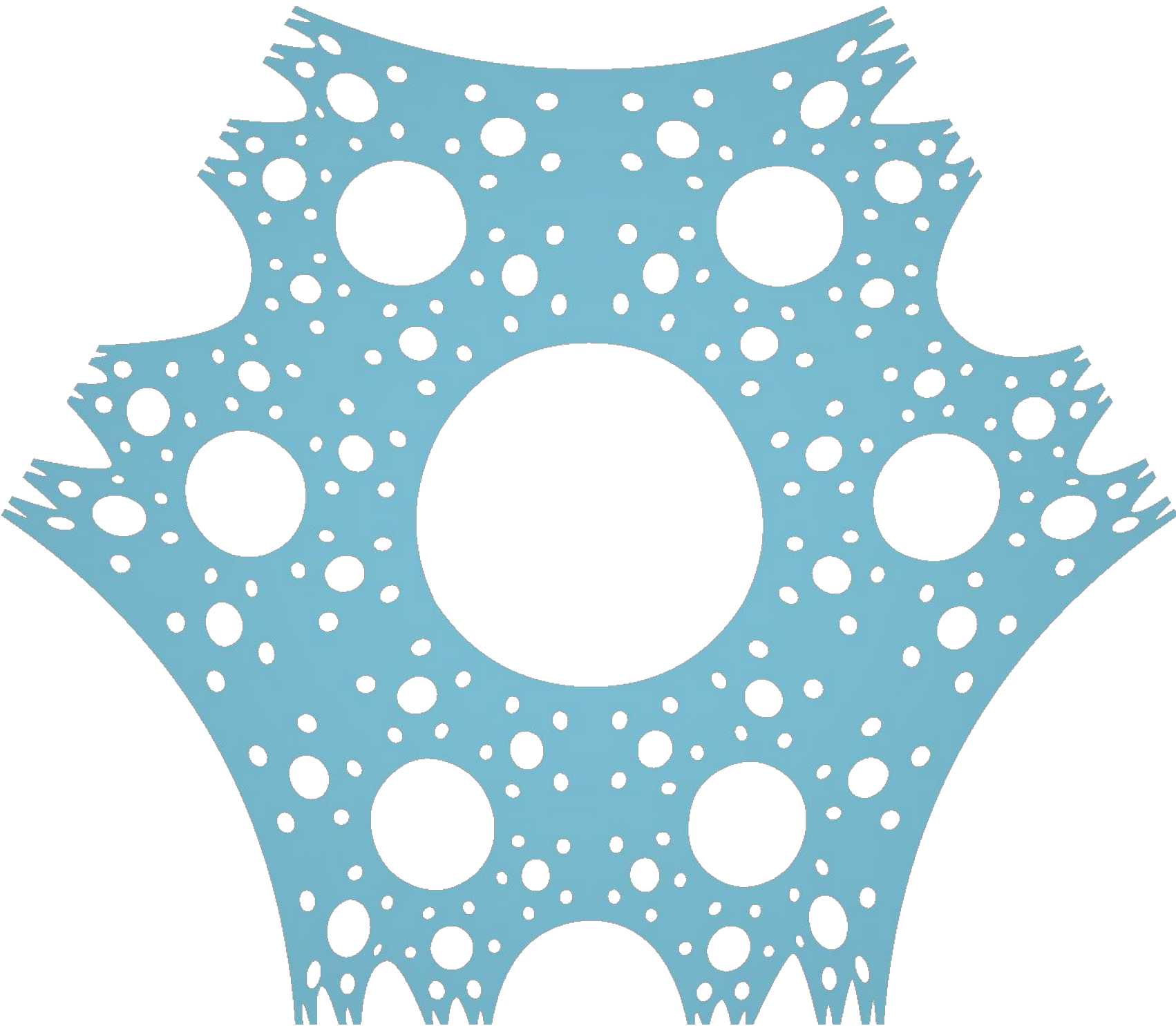
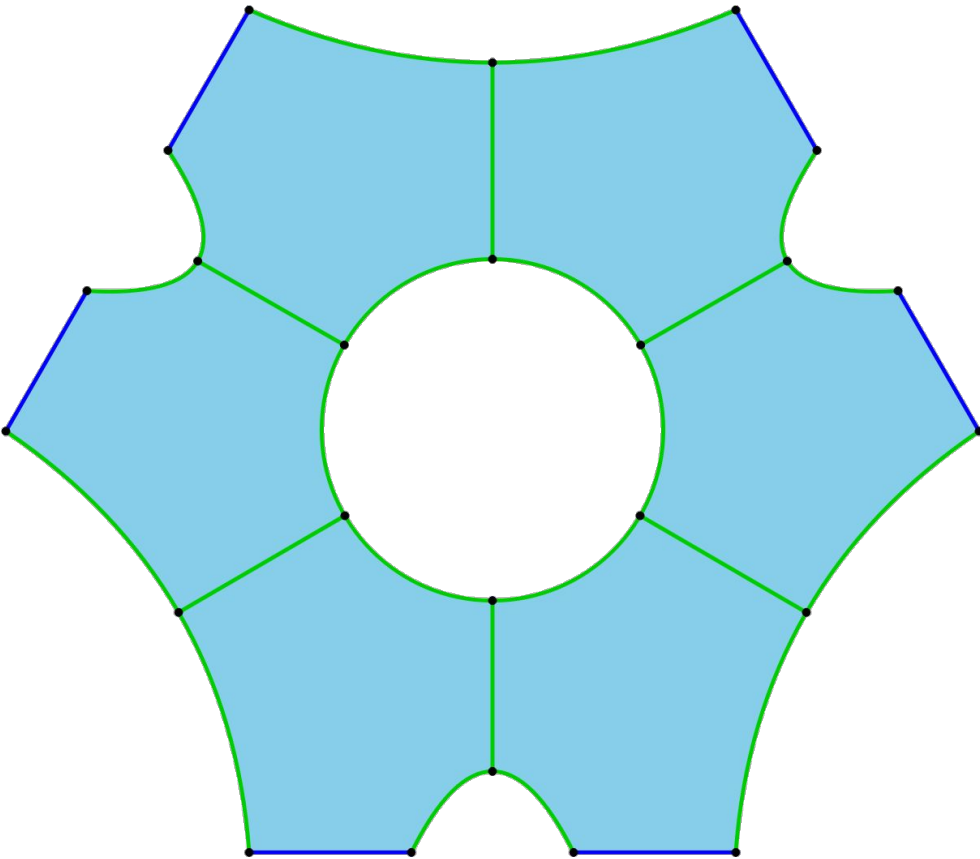
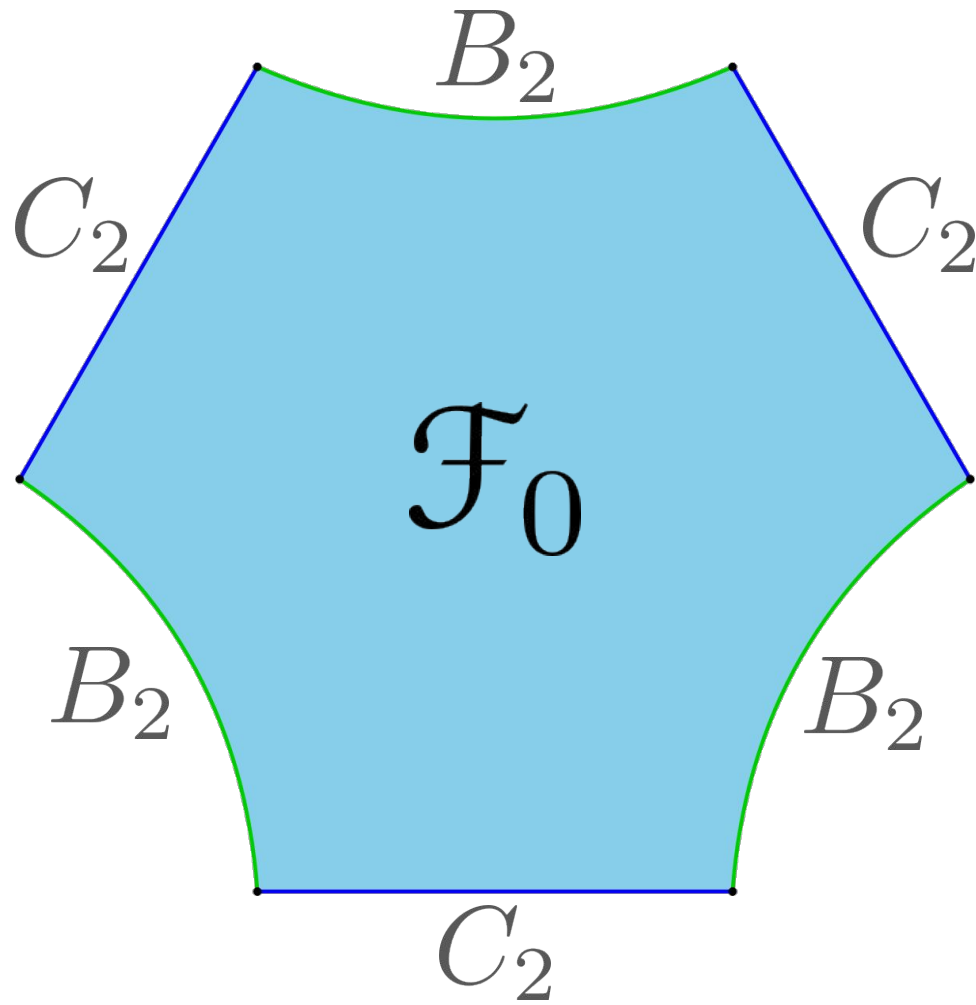
- $E_a: B_2$
- $E_l: C_2$
- $E_c: B_2$



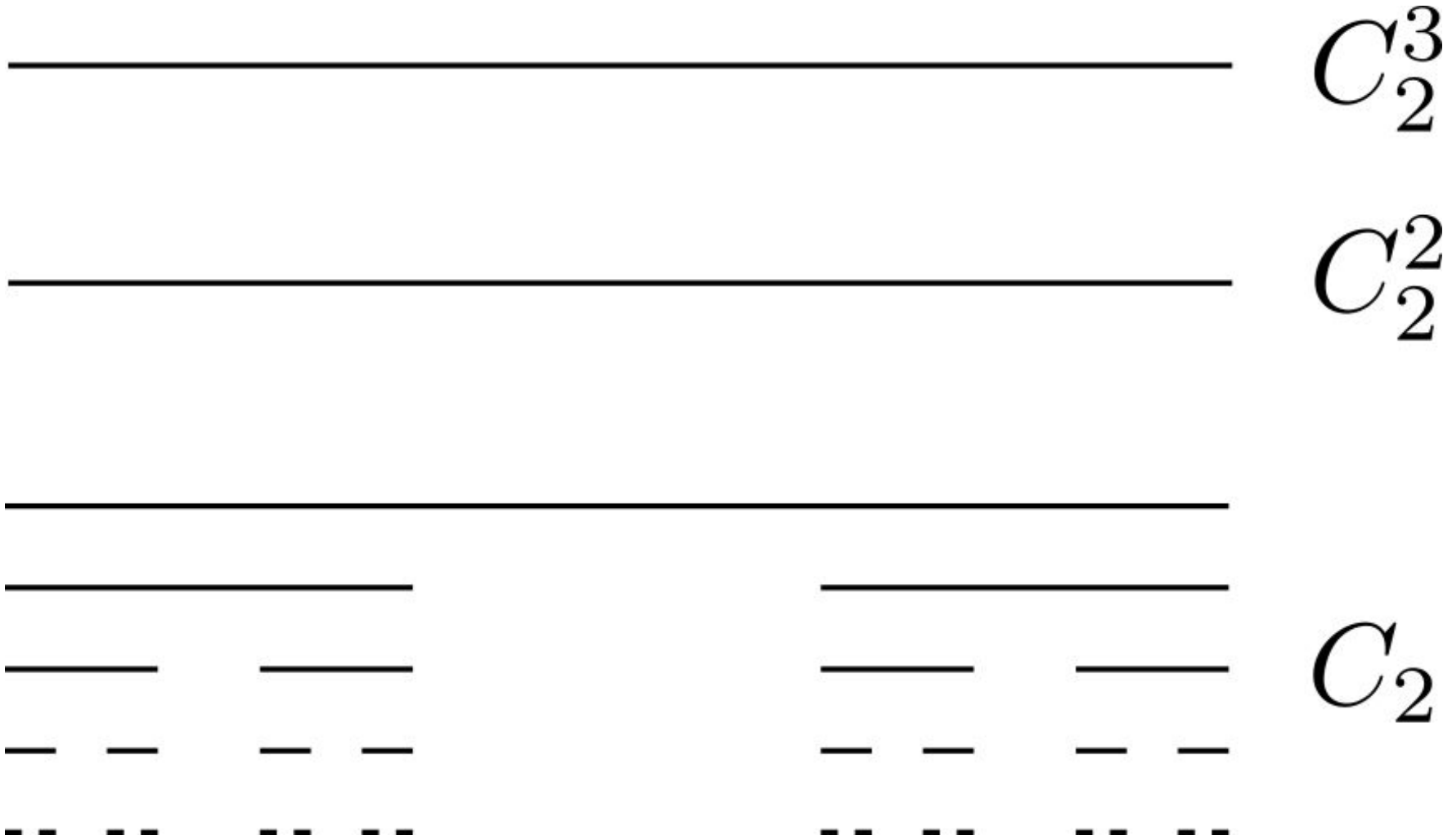
- $E_a: C_2$
- $E_l: B_2$
- $E_c: C_2$



- $E_a: B_2$
- $E_l: B_2$
- $E_c: B_2$

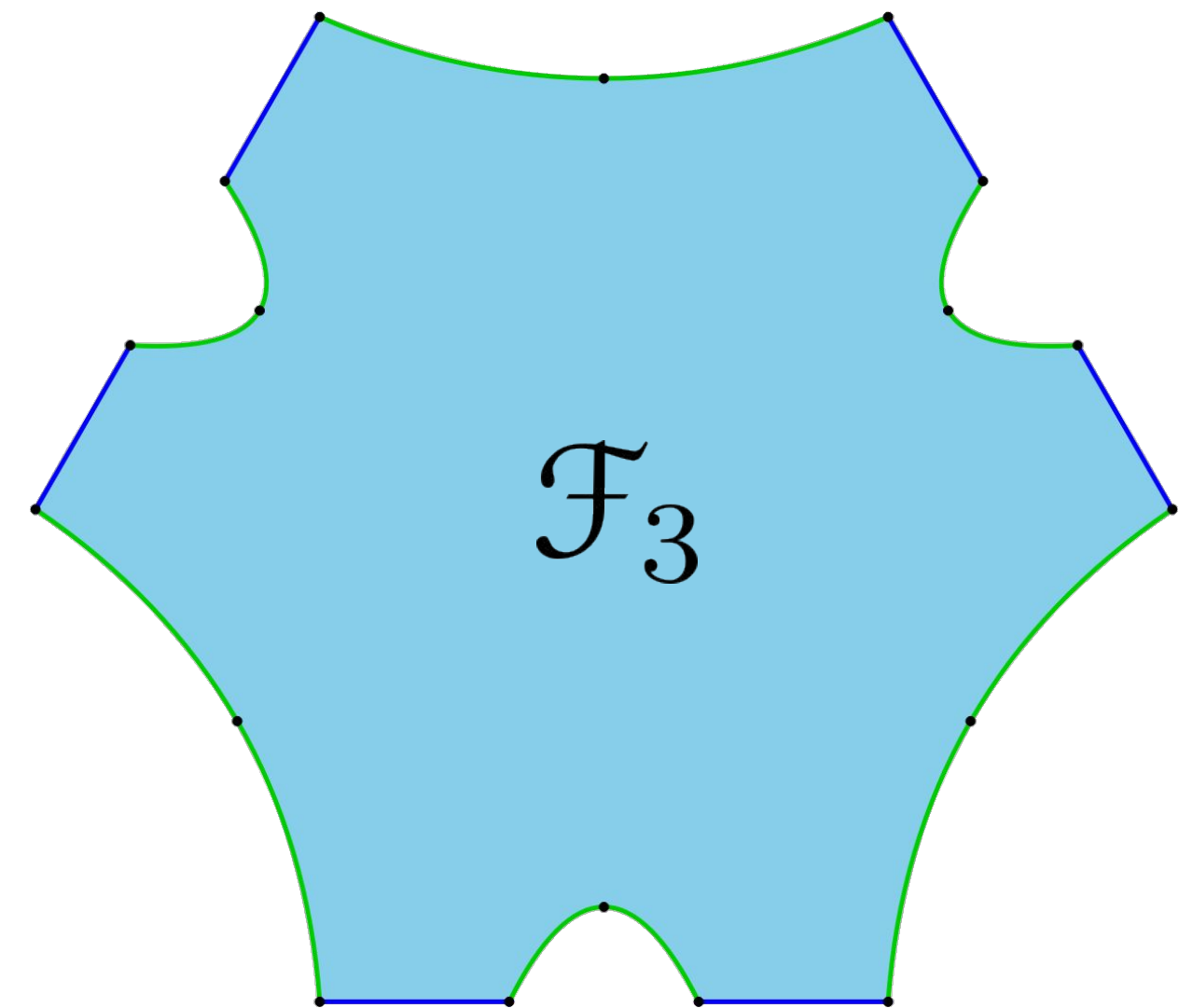
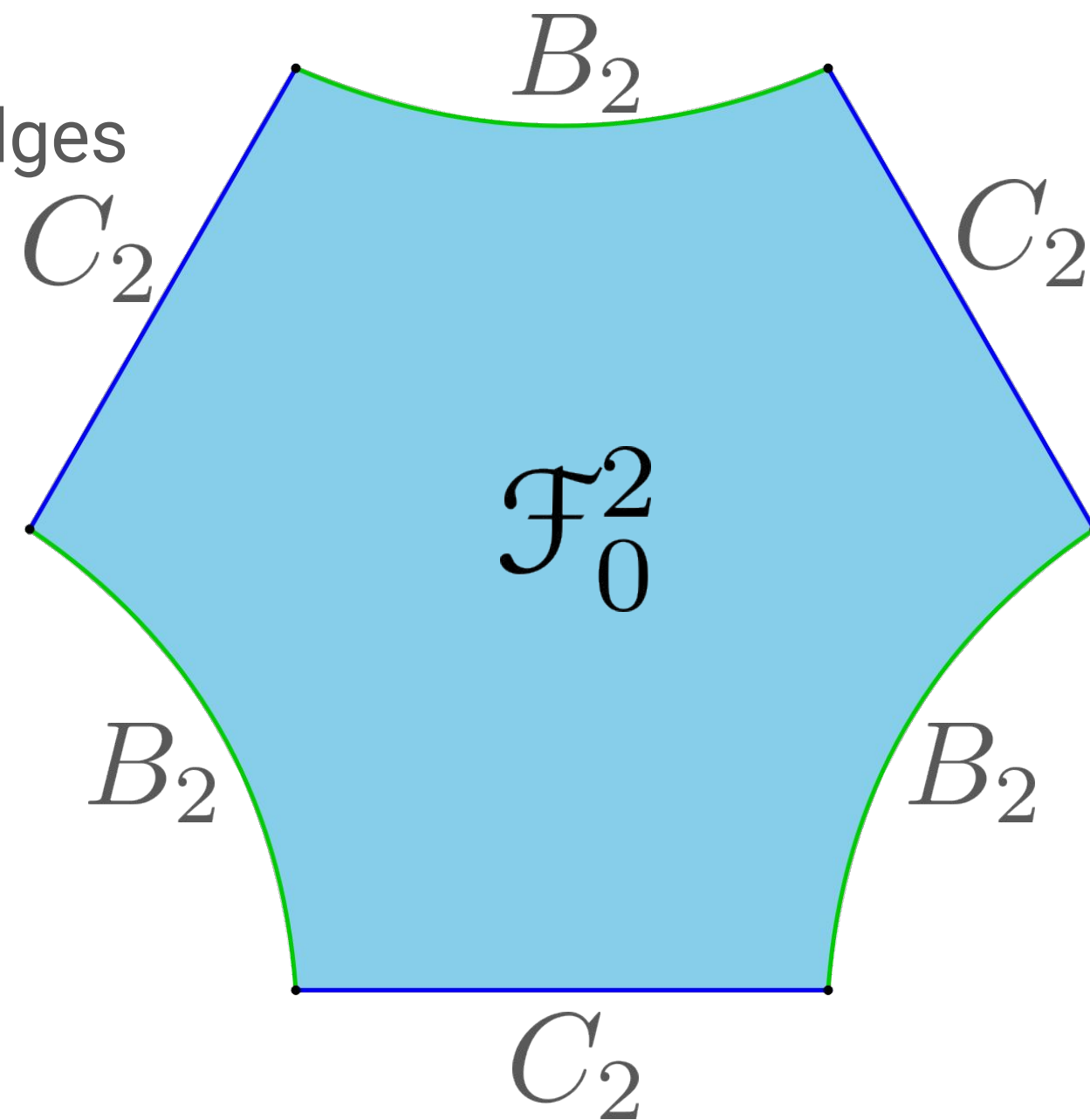


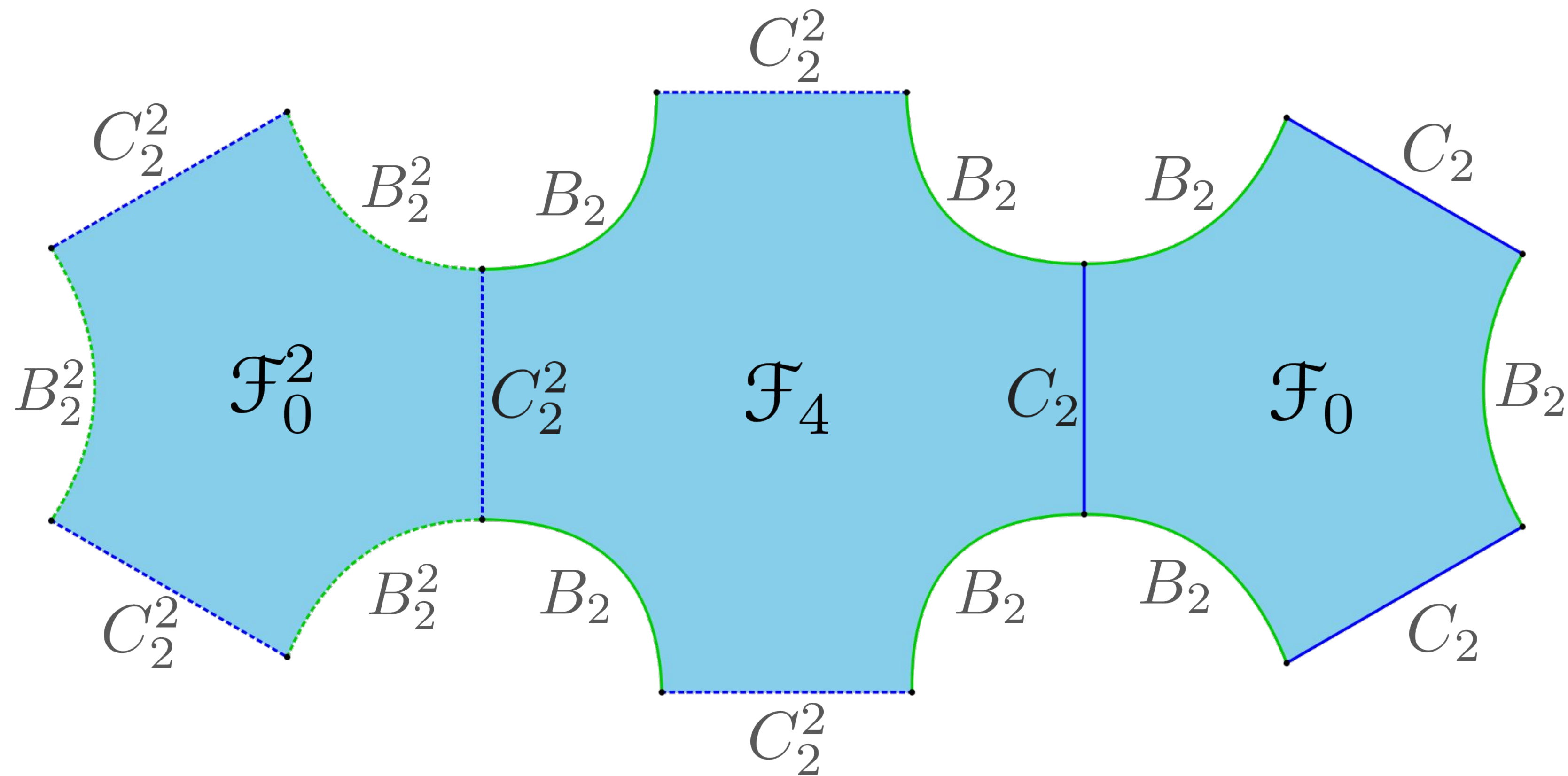
- Edges
 - Subdivide into same edge with decremented delay

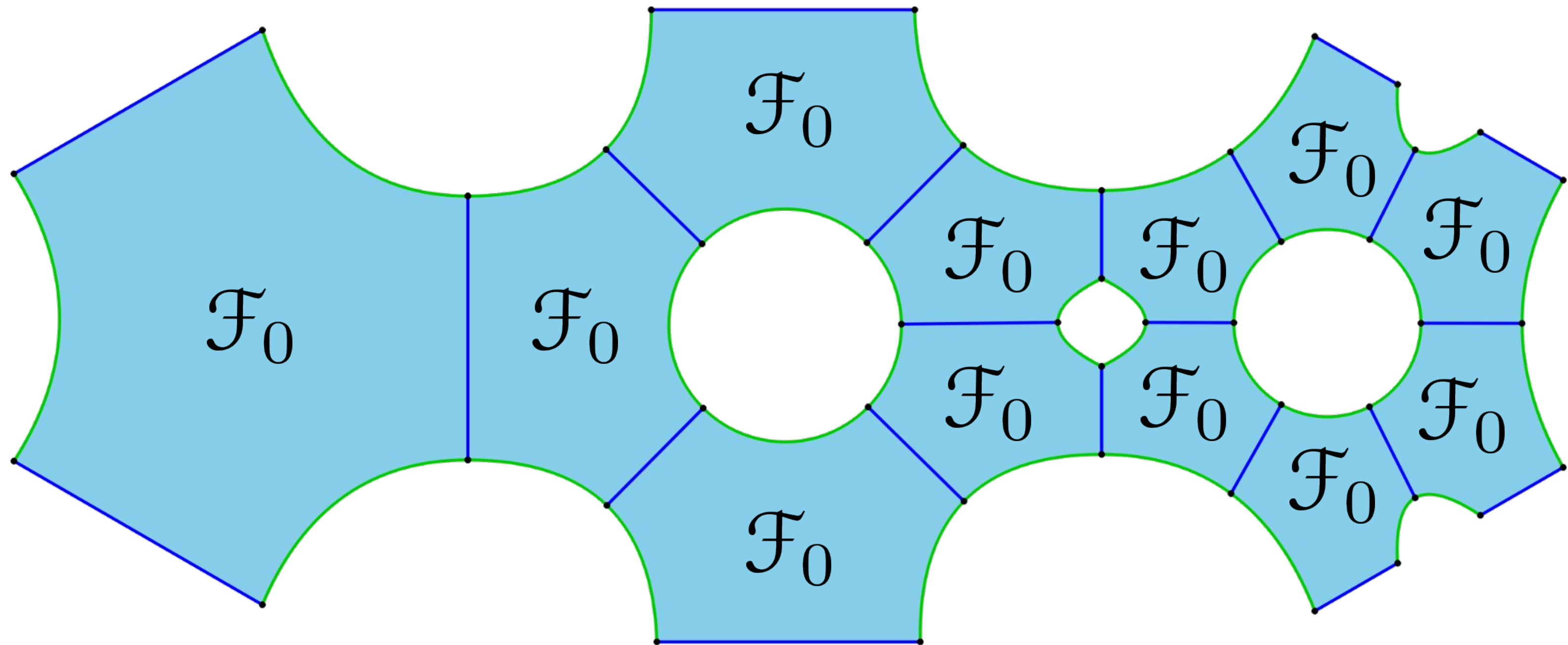


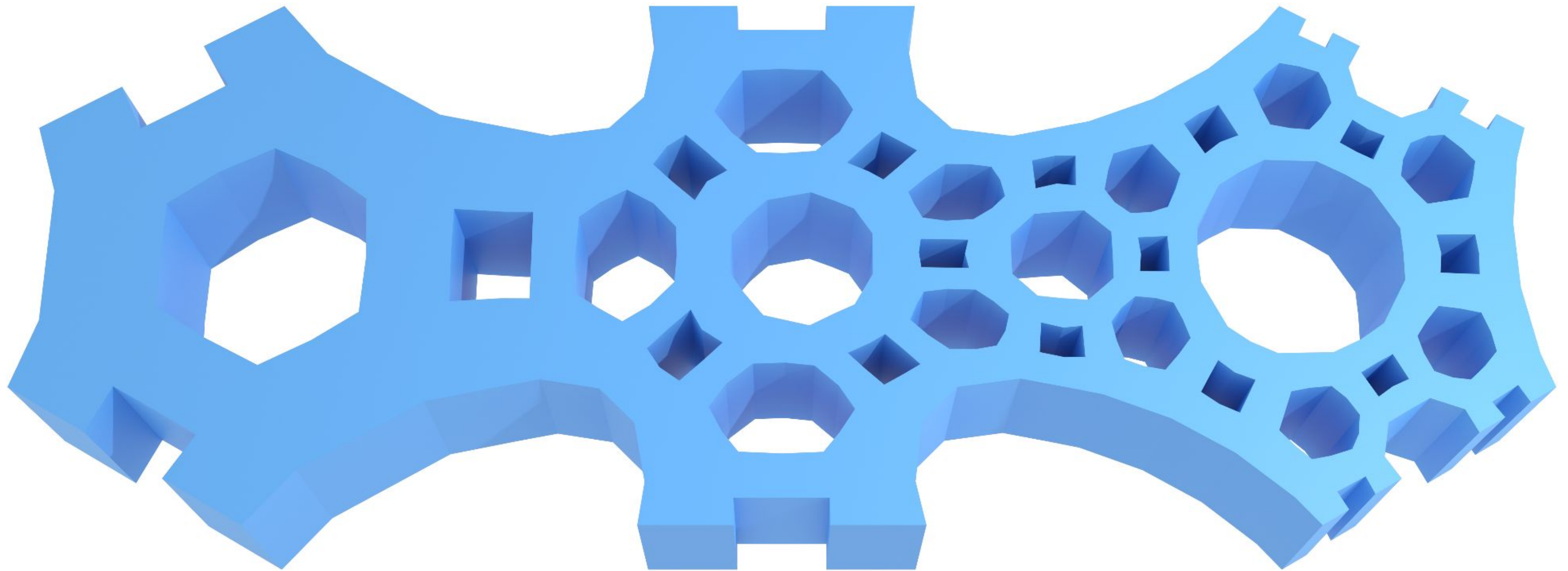
■ Faces

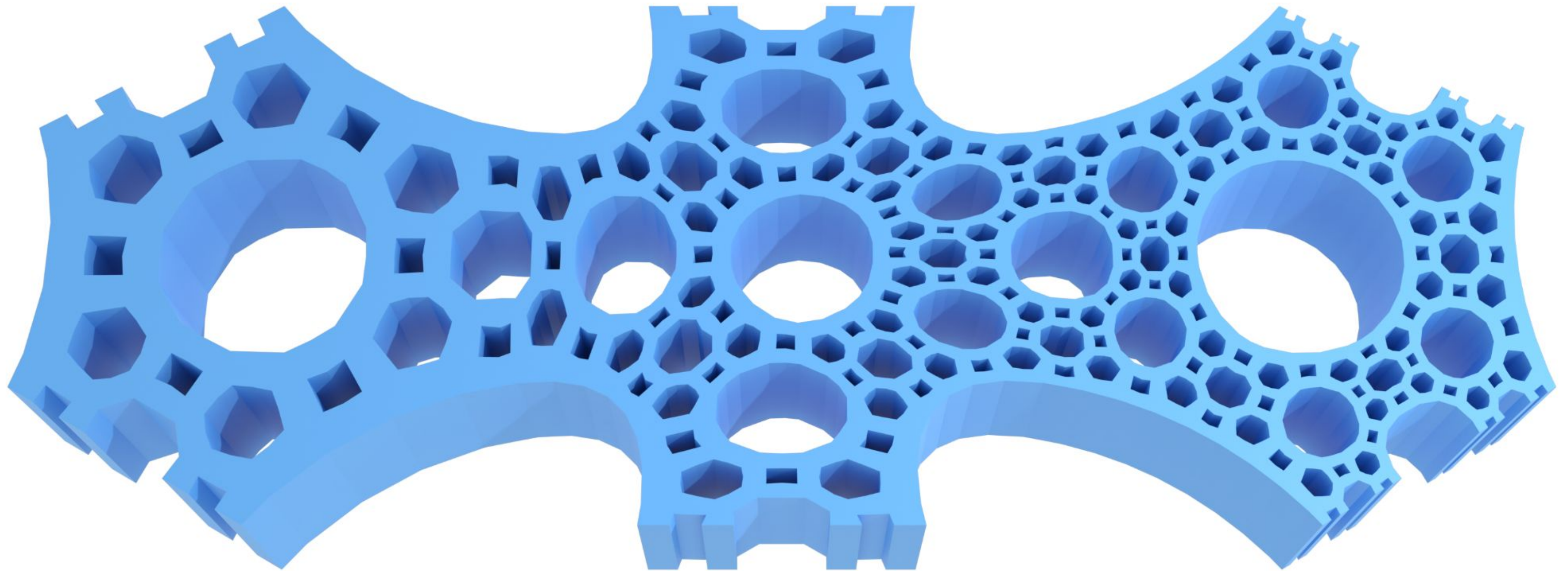
- Subdivide into one face without lacuna
- Edges are subdivided
- Need to connect sub-edges

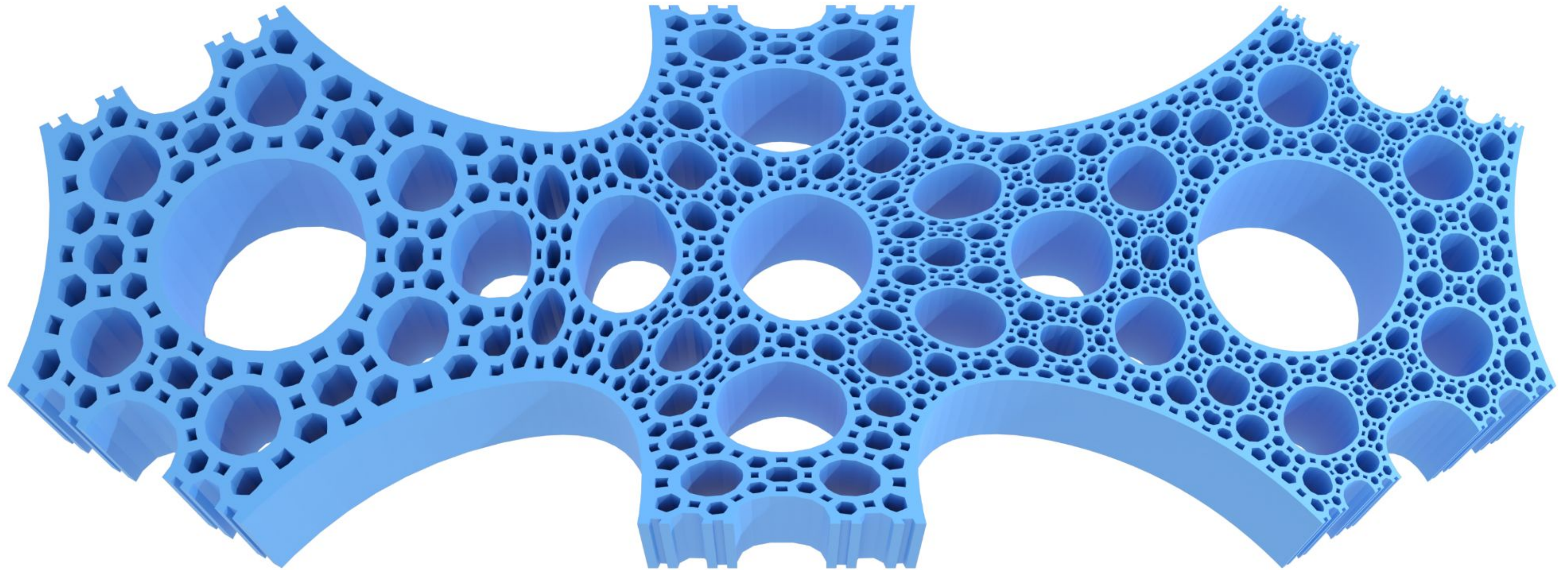


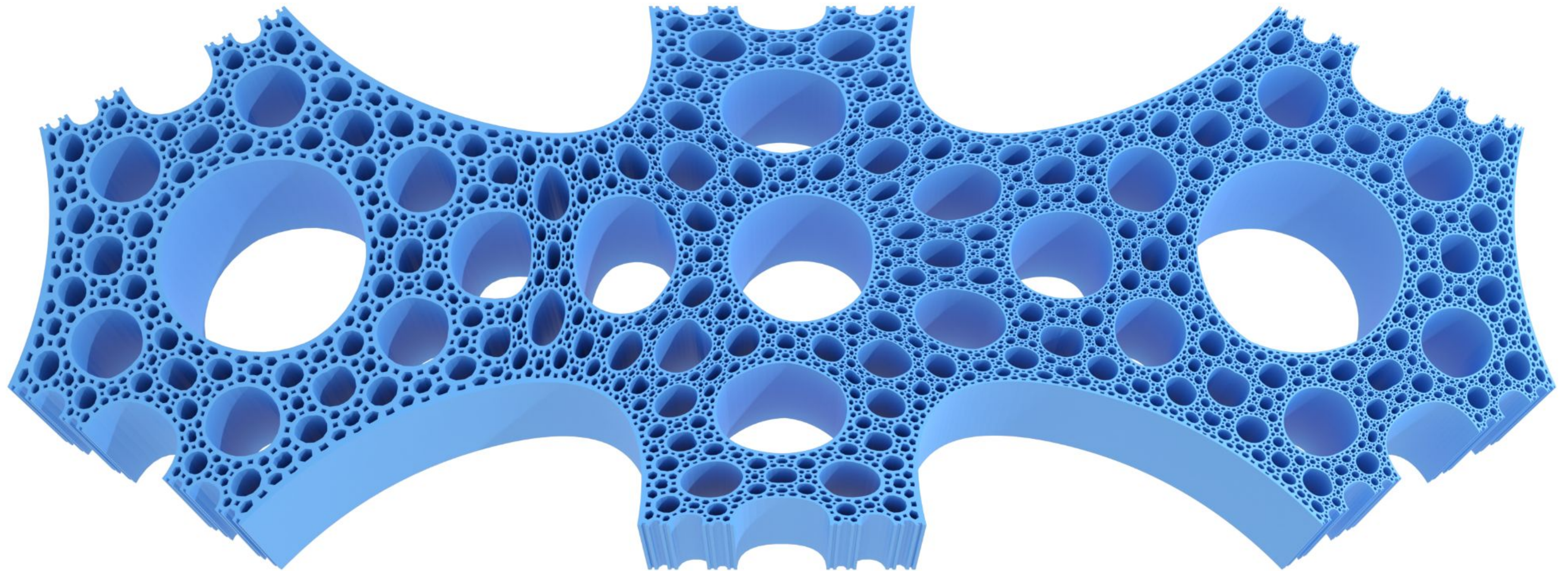


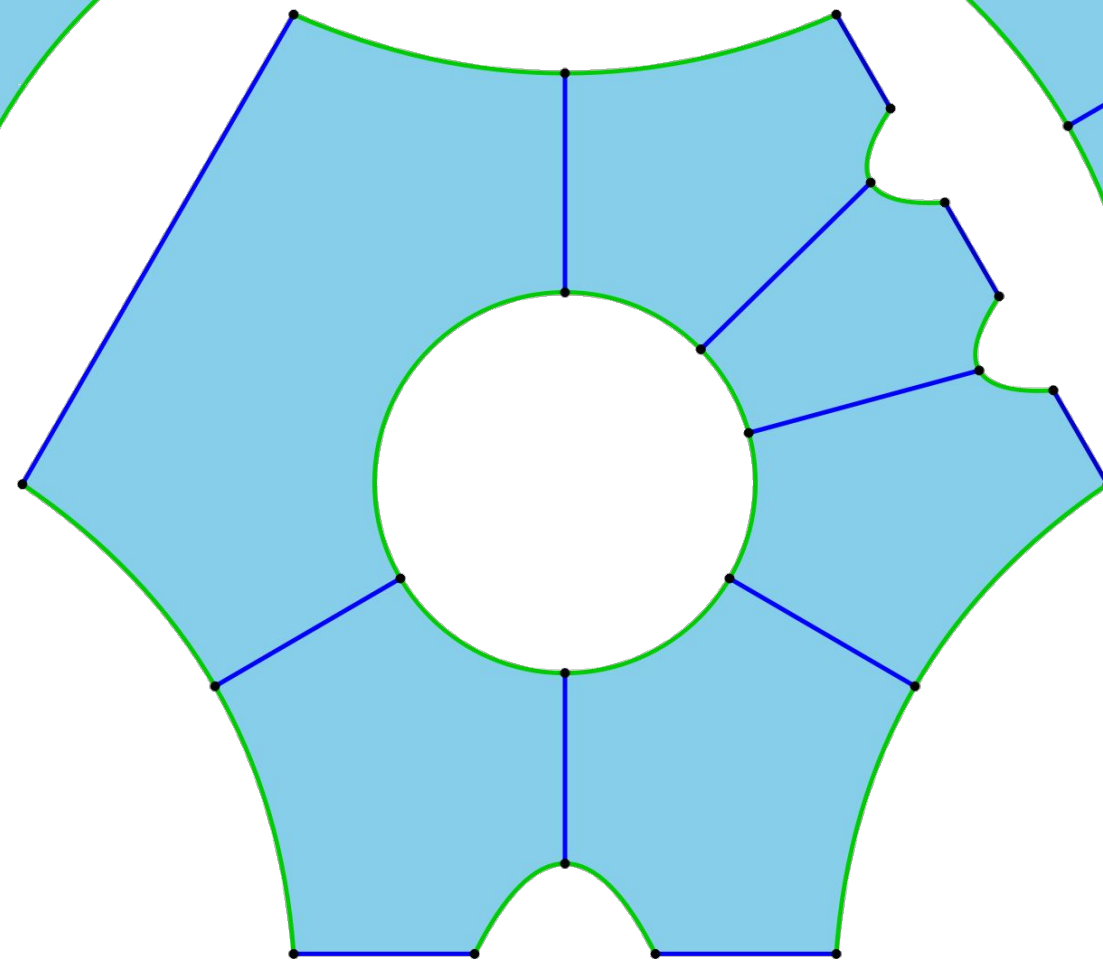
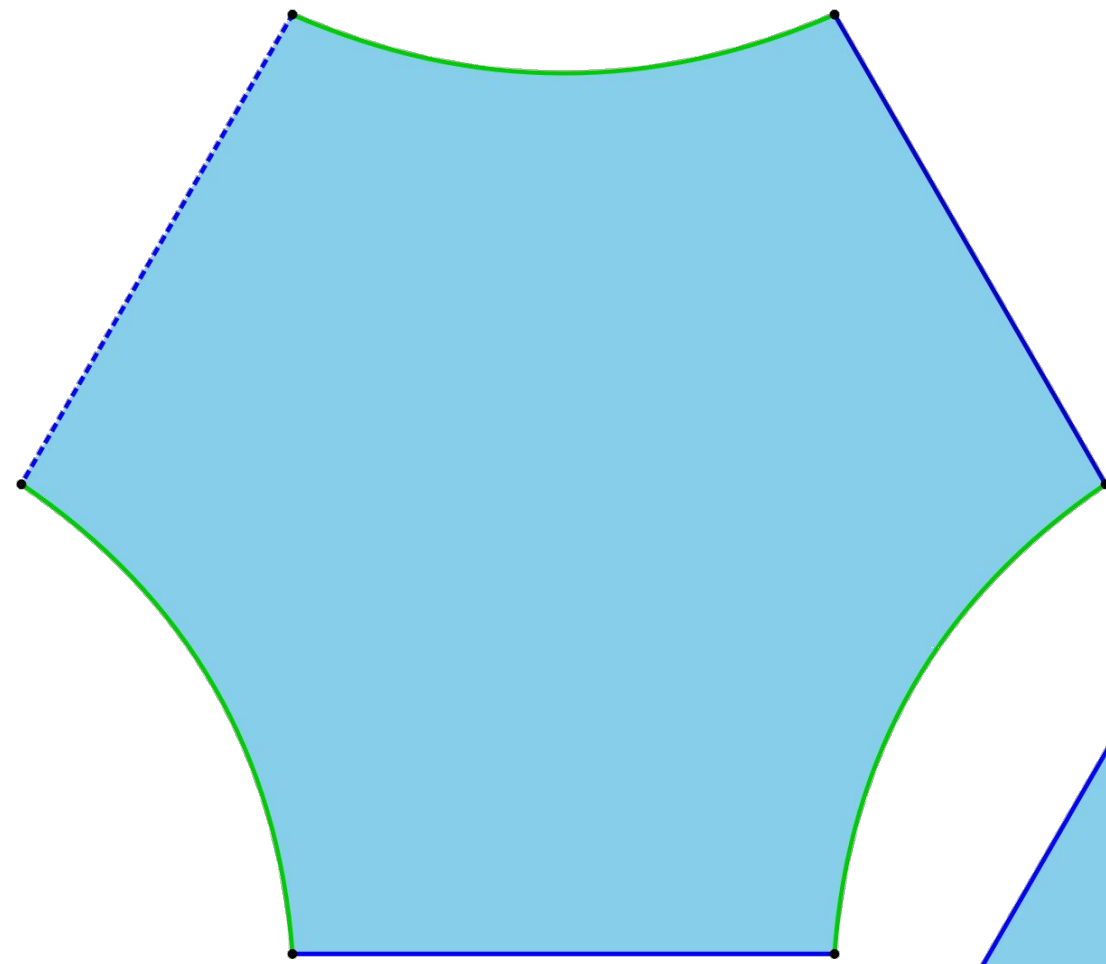




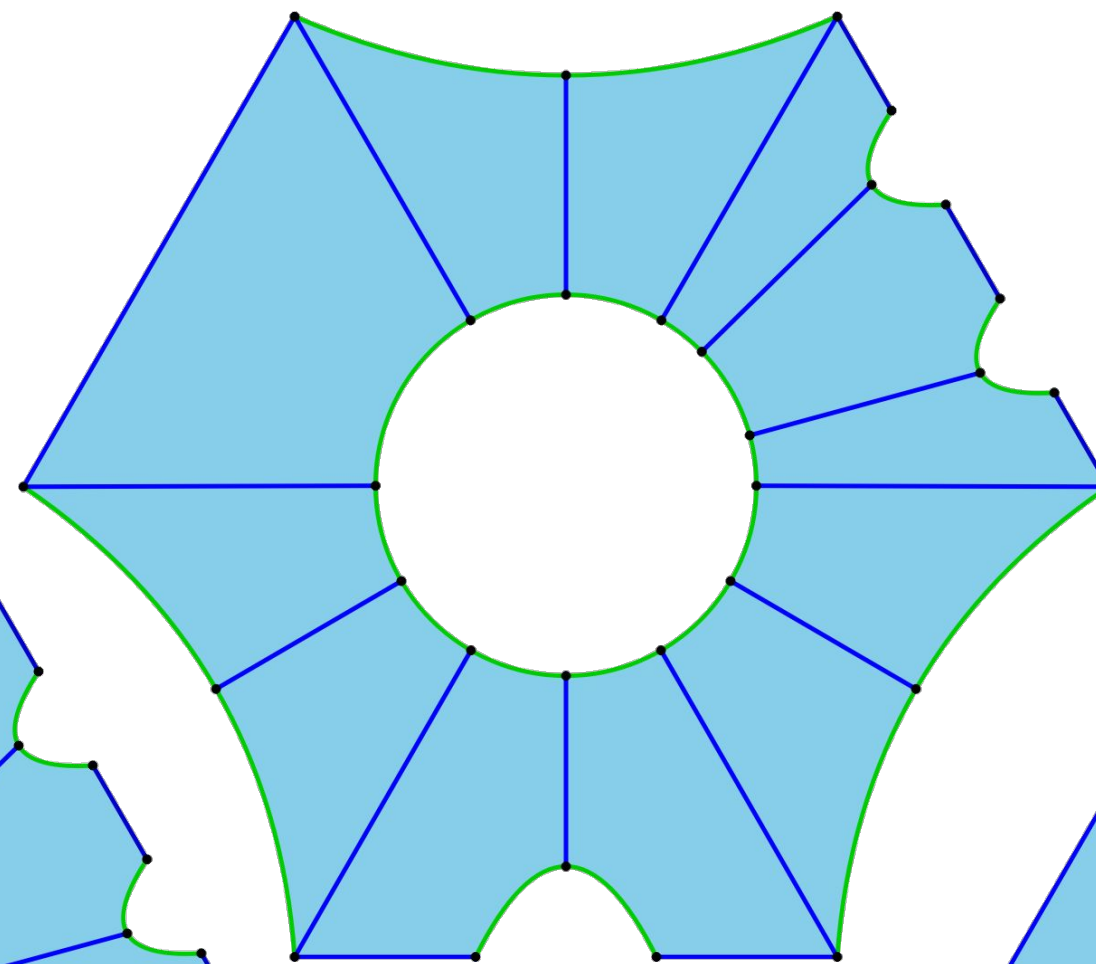




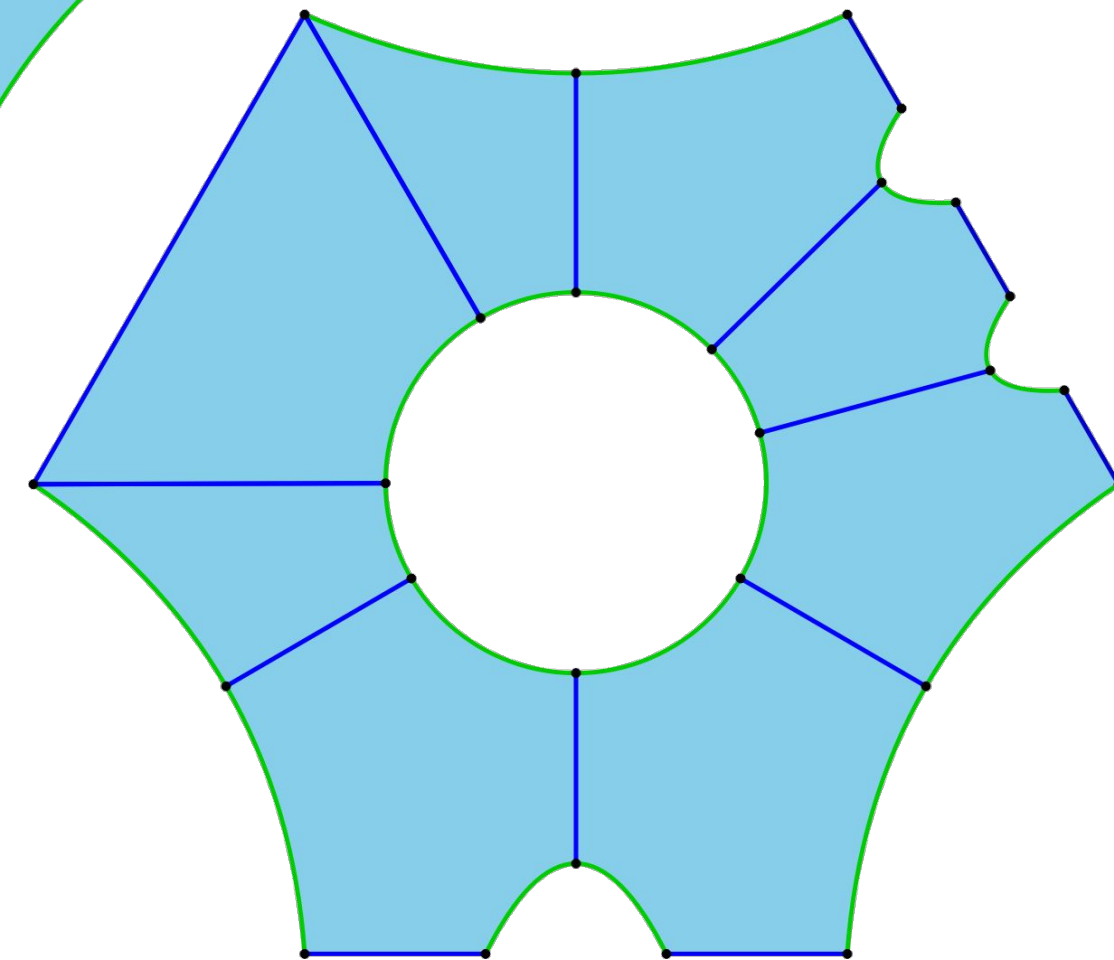




Algorithm 1



Algorithm 2



Algorithm 3

Lacunarity quantification



- Area and perimeter
- Relative density
- Fractal dimension
- **Ratio limit between lacunas and sub-faces**

- Number of sub-faces

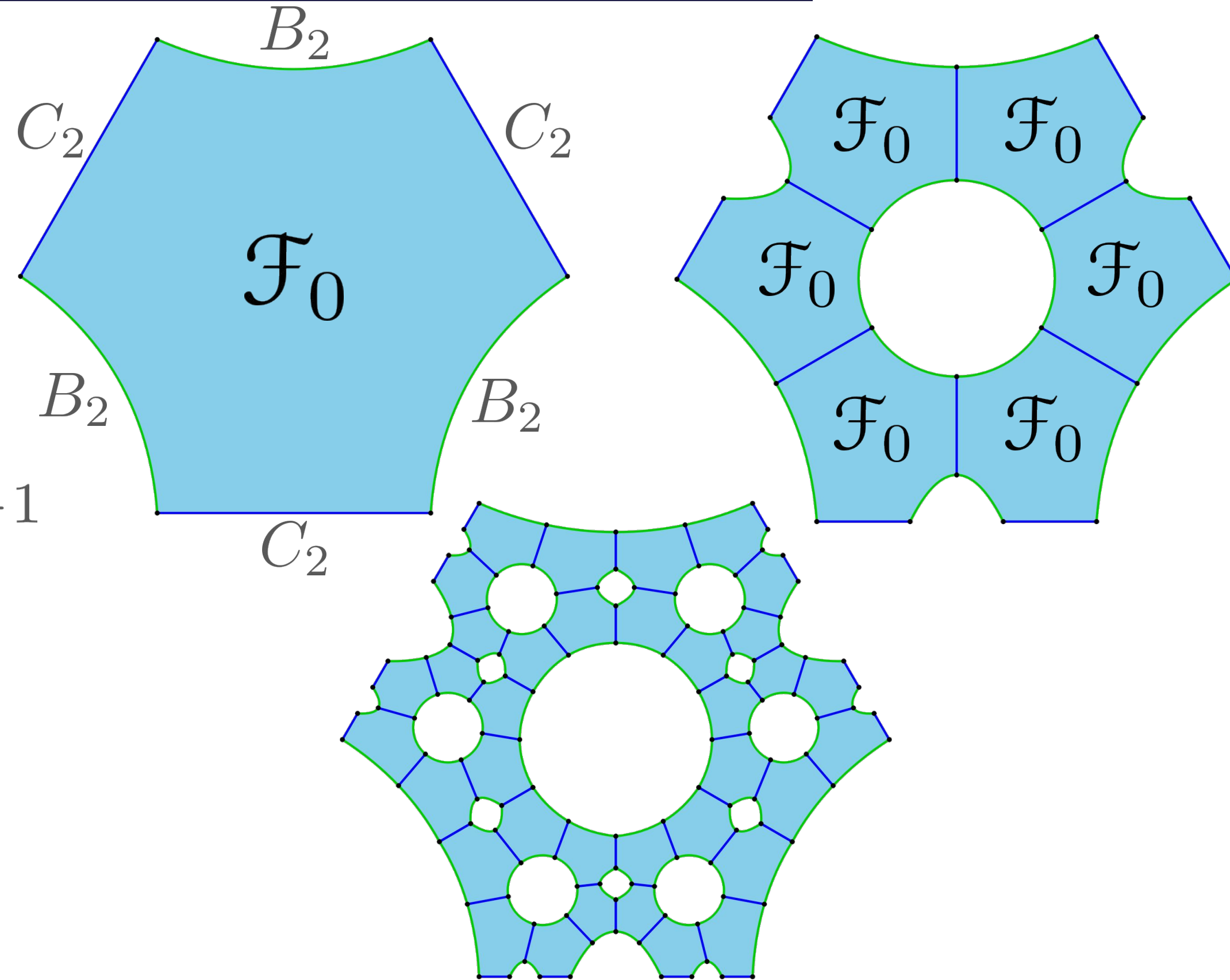
$$N_i^C = n^i$$

- Number of lacunas

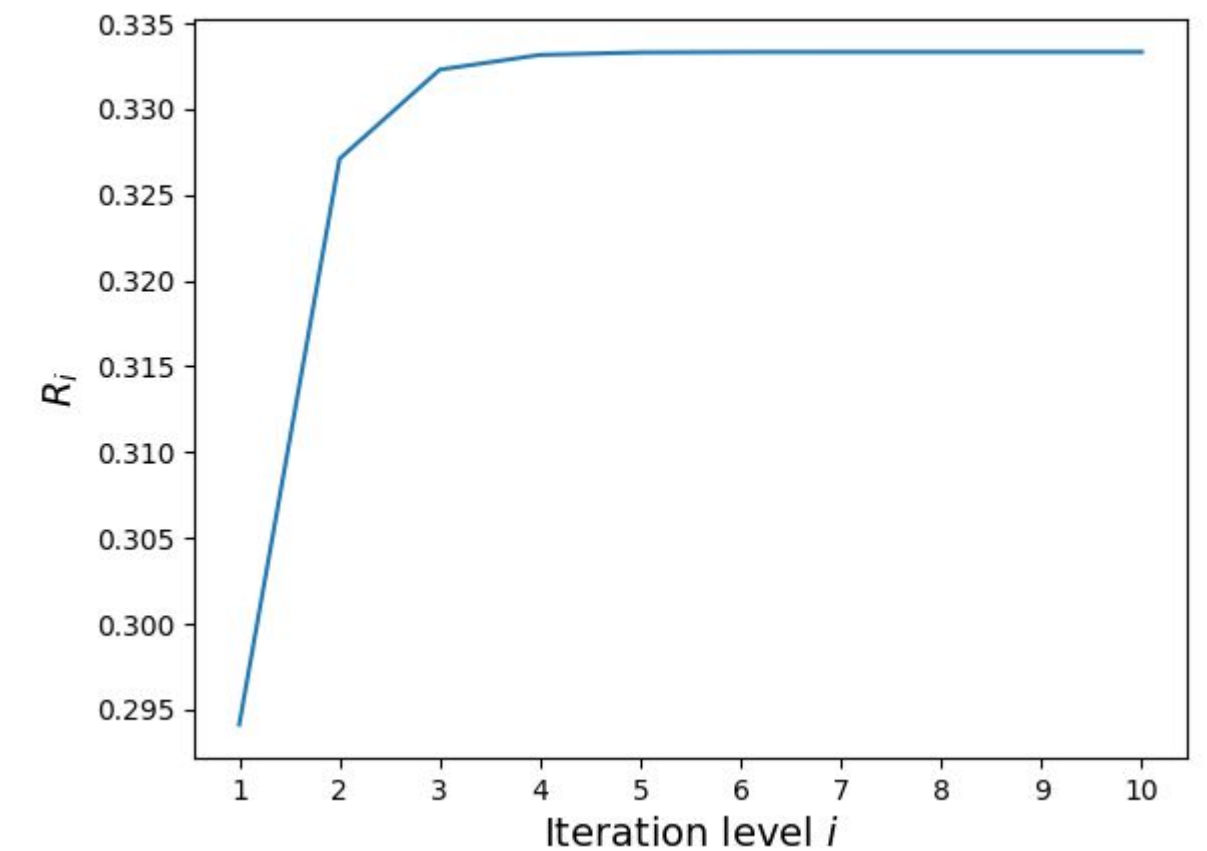
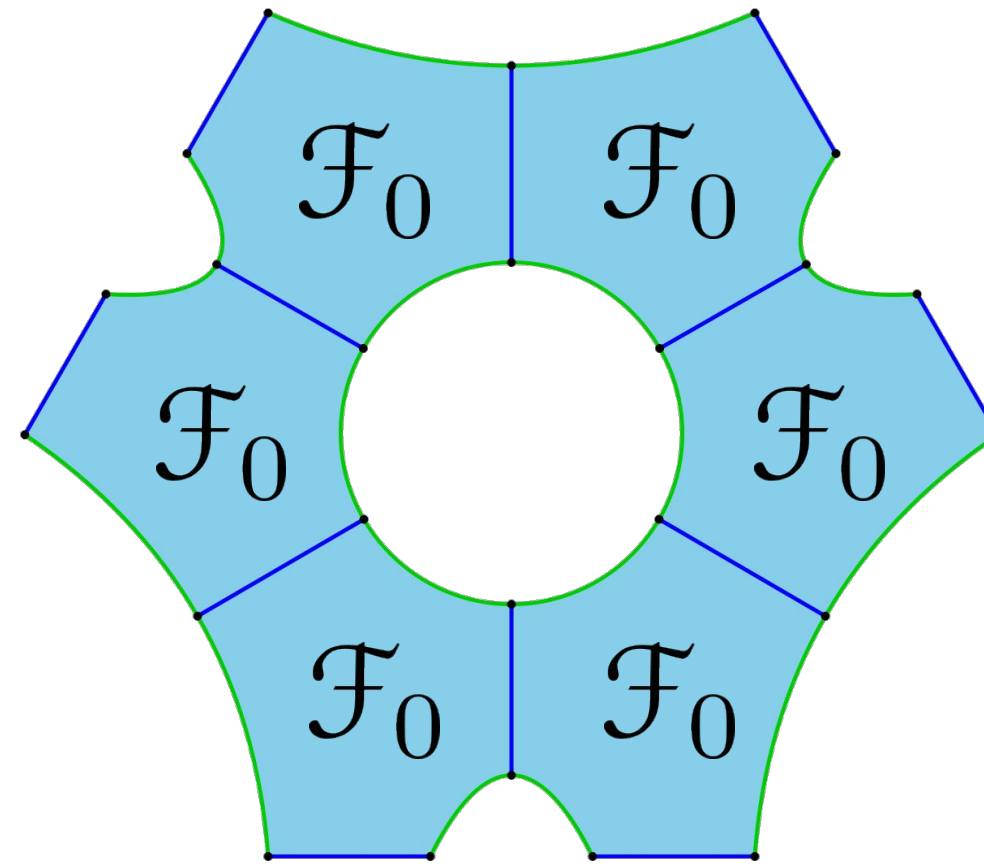
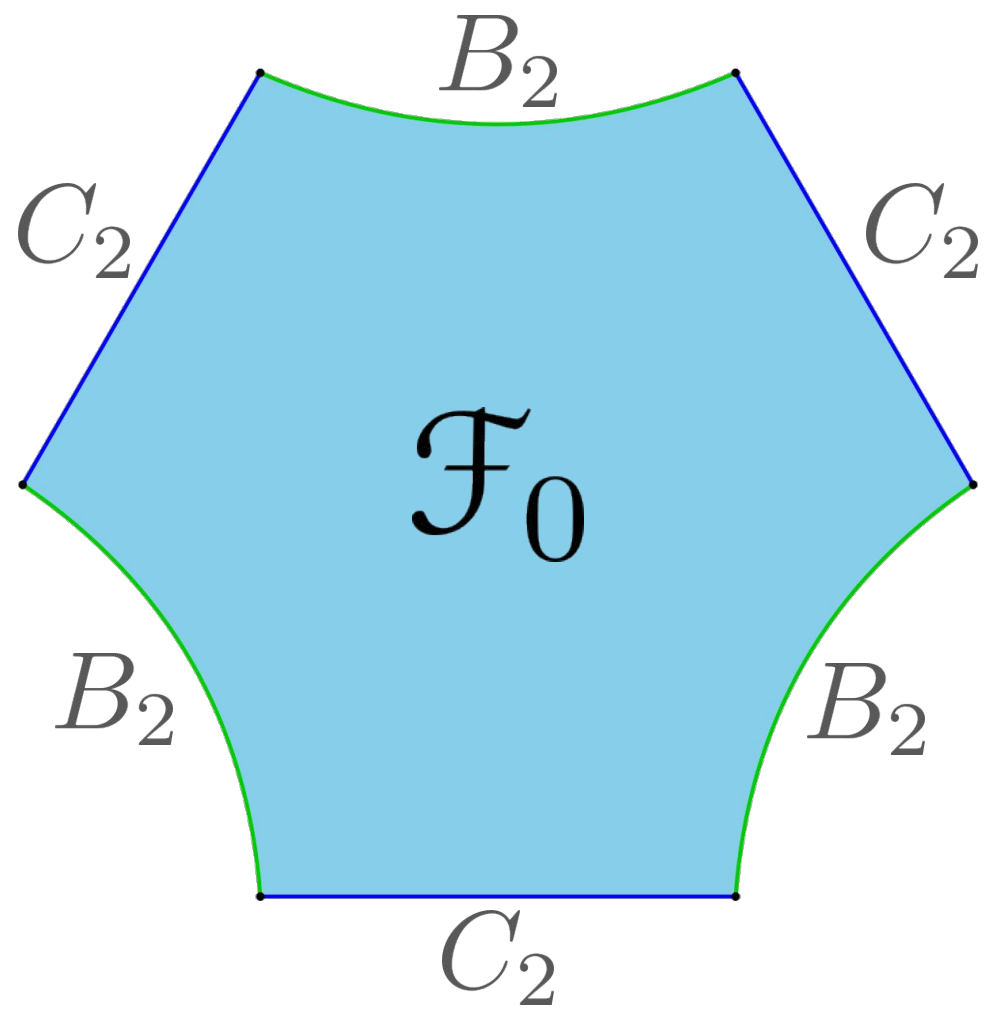
$$N_i^L = N_1^L + n \times N_{i-1}^L$$

- Ratio

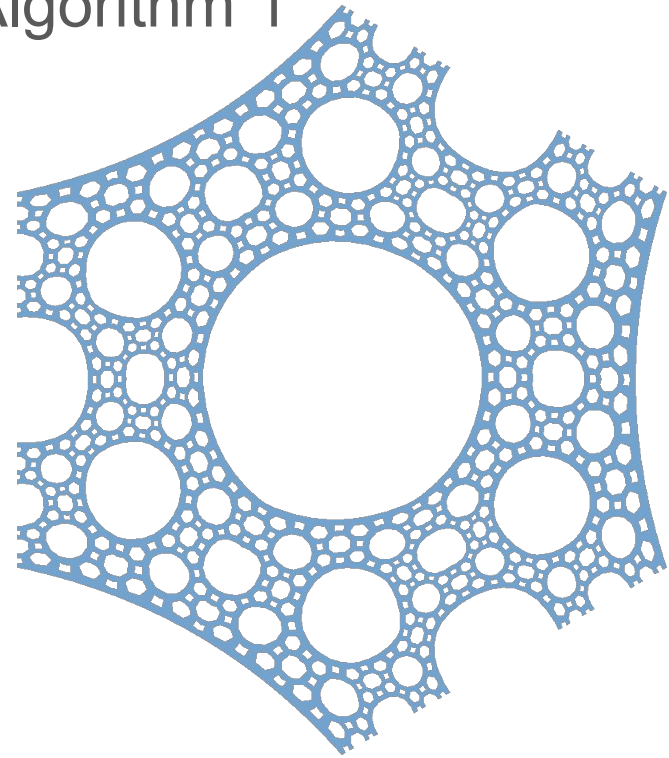
$$R_i = \frac{N_i^L}{N_i^C + N_i^L}$$



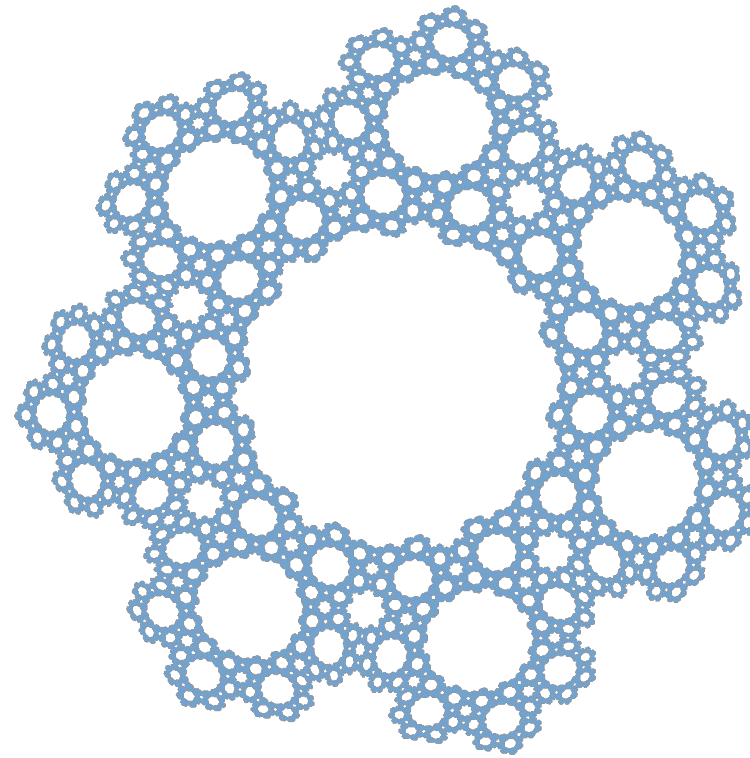
$$R = \lim_{i \rightarrow +\infty} R_i = \frac{N_1^L}{N_1^L + N_1^C - 1}$$



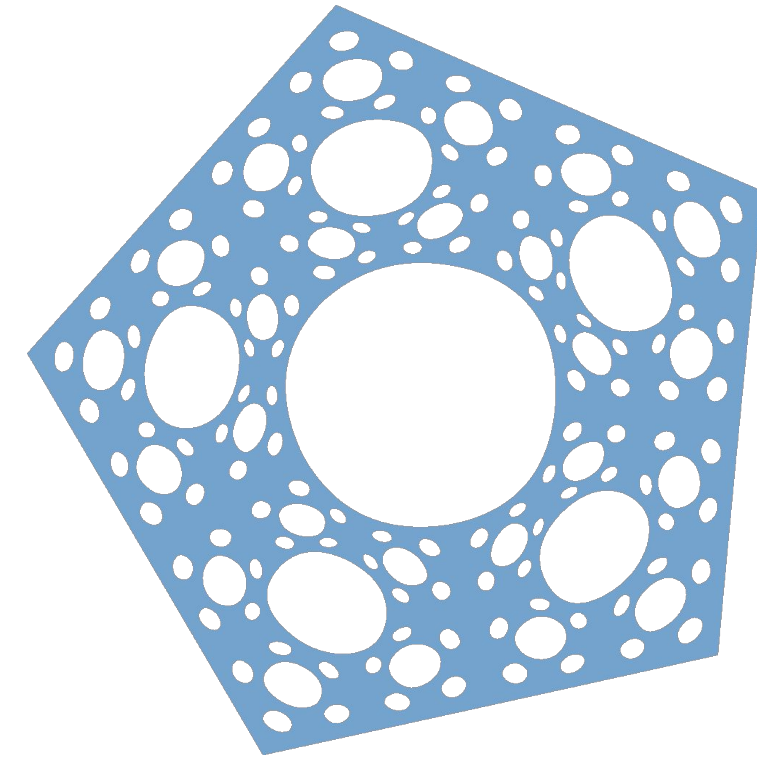
Algorithm 1



$$R = 0.33$$

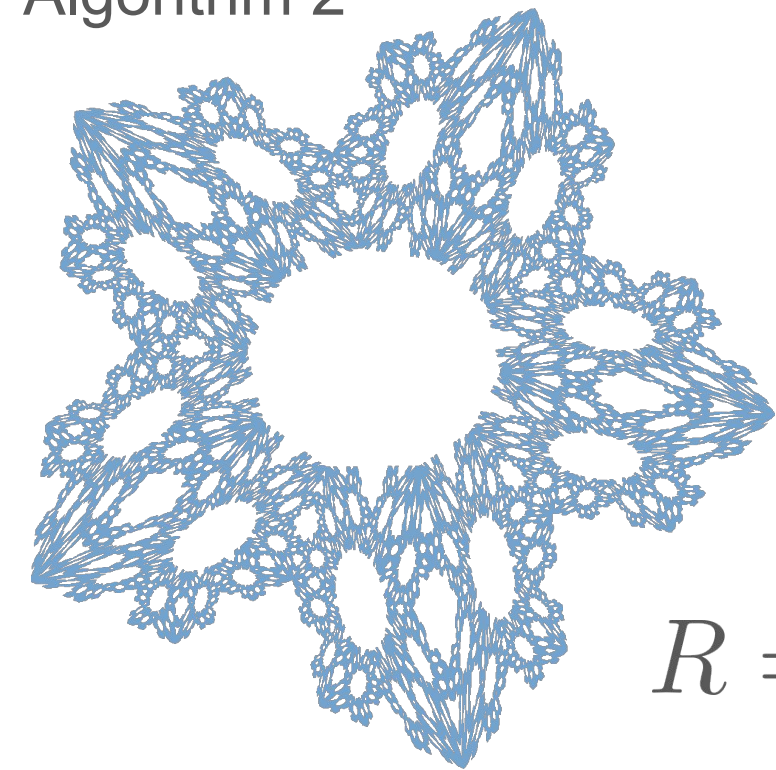


$$R = 0.43$$

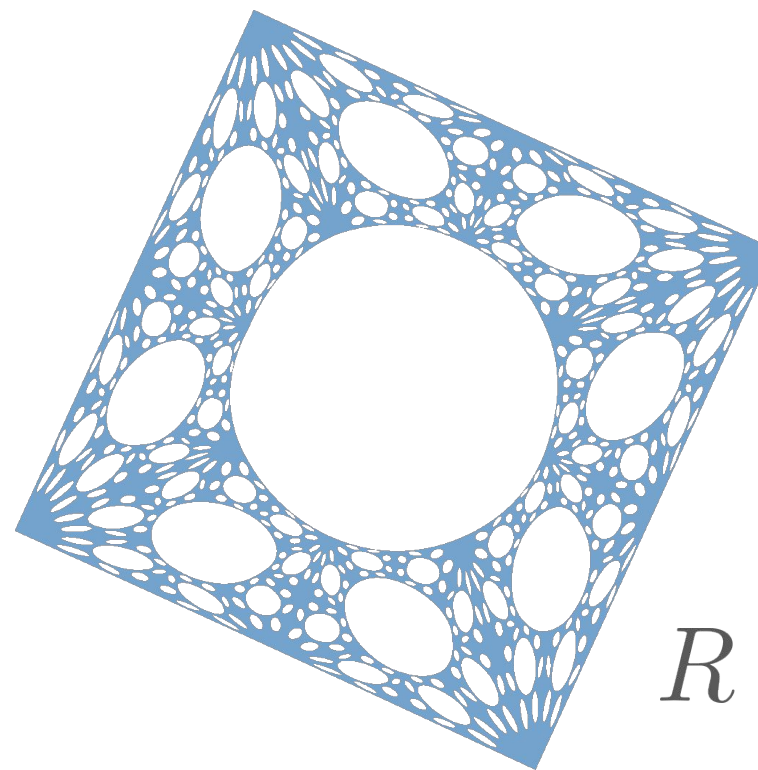


$$R = 0.20$$

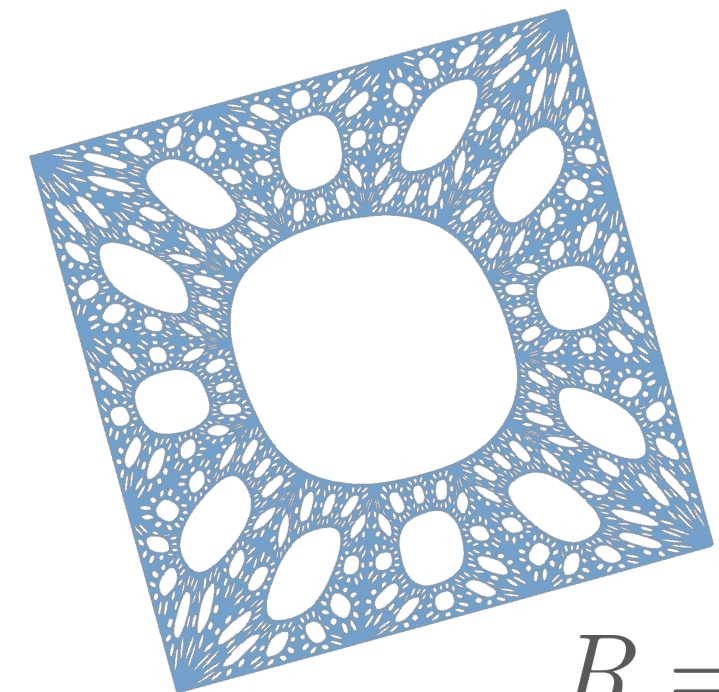
Algorithm 2



$$R = 0.28$$



$$R = 0.12$$



$$R = 0.08$$

Conclusion



- Algorithms to describe subdivision of a 2D face

- Local variation of lacunarity

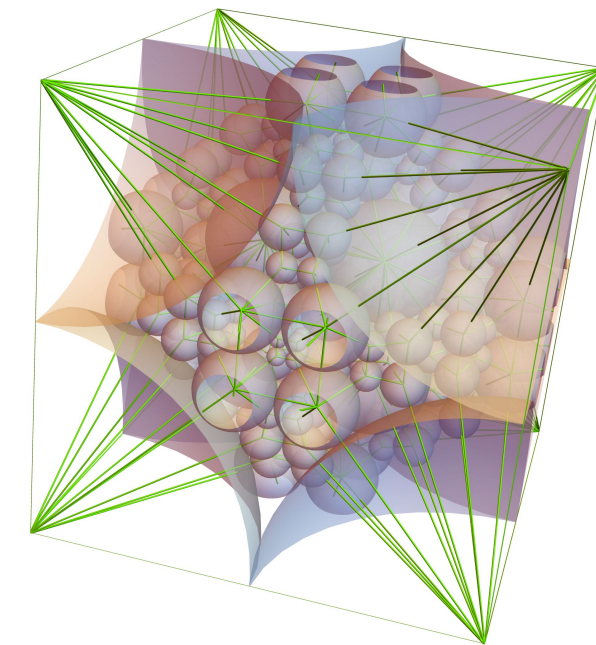
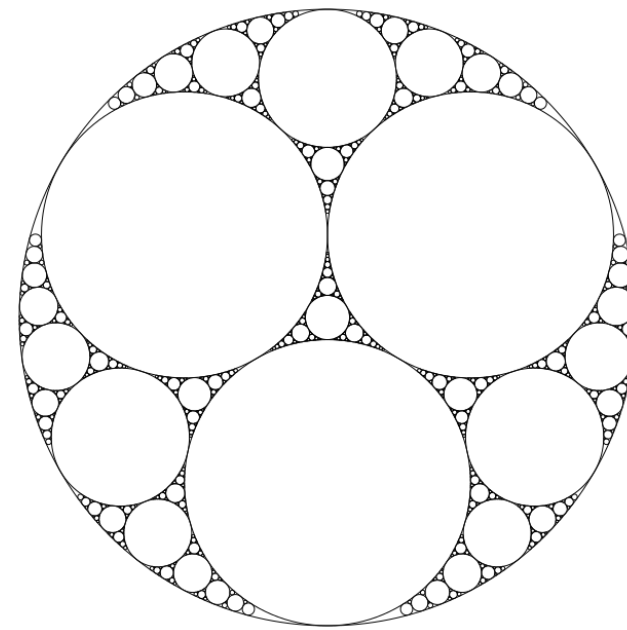
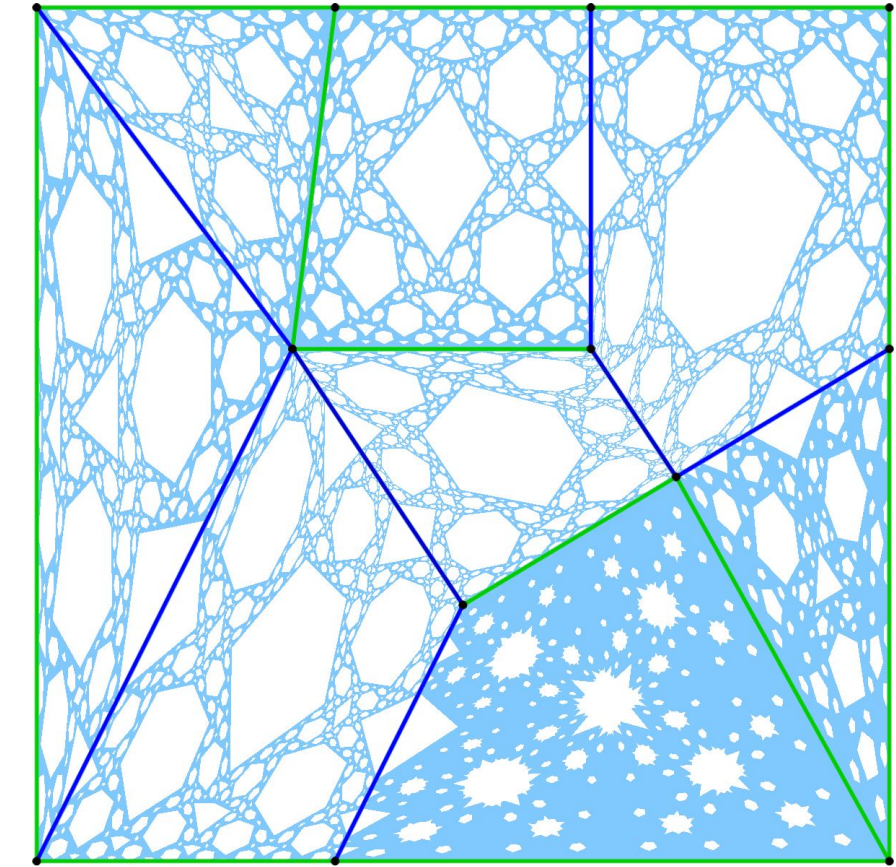
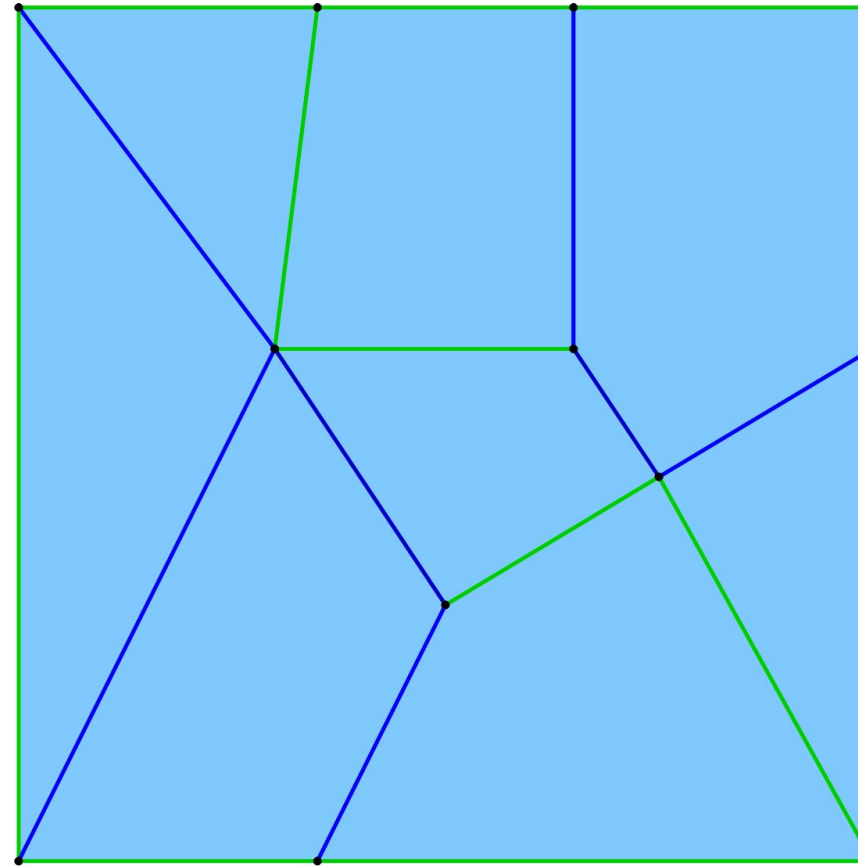
- Future works

- 3D automatic subdivision

- Find other 2D algorithms

- Polytopal sphere packings

- Pipeline for topological optimization



Thanks for your attention



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